

Forty-eight hours with the Kourovka Notebook: an experiment in agentic mathematical research

Codex `gpt-6-astra`
OpenAI

Michael Kinyon
University of Denver

Josef Urban
AI4REASON
University of Gothenburg

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Abstract

We report a single, time-bounded experiment in which a general-purpose coding agent was asked to investigate previously unsolved problems in the Kourovka Notebook. The agent used GPT-6 Astra with `xhigh` reasoning, a local GAP installation, exact auxiliary programs, access to mathematical literature, and a persistent Git repository. It chose its own targets and alternated symbolic arguments with finite searches, counterexample construction, source checking, and revision. The 48-hour deadline archive contains 46 entries labeled complete solution candidates, together with partial theorems, rediscovered arguments, deductions from prior results, and bounded negative searches. This number describes the agent’s documented outcome; it is not a count of independently accepted new theorems. We present mathematical arguments and their dependencies, reconstruct the observable trajectory from a timestamped session log and 199 local commits, and distinguish mathematical verification from artifact integrity and priority assessment. The case study illustrates both the productivity of a persistent tool-using workflow and the review obligations it creates. It supports recommendations for future experiments, including separate novelty and correctness ledgers, independent checking of search predicates, explicit coverage boundaries, and preservation of failures and resource records. Reconciliation of the detailed usage log also separates research from later writing and identifies compaction requests omitted from the session summary counters.

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1 Introduction

Research mathematics rarely arrives as an isolated request for a proof. A researcher must decide which question to pursue, determine what is already known, interpret the statement correctly, choose representations that make computation useful, and decide when an approach has stopped being productive. An experiment that permits these choices studies a different object from a fixed collection of independently graded proof problems. Its output is a research process and a collection of artifacts, and evaluating it requires more than counting final answers.

The Kourovka Notebook [51] provides a demanding setting for such an experiment. It is a long-running collection of problems in group theory and neighboring areas. Its statements vary substantially in generality, technical prerequisites, and computational accessibility. Some have separately posed subparts; some have already been resolved without an immediately obvious indication in the statement; and some admit useful partial results without a complete answer. The supplied September 2026 version of the twenty-first edition was therefore treated as a source to interpret and investigate, rather than as a benchmark with a ready-made binary answer key.

Michael Kinyon and Josef Urban conceived the prompt and oversaw the experiment. They asked the agent to solve as many previously unsolved Notebook problems as possible within 48 hours, using a supplied GAP installation, generally staying within 20 CPU cores and 100 GB of memory, and keeping detailed plans, reports, and frequent local Git commits. The agent was allowed to choose problems, consult the literature, write programs, and change direction. The research window ran from 10 September 2026, 20:56:46 UTC, to 12 September 2026, 20:56:46 UTC. The final archive was committed 16 seconds later, after its administrative integrity check. The manuscript is a separate, later activity.

At the deadline the canonical ledger contained 46 problem entries labeled complete solution candidates. Several entries cover specified subparts rather than an entire multipart problem. Multiple covered subparts of one entry are counted once. The ledger groups Problem 19.63 with Problem 19.62; the former’s square-closedness criterion was already published before the experiment and is credited below. No entry had received outside review, and priority was unestablished. We retain these qualifications throughout: a historically recorded candidate is an observable outcome of the experiment, whereas a correct and new mathematical theorem requires further judgments.

Three aspects of the experiment are of particular interest. First, the artifact collection contains explicit mathematical constructions whose correctness can be considered separately from the agent that produced them. Second, the session and Git history expose practical interactions between proof development, computation, and correction. Third, the experiment produced substantial work that a binary solved/unsolved classification would obscure: smaller examples of known phenomena, alternative deductions, sharp bounds in restricted classes, and complete searches within clearly stated finite ranges.

1.1 Motivation

Urban’s motivation for the shared time budget was the 2006/7 MPTPChallenge: attempt many related problems and reuse discoveries across them [94]. MaLAREa implemented this approach through repeated proving and learned premise selection [49]. The present experiment adopts the broad portfolio setting, with a 48-hour wall-clock limit, informal mathematical targets, and adaptation through the agent’s context and files. It does not include observed online training of the model’s weights. Given the breadth and rapid development of learning-guided proving and LLM-based AI for mathematics, we do not attempt a comprehensive survey of related work here.

1.2 Reading the paper

Section 2 describes the setting and the limits of the retained evidence. Section 3 reconstructs the workflow and selected changes of direction. Section 4 gives complete representative arguments, including a definability counterexample and two finite group constructions. The appendices organize the wider candidate portfolio, partial results, prior-result deductions, and reproducibility material. Section 5 discusses what this single run can support and what a more controlled follow-up should measure.

2 Setting, protocol, and evidence

2.1 Task and problem source

The full user objective is reproduced in Appendix A. It specifies an overall time limit and a research objective, rather than a list of preselected questions or a target number of outputs. The agent’s initial plan explicitly prioritized precise statements, literature checks, finite

counterexamples, elementary reductions, and independent implementations of computational checks. It also required partial results and unsuccessful searches to be recorded.

The input file `docs/21tkr.pdf` has SHA-256 digest

`2fcce9b98a4df10267fe120229217bfe556c70510704e311540da0cef438f911.`

The agent extracted a main-body index with 1,308 entries, including all 150 entries of Issue 21. This is an index of problem statements, not a verified count of open problems. Editorial stars, solved subparts, duplicates, and literature published after a problem was posed all affect the open status of individual assertions. The source PDF, rather than its text extraction, was treated as authoritative for mathematical notation and scope.

2.2 Agent and computational environment

The experiment used Codex 0.153.4 with `gpt-6-astra` and reasoning effort `xhigh`. The full JSONL record corroborates the settings supplied during manuscript preparation: all 139 retained turn-context records identify this model and effort, including 133 in the research window. The session metadata records the OpenAI provider and the CLI application version. These observations identify the recorded configuration, without establishing every provider-side implementation detail.

The agent worked in a persistent Linux workspace with shell access, file editing, Git, internet source retrieval, and tools for inspecting PDF pages. GAP 4.16.1 supplied the main finite-group computations. The retained scripts also use Python and C/C++ for exact arithmetic, word calculations, independent finite models, enumeration, and certificate parsing. The source and report files record which libraries or catalogues each computation depends on. Versions not retained in the experimental records cannot be reconstructed merely by inspecting the later manuscript environment.

The user initially supplied an inaccessible GAP path and repaired it during setup. The visible human message is *sorry, gap is now here*. The subsequent reports establish that GAP was available with the required packages. The initial suggestion that a replacement installation might be needed is therefore not evidence that such an installation actually occurred.

The working limits were generally 20 CPU cores and 100 GB of memory. The repository contains bounded launchers and process observations, but not a continuous, comprehensive record of global peak resource use. In the largest retained certificate check, six GAP workers each had a 14 GiB workspace ceiling, for a combined configured ceiling of 84 GiB. A workspace ceiling is not an observed resident-memory peak. Likewise, local CPU limits do not measure the remote compute used by the language model.

The research plan prohibited delegation to other agents without authorization, and the reports record no delegated agents. Parallel GAP or C++ processes were computational workers, not separate language model researchers. The same agent developed the arguments, wrote checkers, and conducted the internal reviews. The phrase *independent implementation* below therefore refers to a separate representation or algorithm, not to an independent human author or an independently trained reviewing model.

2.3 Persistence and local records

The workspace served as external memory. Plans and status files recorded the next steps; proof drafts preserved mathematical arguments; scripts and logs preserved computations; and Git recorded successive states. The research window contains 54 task-start events, 52 task-completion events, one aborted turn, and 79 context compactions. One further task completion occurs during closeout. The opening task was paused during setup; after the

GAP correction at 20:58:51 UTC it resumed at 20:59:58 UTC. The wall-clock budget includes this interval. Repeated continuations and compactions are visible in the record; the work was not a single uninterrupted model response.

There were 198 commits during the 48-hour window. The first, at 21:00:37 UTC on 10 September, initialized the investigation. The 199th commit, `cff2c37`, completed the local handoff at 20:57:02 UTC on 12 September. Commit times date repository checkpoints; they do not directly date the discovery of an idea or establish when its full final scope was proved.

At `cff2c37`, the Git tree contains 4,147 tracked files with a total logical size of 2,440,677,642 bytes. This includes 430 files under `research/`, 456 under `scripts/`, and 2,755 under `results/`, as well as references, the source Notebook, and other records. Logical file size is not compressed repository size. The manuscript analysis reads this fixed revision, so later writing and review do not increase the experimental counts.

2.4 Session records and observable operations

The full JSONL session supplied on 15 September contains 36,016 records, from 10 September 20:56:46.325 UTC through 14 September 17:05:09.077 UTC. It includes the research run, its closeout, and initial manuscript preparation. We separate these phases before counting operations or usage. Manuscript preparation begins with the new goal recorded at 14 September 14:12:56.892 UTC.

The earlier `session.html` and `session.txt` files are projections ending at research closeout. The HTML advertises 34,102 events and retains 4,439 event elements, including 3,616 token notifications. Every one of those token observations matches the corresponding JSONL line, including all cumulative and last-use fields. The full record adds individual timestamps, tool observations, configuration records and request-level usage. Its digest and the reconciliation procedure are supplied in Appendix G.

Within the research window, completed-item notifications record 4,350 command or polling observations, 676 file-change items, 357 image views, and 1,293 web-tool observations. The web count includes searches and page retrieval; it is not a count of search queries. There are also four completed sleep-tool observations. We count each completed item once by its identifier, keeping these counts separate from the outer code-tool calls that can contain several operations. Of the command observations, 251 have nonzero exit status; these include expected no-match results and interrupted processes, and cannot be interpreted as 251 failed mathematical experiments.

The dated visible messages support the narrative below. Completed human-message items record the GAP correction during research and the model confirmation during manuscript preparation; the original objectives also occur in goal records. These counts do not measure oversight outside the recorded interaction. The derived public data omit internal reasoning and system/developer instruction bodies.

2.5 Token and cost accounting

The JSONL contains two distinct usage ledgers. Its 3,829 unique `token_usage_record` responses sum exactly to every running `thread_token_usage` value. Of these responses, 84 match context-compaction identifiers. Removing those responses gives exactly the final cumulative `event_msg/token_count` counters, which are the counter stream used by ccusage [21]. Embedded copies of history and usage are not new requests.

The organizers supplied the ccusage row reproduced in Table 1. It agrees exactly with the second ledger for the *whole supplied session*, including manuscript preparation. Its input column counts uncached input; the renderer’s input counter includes cached input. Reasoning tokens are a subset of output, and are not added a second time [72, 73].

Table 1: Supplied ccusage row and its full-session reconciliation. The row excludes the separately recorded compaction requests.

Counter	Tokens
Uncached input	21,605,689
Cached input	482,598,400
Output	4,055,235
Of which reasoning output	1,894,719
Total (uncached input plus cached input plus output)	508,259,324
Additional input and output in 84 compaction requests	21,569,265
All recorded requests, including compactations	529,828,589

Using flat rates of USD 10 per million uncached input tokens, USD 1 per million cached input tokens, and USD 50 per million output tokens reproduces the supplied USD 901.42 estimate:

$$\frac{10(21,605,689) + 482,598,400 + 50(4,055,235)}{10^6} = 901.41704.$$

These rates match the model documentation consulted on 15 September 2026 [71]. The maximum recorded request input is 259,244 tokens, below its 272,000-token long-input threshold. We use the same flat rates throughout Table 2; this is a transparent API-equivalent reconstruction, not an invoice or an inference about the organizers’ subscription bill, discounts, or additional tool charges. The exact ccusage version and invocation were not supplied.

Table 2: Usage by phase, assigned by request-record timestamp. “Other” means responses other than context compactations. USD estimates are rounded independently to cents.

	Research	Closeout	Manuscript
Other responses	3,515	5	225
Compaction responses	79	0	5
All input and output tokens	497,160,051	460,875	32,207,663
Other-response estimate (USD)	846.09	0.56	54.77
Compaction estimate (USD)	64.52	0.00	3.85
All-response estimate (USD)	910.61	0.56	58.62

Thus the full-session estimate including compactations is USD 969.78, of which USD 910.61 is assigned to the 48-hour research window. The earlier HTML total of 477,334,181 tokens equals research plus closeout usage excluding compactations. A delayed token notification straddles the deadline, so phase assignment uses request-record times, not notification times. These are logging boundaries, not exact provider execution intervals. Cached input dominates both ledgers; neither total measures distinct text read or newly generated text.

2.6 Outcome categories and checks

We distinguish five questions about an artifact:

1. Does its mathematical statement match the question being addressed?
2. Is the argument correct under its stated assumptions and imported results?
3. Do the computations establish the particular finite claims assigned to them?
4. Is the assertion already known or an immediate consequence of prior work?

5. Has someone independent of the producing agent accepted the argument?

These questions require different evidence. A digest match answers none of the mathematical questions. A finite check cannot establish an unrestricted infinite-group theorem without a proved reduction. Failure to find a paper in a literature search does not establish priority. A second implementation can expose an error in a first program while sharing the same mistaken mathematical specification.

The deadline ledger uses *complete candidate* for an argument presented as answering the recorded scope and surviving the internal checks conducted during the run. A *partial result* proves a restricted assertion while leaving the full question open. A *rediscovery* reproduces a known phenomenon or proof after independent work, whereas a *prior-result deduction* applies an identified theorem. A bounded search without a hit remains *computational evidence* for its exact range. These are reporting categories, not interchangeable levels on a single scale of confidence.

3 What happened during the run

The chronology has two complementary sources. Commits give dated, recoverable states of the repository. The timestamped session messages give an account of what the agent believed it had established at each stage. Neither records the instant at which an idea first occurred. We therefore describe observable transitions—a proof being saved, a certificate being completed, a claim being withdrawn—rather than assigning precise discovery times to unobserved reasoning.

3.1 Selection, short arguments, and early corrections

The initial plan favored finite counterexamples, explicit constructions, linear algebra, and questions that could admit short certificates. This was an adaptive search policy over the whole Notebook. It helps explain the eventual collection: the agent repeatedly moved to another entry when a search became expensive or the required generalization remained unclear. A problem’s age was not a measure of the time spent on it in this experiment, and the initial index was not a promise to attempt every entry.

The first retained mathematical lead was Problem 21.106. Message 95 (10 September, 21:00:51 UTC) announced a possible formula selecting generators of the infinite cyclic center of the integral Heisenberg group; message 152 reported a written candidate and local commit at 21:06:04 UTC. Section 4.1 gives the argument. At the same time the agent launched finite screens for other questions, including transitive groups, character tables, and positive binary words. This pattern continued: short symbolic tasks occupied the foreground while finite searches ran in separate processes.

The early graph-symmetry search exposed a substantive validation error. For Problem 21.52, normalizing a permutation representation of inner automorphisms did not by itself certify that a permutation of involutions came from an automorphism of the original group. Message 653 records the replacement of that condition by construction of an actual group automorphism and verification of its action on every involution. The lesson concerns the mathematical predicate being checked: a correct program for a weaker predicate can still give a misleading answer to the research question.

Literature searches also changed the accounting. A graded nil-algebra construction for Problem 21.132 initially appeared on the candidate list, but was later excluded as prior work. For Problem 13.19, a small explicit quotient construction passed exact checks before a 2018 result was confirmed to imply the negative answer. Message 1155 explicitly removed it from the new-solution count while retaining the construction. The longer episode around

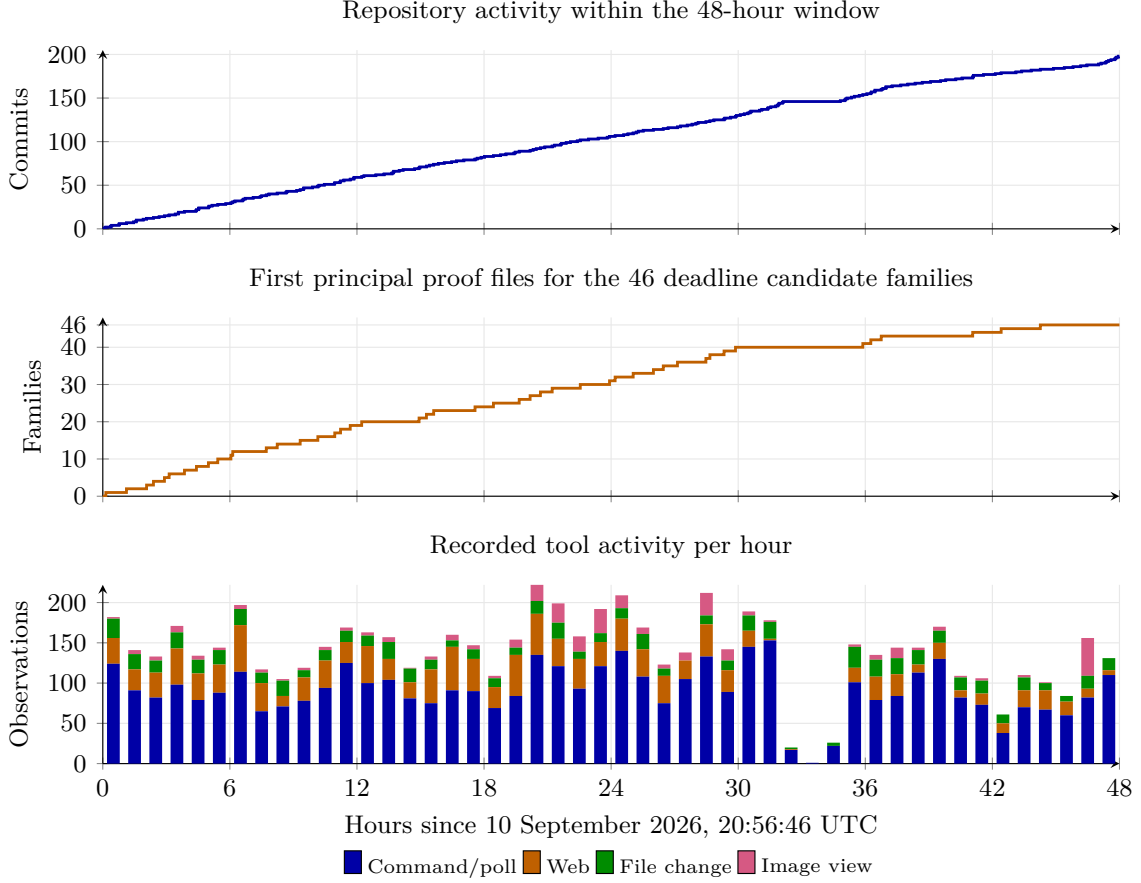


Figure 1: Repository and recorded tool activity during the research window. The upper panel counts the 198 commits within the research window; the closeout commit 16 seconds later is excluded. The middle panel retrospectively follows the first committed principal proof file for each family retained on the final candidate list. It excludes withdrawn candidates and does not measure the contemporaneous candidate count, proof completion, novelty, or outside acceptance. The lower panel counts completed command/polling, web, file-change and image-view observations in one-hour bins. It excludes context compaction and sleep operations; counts are not independent experiments. All panels are derived from the supplied CSV measurements.

Problem 17.34 is particularly informative. The agent wrote a PBW-based argument and ran thirteen exact controls; it first found prior coverage in nilpotency class two, then found a strong-amalgamation theorem with the full required scope. Messages 2729 and 2859 document the intervening increase and subsequent decrease in the working candidate count. These are rediscoveries or alternative deductions with potential expository value, but they cannot support a claim of new priority.

3.2 Constructions guided by the literature

Two examples in Section 4 illustrate how source reading and explicit calculation interacted. For Problem 21.68, the order-96 configuration in Kida’s work suggested an extension by an abelian group that could preserve a nonmonomial character. The resulting group has order 2592. Its membership in the class of semi-abelian groups and its failure to be monomial require different arguments; checking only one would not settle the conjecture.

For Problem 20.108, exceptions in a multiple-holomorph theorem suggested a group of order 605. The retained proof describes its entire automorphism group, its holomorph as explicit permutations, and a normalizing permutation whose coset has order four. The source supplied a productive place to look; the final certificate does not ask a reader to trust a SmallGroups identifier or an unexplained normalizer calculation. In both examples the finite checks are useful controls on conventions and coordinates, while the displayed construction explains why the claimed property holds.

Other entries depended more substantially on imported theorems. The constructions for Problems 10.62 and 4.75 share the same geometric existence theorem, and the carpet arguments for Problems 19.61 and 19.62 share structural machinery. The inventory retains separate Notebook questions, but they are not independent pieces of evidence about discovery ability. The appendices identify these dependencies.

3.3 A long computation and the meaning of completion

Problem 20.100 became the largest verification task. A structural reduction transformed a statement about arbitrary groups into an exact finite problem for each fixed value of n . Small values could then be established by certificates of integer relations, without imposing a bound on the order of the original group. Messages 6458–6763 distinguish the reduction, prototype searches, and completed checks for $n = 4, 5, 6$.

For $n = 7$, generation produced a certificate with 42,891,332 nodes. Message 10686 reports its completion and immediately notes that verification remained outstanding at 11 September, 11:08:29 UTC. Six GAP processes subsequently checked the separate parts. The final records account for 786,577,344 collision cases and 5,447,182,055 relation implications; the complete compressed certificate occupies 1,271,256,410 bytes. The retained full verification took approximately 26 hours of elapsed time. A parser restricted to the certificate’s data grammar was also used, avoiding reliance on executing arbitrary certificate text as GAP code.

The last workers were still active on 12 September at 12:45 UTC. Message 28901, at 13:10:31 UTC, reports all six workers finished with zero exits and a successful aggregate check, while explicitly retaining a discrepant exit from the historical launcher. Message 29167 reports the committed proof packet at 13:31:41 UTC. These dated observations connect the long-running process records to the final mathematical claim.

This distinction between generation and checking matters. Intermediate messages report how many nodes had been checked, how many workers were still running, and which shards had completed. They do not convert partial progress into a full theorem. At the deadline the

fixed cases $n = 4, 5, 6, 7$ were recorded as established for all eligible groups. The general problem for arbitrary n remained open and contributed zero entries to the complete-candidate count.

Several other large searches ended with bounded information. The endomorphism-count investigation of Problem 19.20 checked 91,774 nonabelian groups of order at most 511, found no equality example, and found 469 cases contradicting a stronger inequality. The character-table screen for Problem 21.113 involved 9,850 cases: 7,573 had the needed modular tables and 2,277 did not. For Problem 21.99, all 4,722 transitive groups of degrees 2 through 20 in the installed catalogue were screened, with a separate direct control on 86 groups in degrees 2 through 8. The word search associated with Problem 18.43 processed 3,933,931,043 positive binary necklaces of lengths 1 through 36 without producing an identity candidate. These are coverage statements. They do not establish the unrestricted assertions.

3.4 Late structural results and deadline review

The later work included structural progress on almost Engel elements and p -Jordan bounds. For Problems 20.89/19.83 the deadline result covers all linear groups in characteristic zero and several further classes; the unrestricted positive-characteristic question remains unresolved. For Problem 21.121(b), the archive contains bounds for soluble finite subgroups, examples with exactly computed exponents, and permanence results. The universal proposed bound of three remains unproved. These partials illustrate why a single binary solved/unsolved label would discard much of the experiment's mathematical output.

Statement checking remained necessary even when a construction was correct. For Problem 20.122, an early treatment of inclusion-minimal intersections did not settle the question about intersections of minimum order. The final construction addresses the latter condition. For Problem 15.76(b), the retained claim concerns the full varieties of fixed derived length, with a material distinction from arbitrary proper subvarieties. The fixed representation degree in Problem 10.35 and the minimal-polynomial convention in Problem 15.65 are similarly essential parts of the claims.

The final review separated statement coverage from deeper proof review. All 46 candidate entries received a fresh comparison with the supplied PDF; 22 selected candidate proofs received additional late rereading. This is not evidence that all 46 underwent a new, exhaustive mathematical and literature audit in the final phase. The closeout audit also checked 4,231 file bindings, covering 2,242 distinct files totaling 2,008,448,358 bytes, and 92 local links. These integrity checks establish that the named artifacts existed and matched recorded digests. They do not establish the truth or novelty of their contents. The research deadline and the subsequent administrative closeout are kept separate in the chronology.

4 Representative mathematical arguments

The following arguments illustrate three different kinds of output. Each can be assessed without accepting the agent's overall candidate count. Their relation to earlier work is stated separately from their mathematical content. Throughout, $[x, y] = x^{-1}y^{-1}xy$.

4.1 A finite definable set generating an infinite subgroup

Following Conte and Petschick [26], a formula $\varphi(x)$ is *concise* in a class of groups if every finite truth set in that class generates a finite subgroup. Petschick's Problem 21.106 asks whether every first-order formula with one free variable is concise in residually finite groups. Here formulas may have alternating quantifiers; restricting to group words or to existential formulas gives different questions.

Theorem 4.1. *There is a parameter-free first-order formula whose truth set in the integral Heisenberg group is $\{z, z^{-1}\}$, where z generates its infinite cyclic center. In particular, the formula is not concise in residually finite groups.*

Proof. Use the following abbreviations in the language of groups:

$$\begin{aligned} \text{Cent}(z) : & \quad \forall t [z, t] = 1, \\ P(u) : & \quad \forall z (\text{Cent}(z) \implies \exists t [u, t] = z), \\ Q(u, v) : & \quad \forall g \exists h \exists k (g = hk \wedge [h, u] = 1 \wedge [k, v] = 1). \end{aligned}$$

Define

$$\varphi(x) : \quad \exists u \exists v (x = [u, v] \wedge P(u) \wedge P(v) \wedge Q(u, v)).$$

After renaming bound variables and expanding the abbreviations, this is a parameter-free formula with exactly one free variable. All its quantifiers range over group elements.

Write the integral Heisenberg group as

$$H = \mathbb{Z}^3, \quad (a, b, c)(d, e, f) = (a + d, b + e, c + f + ae).$$

Matrix multiplication, or direct calculation, gives

$$(a, b, c)^{-1} = (-a, -b, ab - c), \quad [(a, b, c), (d, e, f)] = (0, 0, ae - db).$$

Thus $Z(H) = \{(0, 0, c) : c \in \mathbb{Z}\} = \langle z \rangle$ with $z = (0, 0, 1)$. Reduction of coordinates modulo m is a homomorphism to a finite group of order m^3 . Choosing m larger than the absolute value of a nonzero coordinate separates any nonidentity element, so H is residually finite.

Let $\pi : H \rightarrow \mathbb{Z}^2$ be projection onto the first two coordinates. For $u = (a, b, c)$, the third coordinates of $[u, t]$ form the ideal $a\mathbb{Z} + b\mathbb{Z}$. Consequently

$$H \models P(u) \iff \gcd(a, b) = 1.$$

If (a, b) is primitive, the solutions of $as - br = 0$ in $(r, s) \in \mathbb{Z}^2$ are precisely the multiples of (a, b) . For example, choose $\alpha a + \beta b = 1$ and put $k = \alpha r + \beta s$; the determinant relation gives $ka = r$ and $kb = s$. Hence

$$\pi(C_H(u)) = \mathbb{Z}(a, b),$$

and the centralizer is the full inverse image of this subgroup.

Suppose $P(u)$ and $P(v)$ hold, with $\pi(u) = (a, b)$ and $\pi(v) = (d, e)$. Both centralizers contain $Z(H)$, so their set product is all of H exactly when their projected subgroups sum to $\mathbb{Z}^2$. Indeed, a factorization after projection can be lifted, and its central error absorbed into the second factor. It follows that

$$H \models Q(u, v) \iff \mathbb{Z}(a, b) + \mathbb{Z}(d, e) = \mathbb{Z}^2 \iff ae - db = \pm 1.$$

Thus any value satisfying φ is z or z^{-1} . Conversely, the ordered pairs $((1, 0, 0), (0, 1, 0))$ and $((0, 1, 0), (1, 0, 0))$ witness the two values. Therefore $H_\varphi = \{z, z^{-1}\}$, but $\langle H_\varphi \rangle \cong \mathbb{Z}$. $\square$

The counterexample is already two-generated, torsion-free, and nilpotent of class two. It is residually a finite p -group for each prime p , by reduction modulo powers of p . The elements used to describe witnesses do not occur as parameters in the formula. No assertion that first-order truth commutes with finite quotients is needed. The existential-formula theorem of Conte and Petschick does not apply to this alternating formula. The experimental controls checked commutator arithmetic and finite interpretations, while the proof above establishes the infinite statement.

4.2 A semi-abelian group which is not monomial

A finite group is *monomial* if each irreducible complex representation is induced from a one-dimensional representation of a subgroup. In the definition used in Problem 21.68, semi-abelian groups are obtained along a subgroup chain by abelian semidirect extensions and quotients. The construction below uses only split extensions with abelian kernels.

Theorem 4.2. *There is a semi-abelian group of order 2,592 with a non-monomial irreducible complex representation of degree eight.*

Proof. Let E be the even-weight subspace of $\mathbb{F}_2^4$ and let A_4 permute its coordinates. Put $K = E \rtimes A_4$, with multiplication $(v, g)(w, h) = (v + g(w), gh)$. In K , set

$$z = (1111, 1), \quad x = (1010, (12)(34)), \quad y = (1001, (13)(24)), \quad r = (0000, (234)).$$

Direct multiplication gives

$$x^2 = y^2 = (xy)^2 = z, \quad z^2 = 1, \quad r^3 = 1, \quad rxr^{-1} = y, \quad ryr^{-1} = xy.$$

The eight elements $1, z, x, zx, y, zy, xy, zxy$ are distinct: their permutation parts form the Klein four group, with two distinct vectors above each part. Thus $\langle x, y \rangle$ is quaternion of order eight and $H = \langle x, y, r \rangle$ has order 24. Its abelianization is C_3 : in an abelian quotient the conjugation relations give $x = y = xy$, whence $x = y = 1$. In particular, H has no subgroup of index two.

The group H has an irreducible complex representation ρ of degree two. To see this explicitly, represent x, y by the quaternion units i, j and r by $(-1 - i - j - k)/2$. The latter has order three and conjugates i to j and j to $k = ij$. The usual two-dimensional complex representation of the quaternions is irreducible: i has the two coordinate eigenlines and j interchanges them.

Let $\Omega = K/H$, a set of four left cosets, and put

$$B = \left\{ b \in \mathbb{F}_3^\Omega : \sum_{\omega \in \Omega} b_\omega = 0 \right\}, \quad G = B \rtimes K.$$

Then $|G| = 27 \cdot 96 = 2,592$. The chain

$$1 < C_3 < A_4 < K < G$$

exhibits the required abelian split extensions. Fix $\omega_0 = H$ and a primitive cube root of unity ζ . The character $\lambda(b) = \zeta^{b_{\omega_0}}$ has stabilizer exactly H in K . Indeed, the four coordinate characters on B are distinct: for distinct coordinates u, v , choose a third coordinate w and take $b_u = 1, b_w = -1$, with all other coordinates zero. Hence the inertia subgroup of λ in G is $I = B \rtimes H$.

The representation $\tau(b, h) = \lambda(b)\rho(h)$ of I is well-defined because H fixes λ . Let $V = \text{Ind}_I^G \tau$. Its restriction to B has four distinct weights, each of multiplicity two. The group G permutes their weight spaces transitively, and the stabilizer acts irreducibly on each such space. Every invariant subspace splits into B -weight spaces, using the averaging projections for the finite abelian group B . Therefore V is irreducible and has dimension eight.

Suppose $V \cong \text{Ind}_L^G \mu$ with μ one-dimensional. Then $[G : L] = 8$. Since B is a normal 3-group, $[B : B \cap L] = [BL : L]$ is a power of three dividing eight; thus $B \leq L$. The character $\mu|_B$ must be one of the four weights of $V|_B$. After conjugation, assume it is λ . A linear character is invariant under conjugation by its own domain, so L stabilizes λ . Thus $L \leq I$ and

$$[I : L] = [G : L]/[G : I] = 8/4 = 2.$$

Because $B \leq L$, this gives an index-two subgroup of $I/B \cong H$, a contradiction. $\square$

Kida's paper [52] provided the small order-96 configuration and posed the monomiality conjecture. The abelian extension and the character obstruction are the proposed additional argument in the experimental archive. The proof does not rely on a SmallGroups identifier. That identifier and separate finite checks are useful for reproduction, not for the logical existence of the example.

4.3 An order-four element in a multiple holomorph quotient

For a group G , write $\text{Hol}(G)$ for the normalizer of its left regular action in the permutation group on its underlying set, and $\text{NHol}(G)$ for the normalizer of $\text{Hol}(G)$. Tsang's Problem 20.108(a,b) asks whether a centerless group, and specifically a finite one, can have $T(G) = \text{NHol}(G)/\text{Hol}(G)$ which is not elementary abelian of exponent two.

Theorem 4.3. *For $G = (C_{11} \times C_{11}) \rtimes C_5$, where a generator of C_5 acts by $\text{diag}(3, 9)$ over $\mathbb{F}_{11}$, the group G is centerless and $T(G)$ contains an element of order four.*

Proof. Put $\lambda = 3 \in \mathbb{F}_{11}$, of order five, and represent G by triples with multiplication

$$(x, y, k)(x', y', k') = (x + \lambda^k x', y + \lambda^{2k} y', k + k'),$$

where $k, k' \in \mathbb{Z}/5\mathbb{Z}$. Commuting with all translations forces a central element to have $k = 0$; commuting with $(0, 0, 1)$ then forces $x = y = 0$. Thus $Z(G) = 1$.

The unique Sylow 11-subgroup $V = \mathbb{F}_{11}^2$ is characteristic. An automorphism induces a matrix M on V and multiplication by $u \in (\mathbb{Z}/5\mathbb{Z})^\times$ on G/V . Compatibility requires

$$M \text{diag}(\lambda, \lambda^2) = \text{diag}(\lambda^u, \lambda^{2u})M.$$

Equality of the eigenvalue sets gives $\{u, 2u\} = \{1, 2\}$ modulo five, so $u = 1$ and M is diagonal. The image of a quotient generator can have any translation part. Conversely, the resulting maps

$$(x, y, k) \mapsto (ax + b(1 - \lambda^k), dy + e(1 - \lambda^{2k}), k), \quad a, d \neq 0,$$

are automorphisms.

A normalizer of the left regular group, after a translation making it fix the identity, is an automorphism. Consequently the full holomorph consists exactly of the permutations

$$h(x, y, k) = (Ax + B + C\lambda^k, Dy + E + F\lambda^{2k}, k + j), \quad (*)$$

with $A, D \neq 0$ and all remaining parameters arbitrary. Define

$$\theta(x, y, k) = (y, \lambda^{-k}x, 2k), \quad \theta^{-1}(X, Y, K) = (\lambda^{3K}Y, X, 3K).$$

Direct composition gives

$$\begin{aligned} (\theta h \theta^{-1})(X, Y, K) &= (DX + E + F\lambda^K, \\ &\quad \lambda^{-j}AY + \lambda^{-j}C + \lambda^{-j}B\lambda^{2K}, K + 2j). \end{aligned}$$

This again has form (*). Since the holomorph is finite, the resulting containment is equality, so θ normalizes it. Moreover,

$$\theta^2(x, y, k) = (\lambda^{-k}x, \lambda^{-2k}y, -k), \quad \theta^4 = 1.$$

Every holomorph element acts on the final coordinate by a translation, whereas θ^2 acts by negation. These are different functions on $\mathbb{Z}/5\mathbb{Z}$. Therefore $\theta^2 \notin \text{Hol}(G)$ and the coset of θ has order exactly four. $\square$

Tsang’s paper [92] singled out this group among exceptions to its computational coverage. The proof above uses that lead but not the paper’s general classification argument. The experimental checks compare the explicit holomorph with GAP’s automorphism group and evaluate the normalizer identity separately. Determining all of $T(G)$ is unnecessary for the stated conclusion.

5 Interpretation, limitations, and recommendations

The principal outcome is a reviewable mathematical archive produced under a broad, fixed-duration instruction. It contains explicit constructions, arbitrary-size arguments, finite certificates, and records of reclassification. The experiment demonstrates that this particular configured agent could sustain such a workflow across many continuations. It does not by itself establish a general success rate for open problems, a comparison with professional mathematicians, or the causal contribution of any single component.

5.1 What the evidence can support

The strongest local evidence is a short argument whose hypotheses and conclusion can be inspected directly. The examples in Section 4 are included for that reason. Finite computations can add exact evidence, especially when their output is a certificate checked in a different representation. However, two implementations written by the same agent may share a mistaken mathematical interpretation. A passing checker establishes only what its implemented predicate and coverage justify. The corrected Problem 21.52 test supplies a concrete instance of this limitation.

Likewise, an explicit reduction from the printed question to an imported theorem may be more informative than a long new calculation. Its obligations are to cite the correct version, check every hypothesis, and account for conventions. It should be described as a deduction from prior work when that is what it is. The Problem 17.34 episode shows that further literature review can change the novelty classification without invalidating the mathematics. Such changes belong in the results of a research experiment.

The archive also shows practical value in persistent written state. Proofs, process records, and explicit boundaries survived between turns; unfinished checks remained identifiable; later source findings could be propagated into a consolidated ledger. This is an observation about how the run proceeded. Without an ablation, it cannot quantify how much persistence, Git, browsing, or GAP improved the outcome. The repeated user-facing references to GAP in continuation messages also suggest that the conversation’s local cues affected the wording of updates; they should not be read as independent installation events.

5.2 Limits of the case study

Selection was highly nonuniform. The agent had access to the entire public Notebook and chose entries opportunistically. It could recognize a familiar construction, find an implication in a recent paper, or abandon an expensive line. The 46-entry list is therefore an outcome inventory, not 46 successes in a prespecified sample of 1,308 trials. The number of meaningfully attempted targets cannot be recovered simply by counting indexed entries or occurrences of problem numbers in logs.

Training-data overlap is unknown. Many questions and ingredients were public long before the run. Even a proof that does not quote a source may use knowledge learned from it. Conversely, an elementary argument can be independently rediscovered after its ingredients become familiar. Neither possibility can be decided from the wording of an

answer. The experiment had browsing access, so retrieval and mathematical reasoning were deliberately interwoven. We make no claim to have isolated either capability.

The full JSONL improves behavioral reconstruction through dated tool observations, request identifiers, compaction records and corroborated model settings. It remains a client-side record: remote inference hardware, actual peak local resource consumption, provider invoices, and a complete historical sampling configuration are unavailable. The cost reconstruction in Section 2 makes its rates and counter scope explicit; it does not measure actual billing or energy use. Human oversight outside the recorded interaction is also not observable from the trace.

Finally, manuscript review is still review by the same agent that produced the research. Repeated rereading and fresh calculations can find errors, but they do not provide authorial independence. The deadline archive contains zero outside reviews. Neither a preserved PASS message nor the absence of a finding in an internal review substitutes for specialist assessment. Codex `gpt-6-astra` is credited as the principal author for research execution, proof and code drafting, manuscript preparation, and internal review. Michael Kinyon and Josef Urban conceived the prompt and experiment and oversaw the work. These contributions do not constitute outside mathematical review.

5.3 A more informative follow-up experiment

A follow-up should retain the broad research setting while improving the measurements that this run could not provide. A practical protocol would have the following components.

1. **Define the unit of evaluation in advance.** Use a dated set of problems with explicit subparts and a documented editorial status. Record every substantive attempt, including abandoned ones, with the reason for stopping. A randomized target subset or matched target sets would support comparisons that an opportunistic portfolio cannot.
2. **Separate discovery from assessment.** Maintain different fields for statement coverage, proof validity, computational verification, prior-art status, and outside review. Freeze proposed answers before specialist assessment, and preserve later corrections as new versions. Reviewers should receive the original question and the mathematical argument before seeing aggregate success claims.
3. **Record the experimental configuration.** Save model identifiers, reasoning settings, harness and package versions, continuation instructions, tool permissions, and retrieval dates. Capture raw event logs with timestamps and request identifiers, along with local CPU time and peak memory. Retain response-level usage, including compaction, and separate the research interval from subsequent writing. Keep usage estimates and billing records separate until reconciled.
4. **Treat verification as a research task.** For a search, specify the exact predicate and its relation to the question. Include a small direct implementation that checks examples without reusing the search representation. For a large certificate, make coverage and the checking rules explicit, retain malformed-input and negative controls, and distinguish successful generation from successful verification.
5. **Use formalization selectively.** Small algebraic identities, finite certificates, and critical reduction lemmas are plausible first targets. Formalizing the entire archive at once would create a large translation project. Any formal account must also justify that its statement matches the Notebook, including quantifiers and conventions; a kernel-checked proof of a different statement would leave the original task unanswered.

6. **Measure the review burden.** Record expert time, corrections, changes in status, and which claims depend on the same imported theorem. Report a short, fully reviewed core alongside the larger exploratory archive. This would make it possible to compare output volume with the work needed to establish reliable mathematical contributions.

The present archive is suited to such a second phase. Its most useful next product would be a set of independently assessed individual results, with corrected proofs and precise attribution, followed by formal certificates where they materially reduce the remaining trust. The current paper provides the experimental context and mathematical entry points for that work.

A The instruction and experimental record

The original goal record in the JSONL and the first expanded goal directive in the HTML retain the following user-provided objective, reproduced verbatim:

Solve as many previously unsolved problems from THE KOUROVKA NOTEBOOK (docs/21tkr.pdf) as you can within an overall time limit of 48 hours. The gap system is installed in gap-4.16.1 but you can also download and install further SW if useful. In general, try to limit yourself to 20 CPU cores and 100G RAM on this machine while working. Write detailed plans and reports, iterate fast and create a local git repo keeping your work and documenting your progress, commit often there (but don't try to push). Work rigorously - your solutions will be independently reviewed.

The surrounding continuation instruction told the agent to keep the full goal active across turns, use the current worktree and external state as authoritative, and base process status on observed handles and exit records. The JSONL records the continuations and context compactions described in Section 2. The human message during setup says that GAP is now present. The initial availability issue and subsequent working installation are documented separately in the early repository records.

The original local plan, `research/PLAN.md`, divided work into overlapping selection, experiment, concentration, and review phases. It called for exact statement transcription, source checks, bounded searches, second representations, and explicit separation of partials from complete candidates. It prohibited delegation without authorization; no model sub-agents are reported in the retained research record. Independent operating-system processes were allowed for computational partitions.

The original research interval was

2026-09-10 20:56:46 UTC to 2026-09-12 20:56:46 UTC.

The final administrative commit is

`cff2c37b9bf6b737e8ad5f7ead12291a551b5201`.

All measurements of the research Git history stop at this revision; later manuscript commits are excluded. The final research response was logged at 20:57:38 UTC, approximately 52 seconds after the deadline. We distinguish this closeout from the 48-hour interval. The supplied Notebook file has SHA-256 digest

`2fcce9b98a4df10267fe120229217bfe556c70510704e311540da0cef438f911`.

This distinguishes the input from later updates at the editors' public download address.

The paper was prepared in a separate phase beginning 14 September 2026, with the same agent drafting and internally reviewing the manuscript. The model designation `gpt-6-astra`

and reasoning setting `xhigh`, confirmed by the user during that phase, are corroborated by the JSONL turn-context records. The typesetting toolchain is a later addition, not software used to obtain the mathematical results. On 15 September, the organizers supplied the full JSONL, the ccusage row, author names and affiliations, and the historical motivation used in this revision. These later contributions do not change the frozen experimental outcomes.

B The deadline candidate inventory

This table preserves all 46 entries on the deadline ledger, including its covered subparts. It is not a list of independently accepted new results. Several entries share imported ingredients. The mathematical discussions give the relevant scope and dependencies; the frozen ledger supplies the original statement and evidence paths. The historical entry for 19.62 also groups 19.63 with it; the latter’s square-closedness criterion was already published before the experiment, as explained in Section C.38. This grouping contributes one ledger entry.

Problem	Parts	Deadline description	Section
21.106	all	A parameter-free definable two-element set generating an infinite subgroup	4.1
21.121	a	The p-Jordan infimum need not be attained	C.1
21.68	all	A semi-abelian non-monomial group of order 2592	4.2
16.87	a,b	Maximal test rank at every finite rank in a nonabelian nonperiodic variety	C.2
10.35	all	A torsion-free complex linear group not residually rational in the same degree	C.3
16.28	a	A closed set whose every power greater than one is nonclosed	C.4
14.22	all	Failure of the stated finite isolated-radical presentation	C.5
16.20	all	A finite modular nondistributive dominion lattice	C.6
17.33	all	Infinite axiomatic rank of the Klein bottle quasivariety	C.7
17.101	all	Every group representation has a homogeneous extension	C.8
18.76	all	A nonsplit additive division-ring extension	C.9
18.92	a,b	Non-algebraic and non-modular complete lattices of formations	C.10
20.108	a,b	A centerless group of order 605 with a multiple-holomorph quotient element of order four	4.3
20.90	all	A two-generator infinite profinite CA-group mapping onto A5	C.11
11.116	all	Finite subgroup-lattice dimension for the specified Chernikov groups	C.12
14.72	all	A smooth affine surface with a smooth fixed divisor and a singular cyclic quotient	C.13
21.40	all	Virtual nilpotence of rational linear groups with finitely many automorphism orbits	C.14
21.60	all	An order-twelve counterexample to the semiperfectness criterion	C.15
4.55	all	Non-unique indecomposable projective decompositions over $\mathbb{Z}_5[3.A7]$	C.16
21.107	all	A countable resolvable group without an expansive sequence	C.17
15.89	all	An infinite four-regular Cayley graph with invertible adjacency on all functions	C.18
16.9	all	An exact cubic-time minimum-palindrome-factorization algorithm in free groups	C.19
18.18	all	Non-enumerability of both the cofinite finite-group theory and its complement	C.20
19.56	all	Solubility from the restricted coprime-commutator order condition	C.21

Problem	Parts	Deadline description	Section
20.92	a	The printed brace average is pre-Lie for every uniformly sufficiently large prime	C.22
9.47	all	A height-four scattered compact space excluded from the stated E-spaces	C.23
9.45	all	An explicit criterion for orthogonal bases in every dimension	C.24
10.62	all	A three-generated perfect periodic group with one involution class and odd pairwise products	C.25
10.32	all	An explicit logarithmic degree bound for symmetric-group power-word universality	C.26
15.65	all	Nonrationality of the stated classical cyclic and semisimple limits	C.27
15.92	all	Continuum many infinite alternating-residual quotients of every triangle group $\Delta(2,3,r)$, $r>6$	C.28
4.75	all	A periodic group with Sylow two-subgroups of order two and noncentral involutions modulo the odd-order radical	C.29
9.4	all	No finite quasigroup generates the ordinary isotope variety of a nontrivial finite group	C.30
12.40	all	Unbounded Brauer-degree p-parts with the group-order p-part fixed	C.31
13.42	all	A free nilpotent group with a $\mathbb{Z}[t]$ -completion that is not locally nilpotent	C.32
14.26	all	Closure of the torsion-free nilpotent quasivariety under restricted wreath products with $\mathbb{Z}$	C.33
15.76	b	Innerness of category automorphisms for full varieties of groups of fixed derived length	C.34
16.38	all	Periodicity of subgroups contained in a product of periodic subgroups in a soluble group	C.35
19.93	all	Unbounded lower p-central factors in the specified two-generator finite p-groups	C.36
20.1	all	The insoluble totally two-closed group $J1$ times $C13$ with Fitting subgroup $C13$	C.37
19.62	all	Every derived elementary carpet is closed	C.38
19.61	all	Every carpet closure is a carpet with unchanged derived carpet	C.39
21.76	all	Irreducible closed noncompletable nets of every order in every odd characteristic	C.40
11.115	all	Proper-power normal closures with equal commutator subgroups	C.41
20.122	a,b,c	Minimum-order intersections outside the Fitting subgroup in a soluble wreath group	C.42
17.39	all	Unbounded numbers of system normalizers, with fixed Frattini order three	C.43

C Further candidate arguments

The following expositions develop the remaining deadline candidates. The three examples proved in Section 4 are not repeated. Statements below retain their mathematical force, while the experimental classification and unresolved priority status remain those explained in Sections 2 and 5.

C.1 An unattained 2-Jordan exponent: Problem 21.121(a)

A group Γ is p -Jordan if some constants J, e have the following property: every finite $H \leq \Gamma$ has a normal abelian p' -subgroup of index at most $J|H_{(p)}|^e$, where $H_{(p)}$ is a Sylow p -subgroup. The question asks whether the infimum of the admissible exponents must itself be admissible.

Proposition C.1. *There is a countable 2-Jordan group with exponent*

$$e_0 = \frac{\log 24}{\log 8}$$

for which no constant works at exponent e_0 .

Proof. For $k \geq 1$, put $n = 2^k - 1$ and index the coordinates of $\mathbb{F}_2^n$ by nonzero $v \in \mathbb{F}_2^k$. Use the simplex code

$$C_k = \{(a \cdot v)_{v \neq 0} : a \in \mathbb{F}_2^k\}.$$

It has dimension k . We need a small intersection estimate: if L is spanned by r nonzero vectors with disjoint supports and $c = \dim(C_k \cap L)$, then

$$2^c \leq r + 1. \quad (1)$$

Indeed, a basis of the intersection corresponds to c independent linear forms on $\mathbb{F}_2^k$, so its coordinate columns include all $2^c - 1$ nonzero columns in $\mathbb{F}_2^c$. Vectors in L are constant on each of the r disjoint supports and zero elsewhere, giving at most r different nonzero columns.

Let $S = Q_8 \rtimes C_3$, where a generator of C_3 cycles i, j, ij in Q_8 . Identify the center of Q_8^n with $\mathbb{F}_2^n$, and form

$$G_k = S^n / C_k, \quad P_k = Q_8^n / C_k, \quad T_k = C_3^n.$$

The code is central and fixed by T_k , so

$$G_k = P_k \rtimes T_k, \quad |G_k| = 24^n / 2^k, \quad |P_k| = 8^n / 2^k.$$

Write Z_0 for the image of $Z(Q_8^n)$. Then $V = P_k / Z_0 = \bigoplus_{i=1}^n V_i$, with each V_i two-dimensional over $\mathbb{F}_2$; the i -th C_3 factor acts irreducibly on V_i and trivially on the other coordinates.

We claim that every $H \leq G_k$ contains a normal abelian odd-order subgroup A_0 for which, with $u = |H_{(2)}|$,

$$[H : A_0] = u3^r, \quad u \geq \frac{8^r}{r+1} \quad (2)$$

for some integer $r \geq 0$. Put $U = H \cap P_k$, its normal Sylow 2-subgroup. A Sylow 3-subgroup A complements U , simply by comparing orders. Conjugating into the Sylow subgroup T_k , we may assume $H = U \rtimes A$ with $A \leq T_k$. Set $A_0 = C_A(U)$; it is normal abelian of odd order, since it centralizes U and A is abelian. Write $|A/A_0| = 3^r$.

Let W be the image of U in V . The kernel of the action of A on W is exactly A_0 . To check the nontrivial inclusion, an element a in that kernel satisfies $x^a = xz_x$, with $z_x \in U \cap Z_0$, for every $x \in U$. It fixes z_x , and $a^3 = 1$, so $x = x^{a^3} = xz_x^3 = xz_x$, whence $z_x = 1$.

The representation of A on W is completely reducible. Over $\mathbb{F}_4$, its commuting operators diagonalize, since $X^3 - 1$ has distinct roots. Nontrivial irreducibles over $\mathbb{F}_2$ therefore have dimension two and correspond to pairs $\{\alpha, -\alpha\}$ of nonzero homomorphisms $\alpha : A \rightarrow \mathbb{F}_3$. The occurring characters span a space of dimension r , because their common kernel is A_0 . Choose r independent occurring characters and nonzero vectors w_j in their respective isotypic components of W . Their coordinate supports are disjoint: a coordinate V_i belongs to only one character pair.

Choose $a_j \in A$ acting nontrivially on the character of w_j and lift w_j to $x_j \in U$. In each nonzero quaternion coordinate, w_j and $w_j^{a_j}$ are distinct nonzero vectors of $Q_8/Z(Q_8)$, so the commutator of their lifts is the central involution. Thus $[x_j, x_j^{a_j}]$ is the image of the binary support vector d_j of w_j . The d_j are disjoint and independent before quotienting by C_k . If L is their span, (1) shows that their images span a space of dimension at least $r - \log_2(r+1)$ in $[U, U] \leq U \cap Z_0$. Also $\dim W \geq 2r$. Consequently

$$u = |W| |U \cap Z_0| \geq 2^{2r} 2^{r-c} \geq 8^r / (r+1).$$

Together with $[H : A_0] = |U| |A/A_0|$, this proves (2), including $r = 0$.

For $e > e_0 > 1$, it follows that

$$\frac{[H : A_0]}{|H_{(2)}|^e} \leq (r+1)^{e-1} \left(\frac{3}{8^{e-1}} \right)^r.$$

The right-hand side has a finite supremum over $r \geq 0$, independent of k and H , since its exponential base is less than one.

On the other hand, every normal odd-order subgroup of G_k centralizes P_k : their commutator is in their trivial intersection. Its image in T_k must then act trivially on V , because inner automorphisms from P_k act trivially on P_k/Z_0 . The action of T_k is faithful, so that normal subgroup is trivial. Hence the smallest permissible index for G_k itself is $|G_k|$, and

$$\frac{|G_k|}{|P_k|^{e_0}} = 2^{k(e_0-1)} \longrightarrow \infty.$$

Finally take $\Gamma = *_{k \geq 1} G_k$. Every finite subgroup of a free product is conjugate into a factor. One proof uses the free-product tree: a finite group fixes the center of the finite subtree spanned by an orbit, and the action has no edge inversions. Vertex stabilizers are trivial or conjugate to factors. The uniform estimates therefore give every exponent $e > e_0$ for Γ , while its embedded factors rule out e_0 . A smaller admissible exponent would imply admissibility of e_0 , so the infimum is exactly e_0 . $\square$

The use of a free product is material; a direct sum would introduce additional finite products not covered by the argument. No finite generation assumption appears in the question. The construction does not resolve part (b). The archive contains GAP checks on all subgroup conjugacy classes for $k = 1, 2$, and direct code and quaternion controls; the uniform argument above supplies the unbounded quantifiers. Source: `research/21.121a-proof.md`.

C.2 Maximal test rank in a nonabelian variety: Problem 16.87

A test set in a group is a set whose pointwise stabilizer among endomorphisms consists of automorphisms. The test rank is the least size of such a set. Both parts of the question have a negative answer for the variety

$$\mathcal{M}_p = \{G : [[x, y], z] = 1, [x, y]^p = 1\},$$

where p is any prime. It is nonabelian because it contains $\text{UT}_3(\mathbb{F}_p)$, and nonperiodic because it contains $\mathbb{Z}$.

Proposition C.2. *The relatively free group G_r of rank $r \geq 1$ in $\mathcal{M}_p$ has test rank r .*

Proof. Its abelianization is $\mathbb{Z}^r$: the generator maps to and from $\mathbb{Z}^r$ give inverse maps on abelianization, since $\mathbb{Z}^r$ belongs to the variety. Denote the resulting map by $v : G_r \rightarrow \mathbb{Z}^r$. If S has fewer than r elements, choose a primitive nonzero integral row vector w annihilating all $v(s)$, $s \in S$, by solving the corresponding homogeneous linear equations over $\mathbb{Q}$. Then $f = wv : G_r \rightarrow \mathbb{Z}$ is onto and vanishes on S .

Each free generator x_i has central p -th power, because $[x_i^p, g] = [x_i, g]^p = 1$. Choose i with $w_i \neq 0$ and put $c = x_i^{p \operatorname{sign}(w_i)}$. The map

$$\phi(g) = gc^{f(g)}$$

is an endomorphism by centrality of c , and fixes S pointwise. But $f(\phi(G_r)) = (1 + p|w_i|)\mathbb{Z}$ is proper in $\mathbb{Z}$, so ϕ is not surjective. Thus no smaller set is a test set. The r free generators themselves form one. For $r = 1$ the same argument excludes the empty set. $\square$

The free groups here have torsion in their commutator subgroups; nonperiodicity of the variety does not mean that its free groups are torsion-free. No such torsion-free hypothesis is printed in the problem. The proof uses only central powers and integer linear algebra. Source: `research/16.87-proof.md`.

C.3 Residual rationality in a fixed degree: Problem 10.35

The same representation degree occurs on both sides of this question. Consider

$$U(z) = \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}, \quad T = \operatorname{diag}(2i, 2), \quad G = \langle U(1), T \rangle \leq \operatorname{GL}_2(\mathbb{C}).$$

Since $TU(z)T^{-1} = U(iz)$, its unique normal forms are $U(z)T^k$, $z \in \mathbb{Z}[i]$, $k \in \mathbb{Z}$. Uniqueness follows from the lower diagonal entry 2^k and then the upper-right entry. This is $\mathbb{Z}^2 \rtimes \mathbb{Z}$, with the generator acting by a quarter turn, and is torsion-free because its kernel $\mathbb{Z}^2$ and quotient $\mathbb{Z}$ are torsion-free.

Lemma C.3. *Let F have characteristic zero and contain no square root of -1 . If $X, Y \in \operatorname{GL}_2(F)$ satisfy*

$$[X, YXY^{-1}] = I, \quad Y^2XY^{-2} = X^{-1},$$

then $X^2 = I$.

Proof. Assume $X^2 \neq I$. A scalar X contradicts the second relation, so X is nonscalar. Conjugacy to its inverse gives $\det(X)^2 = 1$. If $\det(X) = -1$, equality of traces gives $\operatorname{tr} X = 0$, and Cayley–Hamilton gives $X^2 = I$. Thus $\det(X) = 1$ and $X^{-1} = (\operatorname{tr} X)I - X$.

The centralizer algebra of a nonscalar 2-by-2 matrix is $E = F[I, X]$. Indeed, a vector v with v, Xv independent is cyclic, and a commuting operator is determined by its value on v . Now YXY^{-1} is a nonscalar member of E , so it too generates E with I . Conjugation by Y therefore normalizes E and fixes its scalar subspace. On the one-dimensional quotient E/FI it acts by a scalar $q \in F^\times$. Its square sends the nonzero class of X to its negative, so $q^2 = -1$, a contradiction. $\square$

Put $a = U(1)$. The relations of the lemma hold for a, T in G . Consequently every homomorphism $\phi : G \rightarrow \operatorname{GL}_2(\mathbb{Q})$, and even every homomorphism into $\operatorname{GL}_2(\mathbb{R})$, kills the fixed nonidentity element $a^2 = U(2)$. This disproves the required residual property. If the image is torsion-free it kills a itself, and hence the whole normal subgroup $U(\mathbb{Z}[i])$. The common scalar factor in T is essential to torsion-freeness: $T^4 = 16I$. The example does embed in $\operatorname{GL}_3(\mathbb{Q})$, so it does not address a version allowing the target degree to increase. Source: `research/10.35-proof.md`.

C.4 A closed set with no closed higher power: Problem 16.28(a)

Let $K = \overline{\mathbb{F}_5(t)}$, $H = \mathrm{SL}_2(K)$, and $Y = \{A \in H : \mathrm{tr} A = 1\}$. This is a closed conjugacy class: the polynomial $X^2 - X + 1$ has distinct roots, and a $\mathrm{GL}_2(K)$ -conjugating matrix can be made to have determinant one by multiplying it by a scalar.

The square Y^2 omits $-I$, since $AB = -I$ would give $B = -A^{-1}$ and $\mathrm{tr} B = -1$. Nevertheless, for $s \neq 0, -1$, put

$$u = \frac{s}{1+s}, \quad w = u(1-u) - 1, \quad A_s = \begin{pmatrix} u & 1 \\ w & 1-u \end{pmatrix}, \quad D_s = \mathrm{diag}(s, s^{-1}).$$

Both A_s and $A_s^{-1}D_s$ have determinant one and trace one; for the latter the trace is $(1-u)s + us^{-1} = 1$. Thus $D_s \in Y^2$. The curve $s \mapsto D_s$ shows that $-I = D_{-1}$ is in its closure, and Y^2 is not closed.

In the connected reductive group $G = H \times K^\times$, take the closed set

$$X = (Y \times \{1\}) \cup \{(I, t)\}.$$

Since (I, t) is central,

$$X^n = \bigcup_{k=0}^n Y^k \times \{t^{n-k}\}.$$

The powers of t are distinct. For every $n \geq 2$, intersection with the closed fiber $H \times \{t^{n-2}\}$ is therefore exactly $Y^2 \times \{t^{n-2}\}$, which is not closed in that fiber. Hence no X^n , $n > 1$, is closed.

This answers part (a) as printed. It uses a reducible X and a field with an element of infinite multiplicative order; neither is excluded by the statement. The same class Y also gives the negative answer to part (b), since Y^2 contains the open set $\mathrm{tr}(C)^2 \neq 4$ and omits $-I$. The experiment classified that part as already implied by prior class-product results, so it is excluded from this candidate's counted coverage. Source: `research/16.28-proof.md`.

C.5 A coefficient-support obstruction: Problem 14.22

Write the additive rational group multiplicatively as $G = \{a_q : q \in \mathbb{Q}\}$, with $a_q a_r = a_{q+r}$. It is torsion-free and linear via $a_q \mapsto U(q)$ in $\mathrm{SL}_2(\mathbb{Q})$. The solution set of $S = \{[x, a_1] = 1\}$ is all of G , and is irreducible in the group-equation Zariski topology. Indeed, evaluating a word $w \in G * \langle x \rangle$ at $x = a_t$ gives $a_{m(w)t+b(w)}$, where $m(w) \in \mathbb{Z}$ is its exponent sum and $b(w) \in \mathbb{Q}$ its coefficient sum. An equation therefore has zero, one, or all points as solutions. Proper closed sets in this one-variable topology are finite, and G is infinite.

Let T be any finite system equivalent to S . Every word in T has $m(w) = b(w) = 0$. Choose $d > 0$ such that all its coefficients lie in the cyclic group $C = \langle a_{1/d} \rangle$. Form

$$K = G *_C (C \times \langle x \rangle)$$

and the natural epimorphism $\pi : G * \langle x \rangle \rightarrow K$. Each word of T vanishes in the second factor, since both exponent sums are zero. The group K is torsion-free: normal form in an amalgamated free product conjugates a torsion element into a factor, and both factors are torsion-free. Thus $\ker \pi$ is a normal isolated subgroup containing T , and contains the least such subgroup $\sqrt{T}$.

But $[x, a_{1/(2d)}]$ vanishes at every point of G , so belongs to $\mathrm{Rad}_G(T)$. Its image in K is a reduced alternating word of length four: $x \notin C$ and $a_{1/(2d)} \notin C$. It is nontrivial by normal form. Hence

$$\sqrt{T} < \mathrm{Rad}_G(T)$$

for every finite equivalent T . The radical is isolated because it is an intersection of kernels of maps into a torsion-free group, which also justifies the displayed inclusion.

The coefficient group is not finitely generated, as permitted by the printed question. The separating quotient need only be torsion-free; isolatedness does not mean unique roots. In particular, commuting with a square does not imply commuting with its root in that quotient.

For completeness, the archive also avoids the possible concern that the intended geometry uses nonabelian coefficients. Take $G = \mathbb{Q} * \langle b \rangle$, represented faithfully by $a_q \mapsto U(q)$, $b \mapsto \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$ over $\mathbb{Q}(t)$. For a cyclically reduced alternating word, the trace has nonzero leading coefficient $\prod_j q_j n_j$, where the nonzero rational syllables are q_j and the b -syllables are b^{n_j} . Thus the representation is faithful. The solution set of $[x, a_1] = 1$ is the factor $\mathbb{Q}$, since free factors are malnormal.

The coordinate group is $(\mathbb{Q} \times \langle x \rangle) * \langle b \rangle$. To verify this and irreducibility directly, represent x by $U(z)$ with z a second indeterminate. The same leading-coefficient argument is faithful. Every nontrivial word remains nontrivial under all but finitely many rational specializations $z = q$: a nonzero coefficient polynomial has only finitely many roots. These evaluations discriminate the coordinate group and make every proper relative algebraic subset of the solution set finite.

For a finite equivalent system, choose a common denominator for all rational syllables in its coefficients. Its words already vanish in $(C \times \langle x \rangle) * \langle b \rangle$, which embeds in the coordinate group by normal form. They therefore vanish in the torsion-free group

$$(\mathbb{Q} *_C (C \times \langle x \rangle)) * \langle b \rangle,$$

while $[x, a_{1/(2d)}]$ again survives. This gives the same strict radical obstruction with non-abelian linear coefficients. Source: `research/14.22-proof.md`.

C.6 A modular nondistributive dominion lattice: Problem 16.20

For $H \leq A$ and a class $\mathcal{N}$ of target groups, the dominion consists of those $a \in A$ on which every pair of homomorphisms from A into a member of $\mathcal{N}$ agree whenever they agree on H . The literature convention allows $A \notin \mathcal{N}$; see [83]. Here is the smaller of the archive's two constructions under that convention.

Let

$$U_1 = \mathrm{SL}_2(5), \quad U_2 = \mathrm{SL}_2(7), \quad U_3 = \mathrm{SL}_2(13), \quad A = (U_1 \times U_2 \times U_3) / \langle z_1 z_2 z_3 \rangle,$$

where $z_i = -I$, and take $H = 1$. These standard perfect central covers have simple quotients $S_i = \mathrm{PSL}_2(p_i)$ of orders 60, 168, 1092. No one of these three orders divides another. The center $W = Z(A) \cong C_2^2$ has as its three nonzero vectors the images v_i of the z_i . The group has order 44,029,440.

Every normal subgroup R of A is $A_I K$, where A_I is the product of the factors indexed by some $I \subseteq \{1, 2, 3\}$, and

$$K = R \cap W, \quad \langle v_i : i \in I \rangle \leq K \leq W.$$

To see this, its image in the product of the three simple quotients is a product of factors. For any occurring factor, $[R, U_i]$ projects onto S_i . A normal subgroup of U_i projecting onto S_i is all of U_i : if $U_i = JZ(U_i)$, perfectness gives $U_i \leq J$. Removing those factors leaves precisely a central subgroup.

The key fact is that every homomorphism $f : A \rightarrow A/R$ kills R . Project its restriction on U_i to each surviving S_j . For $j \neq i$ that map is trivial. Indeed, every proper normal

subgroup of U_i is central, so every nontrivial image has order divisible by $|S_i|$, which cannot divide $|S_j|$. Thus a removed factor has central image and is killed by perfectness. A surviving factor maps into its own image D_i times the center; perfectness puts its image in $[D_i Z(A/R), D_i Z(A/R)] = D_i$. When $v_i \in K$, this D_i is S_i , and every homomorphism $U_i \rightarrow S_i$ kills z_i . Every nonzero vector of K is some v_i , so f kills K as well as the removed factors, proving the fact.

Let $q(\mathcal{C})$ denote the quasivariety generated by $\mathcal{C}$, and put $\mathcal{M} = q(\{A/R : R \trianglelefteq A\})$. For a finite domain A , the condition defining membership of an element in a dominion is a finite quasiidentity in the target: use two variables for each element of A , impose both multiplication tables and agreement on H , and conclude agreement at that element. It follows that the dominion for $q(\mathcal{C})$ is exactly the intersection of the dominions for the generating targets.

For $H = 1$, a dominion is simply an intersection of kernels of maps into the target class, by comparison with the trivial map. It is therefore normal. For each R , all maps into A/R kill R , while the canonical quotient has kernel exactly R . The preceding transfer gives

$$\text{dom}_{q(A/R)}^A(1) = R.$$

Every normal subgroup occurs, and the entire dominion family is the normal subgroup lattice of A . That lattice is modular by the Dedekind identity, and its interval $[1, W]$ is the five-element diamond M_3 , so it is not distributive.

The larger archived example also handles the extra convention that every admissible target quasivariety contain A . It uses seven covers $\text{SL}_2(p)$, $p = 7, 11, 13, 17, 19, 23, 29$, with their central involutions mapped onto all nonzero vectors of $\mathbb{F}_2^3$, and a fixed central line H . Under that convention the dominions are exactly the five subspaces between H and $\mathbb{F}_2^3$. The smaller construction above suffices for the documented convention of the printed question. Source: `research/16.20-proof.md`.

C.7 The Klein bottle quasivariety: Problem 17.33

Let $K = \mathbb{Z} \rtimes \mathbb{Z}$ with $(u, k)(v, l) = (u + (-1)^k v, k + l)$, the Klein bottle group. Its quasivariety $q(K)$ cannot be axiomatized by quasiidentities with any fixed finite bound on the number of variables.

For $n \geq 2$, put $E = \mathbb{F}_2^n$, and index $m = n(2^n - 1)$ integer coordinates by pairs (χ, j) , where $0 \neq \chi \in E^*$ and $1 \leq j \leq n$. Let D_e multiply coordinate (χ, j) by $(-1)^{\chi(e)}$. Define

$$\Gamma_n = \{(e, c) \in E \times \mathbb{Z}^m : c_{\chi, j} \equiv e_j \pmod{2}\}, \quad (e, c)(f, d) = (e + f, c + D_e d).$$

Parity makes this a subgroup of the affine semidirect product. Its translation subgroup $N = \{(0, 2z) : z \in \mathbb{Z}^m\}$ has quotient E .

The group is torsion-free. If $e \neq 0$, choose a nonzero χ vanishing on e , possible because $n \geq 2$, and j with $e_j = 1$. Then the (χ, j) -coordinate of $(e, c)^2 = (0, (1 + D_e)c)$ is nonzero. A nonidentity translation also has nonzero square, and any torsion element would have square one since its square lies in the torsion-free group N . On the other hand, the abelianization of Γ_n is finite. Every translation basis vector is inverted by some lift of e_i , since $\chi_i = 1$ for some i , so $2N \leq \Gamma'_n$. The quotient by $2N$ is finite.

Every homomorphism $\Gamma_n \rightarrow K$ is trivial: composition with $K \rightarrow \mathbb{Z}$ is trivial by finite abelianization, and the image is then in the translation subgroup $\mathbb{Z}$, where the same argument applies again.

In contrast, the inverse image Γ_L of every proper subspace $L < E$ embeds in K^m . Choose $0 \neq \psi \in E^*$ vanishing on L , and define, with coefficients $\chi_j \in \{0, 1\}$,

$$\ell_\chi(e, c) = \sum_j \chi_j c_{\psi, j}, \quad F_{\chi, j}(e, c) = (c_{\chi, j}, \ell_\chi(e, c)).$$

The integer-valued ℓ_χ is additive on Γ_L , because D_e fixes the ψ -coordinates for $e \in L$, and it has parity $\chi(e)$. Thus every $F_{\chi,j}$ is a homomorphism into K . Their product is injective: all its first coordinates determine c , and the parities of c determine e . In particular, every subgroup generated by fewer than n elements belongs to $q(K)$.

It remains to justify that $\Gamma_n \notin q(K)$, without assuming a residual characterization that might require additional hypotheses. The group is finitely presented. Explicit generators are translations $t_{\chi,j}$ and lifts $g_i = (e_i, c_i)$ with $(c_i)_{\chi,j} = \delta_{ij}$. A finite presentation is given by commuting translations and the relations

$$\begin{aligned} g_i t_{\chi,j} g_i^{-1} &= t_{\chi,j}^{(-1)^{\chi_i}}, \\ g_i^2 &= \prod_{\chi: \chi_i=0} t_{\chi,i}, \\ g_i g_j &= \left(\prod_{\chi} t_{\chi,i}^{\chi_j} t_{\chi,j}^{-\chi_i} \right) g_j g_i \quad (i < j). \end{aligned}$$

These hold by the affine multiplication law. They collect every word to a translation monomial followed by $g_1^{\epsilon_1} \cdots g_n^{\epsilon_n}$, $\epsilon_i \in \{0, 1\}$. Its affine image is the identity only if all ϵ_i and then all translation exponents vanish, proving the presentation.

For such a presentation, the quasiidentity asserting that its finitely many relations imply that a chosen nontrivial generator is one holds in K , since every homomorphism $\Gamma_n \rightarrow K$ is trivial. It fails in Γ_n . Thus $\Gamma_n \notin q(K)$. If quasiidentities in at most r variables axiomatized $q(K)$, take $n > r$. Every tuple of at most r elements of Γ_n lies in a subgroup of $q(K)$, so all those quasiidentities hold in Γ_n , a contradiction. Source: `research/17.33-proof.md`.

C.8 Homogeneous extensions of representations: Problem 17.101

Fix a commutative ring k with identity. A representation is a left kG -module V with its acting group G ; an embedding is injective on both sorts. The relevant homogeneity extends isomorphisms of subrepresentations whose group is finitely generated and whose module is finitely generated over that group's algebra. The coefficient ring remains fixed; compare the category in [74].

Given any set of partial isomorphisms $(f_i, \theta_i) : (U_i, A_i) \rightarrow (U'_i, B_i)$ in (V, G) , form the multiple HNN extension

$$H = \langle G, t_i : t_i a t_i^{-1} = \theta_i(a), a \in A_i, \text{ all } i \rangle.$$

The HNN embedding theorem embeds G in H [43]. Define

$$M = kH \otimes_{kG} V, \quad E = \bigoplus_i kH \otimes_{kA_i} U_i, \quad \delta_i(h \otimes u) = h \otimes u - h t_i^{-1} \otimes f_i(u),$$

and $W = M / \text{im } \delta$. The map δ_i is well-defined because $a t_i^{-1} = t_i^{-1} \theta_i(a)$ and f_i is equivariant.

We verify that no vector of V is lost. The HNN Bass–Serre tree has vertices hG and edges hA_i with endpoints $hG, h t_i^{-1} G$. Its being a tree follows from HNN normal form: a reduced closed path would be a reduced stable-letter word lying in G . As k -modules, M is the direct sum of one copy of V at each vertex, and E the direct sum of the edge modules. Each edge maps injectively into both incident vertex modules, using the original inclusion and f_i .

If $\delta(z)$ were supported at a single vertex v_0 for a nonzero $z \in E$, the finitely many nonzero edge components of z would form a finite forest with an edge. It has a leaf other than v_0 . The single incident nonzero edge contributes a nonzero vector at that leaf, by injectivity,

a contradiction. Hence $\text{im } \delta$ intersects each vertex module trivially, and $V \hookrightarrow W$. This argument uses no semisimplicity, purity, or finite-dimensionality.

In W , the edge relation gives $t_i u = f_i(u)$. The automorphism of (W, H) with components

$$w \mapsto t_i w, \quad h \mapsto t_i h t_i^{-1}$$

therefore extends the prescribed partial isomorphism. Now repeat the construction at stages $0, 1, 2, \dots$, each time including all isomorphisms between finitely generated subrepresentations, and take the union. The domain and range of any such isomorphism in the union lie in a common stage: finitely many group and module generators suffice. A stable letter at the next stage realizes it, and its action continues to be an automorphism of the union. This proves homogeneity.

If $\kappa = \max(\aleph_0, |k|, |G|, |V|)$, there are at most κ relevant finite tuples and maps at each stage; induction, quotients, and the countable union preserve that bound. Thus a countable input over a countable ring has a countable homogeneous extension. Source: `research/17.101-proof.md`.

C.9 A nonsplit additive division-ring extension: Problem 18.76

Let $F = F(a, b)$, $L = F(c, d)$, and $G = F \times L$. Free groups and their direct products admit two-sided invariant orders. The Mal'cev–Neumann theorem therefore embeds kG in a division ring for $k = \mathbb{Q}$ or any prime field; see [78, 14]. Let D be its division closure there. This is countable, and $G \leq D^\times$. Only elements from different free factors are assumed to commute.

Define a derivation $\delta : F \rightarrow kF$ by

$$\delta(xy) = \delta(x) + x\delta(y), \quad \delta(a) = 0, \quad \delta(b) = 1.$$

The prefix-sum formula on reduced words gives it explicitly; contributions from adjacent inverse letters cancel, with $\delta(b^{-1}) = -b^{-1}$. In $N = (D, +) \rtimes F$, with $(r, x)(s, y) = (r + xs, xy)$, the maps

$$\gamma_c(r, x) = (cr, x), \quad \gamma_d(r, x) = (dr + \delta(x), x)$$

are automorphisms. The derivation law proves the homomorphism identity; multiplication by c^{-1} and the map $(r, x) \mapsto (d^{-1}(r - \delta(x)), x)$ give their inverses. Freeness of L extends them to an action, giving the actual group $E = N \rtimes_\gamma L$.

Its quotient by $(D, +)$ is $G = F \times L$, and the conjugation action on the kernel is left multiplication in D . To check this and nonsplitting, define similarly $\eta : L \rightarrow kL$ by $\eta(c) = 0, \eta(d) = 1$. Then

$$(r, x, h)(s, y, l) = (r + xhs + x\eta(h)\delta(y), xy, hl).$$

This follows by composing the generator automorphisms; the two derivations take values in commuting factor algebras.

If a section existed, its values on a, b, c, d would have arbitrary kernel coordinates $u, v, w, z \in D$. Put $A_0 = a - 1, B_0 = b - 1, C_0 = c - 1, D_0 = d - 1$. The four cross-factor commutations would imply

$$A_0 w - C_0 u = 0, \quad A_0 z - D_0 u = 0, \quad B_0 w - C_0 v = 0, \quad B_0 z - D_0 v = 1_D.$$

Here 1_D is a nonzero additive kernel element. Since A_0, C_0 are nonzero and invertible, the first three equations give

$$w = A_0^{-1} C_0 u, \quad z = A_0^{-1} D_0 u, \quad v = B_0 A_0^{-1} u.$$

The left side of the fourth is then

$$B_0 A_0^{-1} D_0 u - D_0 B_0 A_0^{-1} u = 0,$$

because the two free factors commute, contradicting $1_D \neq 0$. Thus the extension does not split, in any of the stated characteristics. Its description as a nested semidirect product does not split this particular extension by $(D, +)$. Source: `research/18.76-proof.md`.

C.10 Complete lattices of formations: Problem 18.92

The question defines a complete lattice of formations as an intersection-closed nonempty family with a largest member. Joins are internal to that family. Under this definition both requested examples follow from prime sets.

Let $\mathcal{P}$ be the set of primes and let $\mathcal{F}_S$ be the finite groups whose order has all prime divisors in $S \subseteq \mathcal{P}$. This is a formation: quotients and finite subdirect products introduce no new prime divisor. Moreover

$$\mathcal{F}_S \subseteq \mathcal{F}_T \iff S \subseteq T, \quad \bigcap_i \mathcal{F}_{S_i} = \mathcal{F}_{\bigcap_i S_i}.$$

The cyclic group of order p witnesses the reverse implication in the first assertion.

For part (a), take $\Theta = \{\mathcal{F}_S : S \text{ finite}\} \cup \{\mathcal{F}_{\mathcal{P}}\}$. Its joins are union of prime sets when that union is finite and the top otherwise. Every nonbottom element fails to be compact. Choose a prime q in its set and enumerate $\mathcal{P} \setminus \{q\}$. The formations for successive finite initial segments have join the top in Θ , whereas no finite subjoin contains the cyclic group of order q . The bottom is therefore the only compact element, and Θ is not algebraic.

For part (b), use the five formations with prime sets

$$\emptyset, \quad \{2\}, \quad \{2, 3\}, \quad \{5\}, \quad \mathcal{P}.$$

They are closed under intersections, and their order is the pentagon N_5 . Explicitly, writing $A = \mathcal{F}_{\{2\}}, B = \mathcal{F}_{\{2,3\}}, C = \mathcal{F}_{\{5\}}$, we have $A < B$ but

$$A \vee (C \wedge B) = A, \quad (A \vee C) \wedge B = B.$$

Thus the lattice is not modular.

Both examples contain the formation of all finite groups as top, so the empty-intersection convention causes no problem. These statements concern the chosen families; they do not assert either property for the full lattice of all formations. Source: `research/18.92-proof.md`.

C.11 Infinite non-prosoluble profinite CN-groups: Problem 20.90

The question asks for an infinite, topologically finitely generated profinite group with pronilpotent centralizers of nonidentity elements, which is not prosoluble. The archive gives two constructions, counted as one answer. The first has abelian centralizers.

Let $R = \mathbb{F}_4[[t]]$, with $\omega^2 + \omega + 1 = 0$, and put

$$X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 1 & \omega + t \end{pmatrix}, \quad G = \overline{\langle X, Y \rangle} \leq \mathrm{SL}_2(R).$$

The congruence topology makes $\mathrm{SL}_2(R)$ profinite, hence so is its closed subgroup G , which is two-generated by definition. Writing $z = \omega + t$, define $P_0 = 0, P_1 = 1, P_{n+1} = zP_n + P_{n-1}$. Then

$$Y^n = \begin{pmatrix} P_{n-1} & P_n \\ P_n & P_{n+1} \end{pmatrix}.$$

The polynomial P_{n+1} is monic of degree n for $n \geq 1$. Thus no positive power of Y is the identity and G is infinite.

Reduction modulo t is onto $\mathrm{SL}_2(\mathbb{F}_4)$. To check this directly, write x, y for the reduced matrices and $U(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $L(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$. Matrix multiplication gives

$$yxy^{-1} = L(1), \quad d = y^{-1}xy^{-2} = \mathrm{diag}(\omega, \omega^2).$$

Conjugation by d multiplies the parameters of U and L by ω^2 and ω , respectively. Addition of parameters therefore supplies all elementary matrices, which generate $\mathrm{SL}_2(\mathbb{F}_4)$ by row reduction. The commutators

$$dU(a)d^{-1}U(a)^{-1} = U(\omega a), \quad dL(a)d^{-1}L(a)^{-1} = L(\omega^2 a)$$

show that this nontrivial finite group is perfect. Hence the continuous quotient is nonsoluble, and G is not prosoluble. This quotient is the usual group A_5 .

Regard G inside $\mathrm{SL}_2(\mathbb{F}_4((t)))$. In characteristic two, a scalar determinant-one matrix is the identity. Every nonidentity $g \in G$ is therefore nonscalar. The cyclic-vector argument of Appendix C.3 identifies its matrix centralizer with the commutative algebra $\mathbb{F}_4((t))[g]$. Thus $C_G(g)$ is abelian. It is closed, hence an abelian profinite group and in particular pronilpotent. This field-level centralizer argument is classical; its characteristic-two form appears in [35]. We do not claim it as new.

It is not necessary that finite quotients be CN-groups. Indeed, the archive's image modulo t^2 has order 240 and contains the nonidentity scalar $(1+t)I$, whose centralizer is the whole nonsoluble image. The theorem concerns centralizers in the inverse-limit group itself.

A characteristic-zero companion. Let $O = \mathbb{Z}_2[\omega] \subset K = \mathbb{Q}_2(\omega)$ with the same quadratic relation, and take $H = \mathrm{SL}_2(O)/\{I, -I\}$. The four matrices $U(1), U(\omega), L(1), L(\omega)$ topologically generate $\mathrm{SL}_2(O)$: their powers generate parameters in the dense subring $\mathbb{Z} + \mathbb{Z}\omega$, and elementary row reduction works over the local ring O . The image of $U(1)$ has infinite order, and reduction gives the nonsoluble quotient $\mathrm{SL}_2(\mathbb{F}_4)$.

For a nonidentity $h \in H$, a lift $g \in \mathrm{SL}_2(O)$ is nonscalar. Its strict centralizer $C_{\mathrm{SL}_2(O)}(g)$ is abelian by the same matrix lemma. The preimage of $C_H(h)$ consists of s with $sgs^{-1} = g$ or $-g$; the possible sign is a homomorphism with that strict centralizer as kernel. If $\mathrm{tr} g \neq 0$, only the positive sign occurs, so the projective centralizer is abelian.

If $\mathrm{tr} g = 0$, its reduction $\bar{g}$ cannot be the identity. Otherwise $g = I + 2A$ gives $\mathrm{tr} A = -1$, and

$$1 = \det(I + 2A) = 1 + 2 \mathrm{tr} A + 4 \det A$$

would give $2 \det A = 1$ in O . Thus $\bar{g}$ is a nonidentity unipotent of $\mathrm{SL}_2(\mathbb{F}_4)$, with centralizer of order four (the matrices $U(a)$ after a change of basis). The strict centralizer has reduction a 2-group and kernel in the pro-2 congruence subgroup, whose successive quotients embed in $M_2(\mathbb{F}_4)^+$. It is pro-2. The extension by a sign group and subsequent quotient by $\{I, -I\}$ are also pro-2. Every projective centralizer is therefore pronilpotent. Sources: `research/20.90-proof.md` and `research/20.90-2adic-proof.md`.

C.12 Order dimension of Chernikov subgroup lattices: Problem 11.116

Suppose $D \trianglelefteq G$, $[G : D] = m < \infty$, and

$$D = C_{p_1^\infty} \times \cdots \times C_{p_r^\infty},$$

with distinct primes p_i and $r \geq 1$. Then the order dimension of the entire subgroup lattice of G is at most $2rm$.

Here is a coordinate proof. Write C_{p^∞} additively as $\mathbb{Z}[1/p]/\mathbb{Z}$. Its unique terminating base- p expansion $x = \sum_{i \geq 1} d_i p^{-i}$, $0 \leq d_i < p$, defines a bijection to $\mathbb{N}_0$ by

$$b_p(x) = \sum_{i \geq 1} d_i p^{i-1}.$$

A coset of the subgroup of order p^n has all digits beyond n fixed. Its image is an integer interval $[ap^n, (a+1)p^n - 1]$; the whole group maps to $\mathbb{N}_0$. Inclusion is therefore determined by two endpoint inequalities, left endpoints ordered in reverse and right endpoints in the usual order.

Every subgroup of D is the product of its primary components: Chinese remaindering recovers each component as an integer multiple of an element. Thus the product of these bijections sends every coset of a subgroup of D to a nonempty box. Its inclusion order is determined by $2r$ endpoint coordinates. Include the empty set by giving it left endpoints $+\infty$ and right endpoints $-\infty$.

Fix representatives t_q of the m cosets of D . For $H \leq G$, its fiber

$$S_q(H) = \{d \in D : dt_q \in H\}$$

is empty or a coset of $H \cap D$. If d_0 lies in the fiber, multiplying dt_q by $(d_0 t_q)^{-1}$ proves that the fiber is $(H \cap D)d_0$. The fibers partition H , and $H \leq K$ exactly when every fiber of H is contained in the corresponding fiber of K . This embeds the subgroup lattice into a product of $2rm$ chains.

For clarity, a poset embedded in d chains has order dimension at most d : use the full coordinate lexicographic order to break ties, and for each coordinate order first by that coordinate. All d orders extend the poset, and any failure of comparison is detected in at least one coordinate. Their intersection is exactly the original order.

A Chernikov group has such a finite-index divisible subgroup. The question's exclusion of two quasicyclic factors for the same prime ensures the required distinct primes. Finite groups have finite subgroup lattices and cause no exception. The extension need not split. Source: `research/11.116-proof.md`.

C.13 A smooth fixed divisor and a singular quotient: Problem 14.72

Over $\mathbb{C}$, let $X \subset \mathbb{A}^4$ have coordinates (x, y, u, v) and equations

$$y^2 = x^3 - x, \quad yu = xv, \quad (x^2 - 1)u = yv.$$

The automorphism $\sigma(x, y, u, v) = (-x, iy, iu, v)$ has order four. The curve $C : y^2 = x^3 - x$ is smooth and irreducible. The opens $x \neq 0$ and $x^2 - 1 \neq 0$ cover X . On the first, eliminate $v = yu/x$; on the second, eliminate $u = yv/(x^2 - 1)$. They identify X locally with the product of an open subset of C and an affine line, proving it is a smooth irreducible affine surface without hidden components.

The full fixed scheme of $\langle \sigma \rangle$ is $x = y = u = 0$, with v free. It is a smooth affine line. On the second chart its ideal is generated by y , because $x = y^2/(x^2 - 1)$ and $u = yv/(x^2 - 1)$. It is therefore a smooth effective Cartier divisor, of the codimension one required in the question.

Nevertheless the quotient is singular. Put

$$a = x^2, \quad d = xu^2 - v^2.$$

These and v are invariant, and the defining equations give $xd = u^2$ and $(a - 1)d = v^2$. We claim

$$\mathbb{C}[X]^{\langle \sigma \rangle} = \mathbb{C}[a, v, d]/(v^2 - (a - 1)d).$$

After inverting a , a basis of the coordinate ring consists of $x^s y^b u^c$, $s \in \mathbb{Z}$, $b = 0, 1$, $c \geq 0$, with weight i^{2s+b+c} . The invariant monomials have respectively the forms

$$x^s u^{2h} = a^{(s+h)/2} d^h, \quad x^s y u^{2h+1} = a^{(s+h+1)/2} v d^h.$$

Invariance makes the displayed exponents integral. The monomials $a^r d^h, a^r v d^h$ have distinct images in that Laurent basis, so there are no further relations. After inverting $a - 1$, the other chart has invariant ring $\mathbb{C}[a, (a - 1)^{-1}, v]$, since the invariant ring of the curve is $\mathbb{C}[x^2]$. Here $d = v^2/(a - 1)$. The two invariant principal opens cover, so the identifications prove the global assertion. One can also clear powers of a and $a - 1$ and use their comaximality to see that no invariant is omitted.

At $(a, v, d) = (1, 0, 0)$ the local equation is $v^2 = (a - 1)d$. The local dimension is two and the cotangent dimension three, so the quotient is not regular. This point comes from the orbit of $(1, 0, 0, 0)$, whose stabilizer is the proper subgroup $\langle \sigma^2 \rangle$. The full fixed divisor does not control this additional stabilizer. The construction uses cyclic order four; it makes no assertion about a prime-order variant or an action free outside the fixed divisor. Source: `research/14.72-proof.md`.

C.14 Finitely many automorphism orbits in a rational linear group

For Problem 21.40, let $G \leq \mathrm{GL}_n(\mathbb{Q})$ have finitely many orbits under its abstract automorphism group. Then G has a normal torsion-free nilpotent subgroup of class at most $n - 1$ and index at most $(2n + 1)^{n^2}$.

Fix $g \in G$. Two members of $g, g^2, g^4, \dots$ are in one orbit. Thus some $h = g^{2^i}$ and automorphism α satisfy $\alpha(h) = h^r$ for $r = 2^{j-i} > 1$. The matrices $\alpha^{-k}(h)$ are r^k -th roots of h . We first show this forces every eigenvalue of h to be a root of unity.

For an eigenvalue λ of h , choose an eigenvalue μ_k of its r^k -th root with $\mu_k^{r^k} = \lambda$; its degree over $\mathbb{Q}$ is at most n . Choose an integer $D > 0$ clearing the entries of h and let $M \geq 1$ bound the absolute values of all its eigenvalues. Every conjugate of μ_k has absolute value at most M . Also

$$(D\mu_k)^{r^k} = D^{r^k-1}(D\lambda)$$

is an algebraic integer, so $D\mu_k$ is one by transitivity of integrality. Algebraic integers of bounded degree with all conjugates bounded form a finite set: their monic minimal polynomials have integer coefficients bounded by the elementary symmetric-function estimates. Thus some $\mu_k = \mu_l$, $k < l$, giving $\mu_k^{r^l-r^k} = 1$. It follows that λ is a root of unity, and then the same is true for each eigenvalue of g .

Consequently $\mathrm{tr}(g)$ is a rational algebraic integer of absolute value at most n , so $\mathrm{tr}(G) \subseteq \{-n, \dots, n\}$. Let $A = \mathrm{span}_{\mathbb{Q}} G$, a unital algebra of dimension $d \leq n^2$, and define its trace radical

$$R = \{a \in A : \mathrm{tr}(ab) = 0 \text{ for all } b \in A\}.$$

Cyclicity of trace makes R a two-sided ideal. For $a \in R$, all $\mathrm{tr}(a^s)$ vanish. Newton's identities and Cayley–Hamilton give $a^n = 0$.

In fact $R^n = 0$. On $V = \mathbb{Q}^n$, the chain $V \supseteq RV \supseteq R^2V \supseteq \dots$ cannot stabilize at a nonzero module $W = RW$. If it did, choose a smallest finite A -generating list $w_1, \dots, w_s$ of W and express $w_s = \sum_i a_i w_i$ with $a_i \in R$. Since a_s is nilpotent, $1 - a_s$ is invertible, allowing deletion of w_s , a contradiction. Thus there are at most n strict dimension drops, and faithfulness gives $R^n = 0$.

The group $1 + R$ is nilpotent of class at most $n - 1$, using $[1 + R^a, 1 + R^b] \subseteq 1 + R^{a+b}$. It is torsion-free: a finite-order unipotent matrix in characteristic zero has minimal polynomial

dividing both $(X - 1)^n$ and the squarefree polynomial $X^m - 1$, and is the identity. Set $U = G \cap (1 + R)$, a normal subgroup. Choose a basis $g_1, \dots, g_d$ of A from G . The tuples

$$(\mathrm{tr}(gg_1), \dots, \mathrm{tr}(gg_d))$$

take at most $(2n + 1)^d$ values. Equality of tuples for g, h is equivalent to $g - h \in R$, hence to $gh^{-1} \in U$. They distinguish exactly the cosets of U , proving the index bound. No finite-generation hypothesis or matrix realization of the abstract automorphisms was used. Source: `research/21.40-proof.md`.

C.15 Grothendieck classes and semiperfectness: Problem 21.60

Let $G = C_3 \rtimes C_4 = \langle a, b : a^3 = b^4 = 1, bab^{-1} = a^{-1} \rangle$, $R = \mathbb{Z}_{(2)}$, and $k = \mathbb{F}_2$. Every simple kG -module is the reduction of a simple rational module, but RG is not semiperfect. Thus the nonnegative-combination criterion in the question does not suffice.

The central involution b^2 acts trivially on a simple kG -module: its difference from the identity is nilpotent with a nonzero invariant kernel. Every such module factors through S_3 . In kS_3 , the central idempotent $e = 1 + a + a^2$ cuts out kC_2 , whose only simple module is k . The complementary four-dimensional algebra is $M_2(k)$, via

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

These satisfy the relations, and I, A, B, AB are linearly independent, proving the identification. Its only simple module is the natural module V . This lifts to the rational representation with matrices

$$A_{\mathbb{Q}} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad B_{\mathbb{Q}} = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}.$$

The a -matrix has irreducible polynomial $X^2 + X + 1$, proving rational simplicity, and the stable lattice R^2 reduces to V . The trivial module lifts trivially. Hence every finite-dimensional kG -module has Grothendieck class a nonnegative combination of reductions of simple rational modules, by a composition series.

For the obstruction to semiperfectness, let $E = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, and form the quaternion algebra

$$D = E + Ej, \quad j^2 = -1, \quad ju = \bar{u}j.$$

The norm of $u + vj$ is $u\bar{u} + v\bar{v}$. For $u = x + y\omega$, its summand is $x^2 - xy + y^2$; the norm is positive on nonzero elements. The conjugate divided by the norm gives an inverse, so D is a division algebra. The order $O = R[\omega] + R[\omega]j$ is a quotient of RG by $a \mapsto \omega, b \mapsto j$. Its reduction is $M_2(k)$, since ω, j map to the matrices A, B above and their four basis images are independent.

Moreover $J(O) = 2O$. For every $y \in O$, the norm of $1 - 2y$ is congruent to one modulo $2R$, and quaternion conjugation preserves O ; the inverse formula shows $1 - 2y$ is a unit. The radical unit criterion gives $2O \subseteq J(O)$, and the semisimplicity of $O/2O$ gives the reverse inclusion. The nontrivial idempotents of $M_2(k)$ cannot lift to O , because a subring of a division ring has only idempotents zero and one. Thus O is not semiperfect.

A quotient of a semiperfect ring is semiperfect: its quotient by its Jacobson radical is a quotient of the original semisimple quotient; idempotents lift first to that semisimple ring, then to the original ring, and finally to the quotient. Consequently RG is not semiperfect either. The coefficient ring is the rational localization, not the 2-adic completion.

The question is Conjecture 9 in the retained version of [48]. The example also conflicts with its Proposition 3.3 as written. The extension-lifting step there cannot hold for arbitrary chosen lattices: with trivial C_2 -actions,

$$\mathrm{Ext}_{RC_2}^1(R, R) = 0, \quad \mathrm{Ext}_{kC_2}^1(k, k) = k.$$

Directly, an extension with fixed trivial submodule and quotient has generator matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$; its square forces $2t = 0$ over R , whereas $t = 1$ gives a nonsplit extension over k . This concerns the cited version and step, and does not import the proposition as a valid premise. Source: `research/21.60-proof.md`.

C.16 Nonunique decompositions of projectives: Problem 4.55

Let $G = 3.A_7$, the perfect central cover of order 7560, and $R = \mathbb{Z}_{(5)}$. There is an RG -projective module of R -rank 55,440 with two indecomposable decompositions having no isomorphism class of summand in common. This addresses uniqueness itself, beyond the earlier failure of the condition that indecomposable endomorphism rings be local.

We first give the descent facts used. If an R -free G -lattice L has projective reduction modulo five, its restriction to a Sylow 5-subgroup S is free: lift a free basis over the local algebra $\mathbb{F}_5 S$, apply Nakayama's lemma to the resulting RS -map, and use equal R -ranks for injectivity. Averaging over a transversal, with $[G : S]$ invertible in R , splits $RG \otimes_{RS} L \rightarrow L$. Thus L is projective. Also, two finite projective RG -modules with isomorphic reductions are isomorphic: projectivity lifts the reduction map, Nakayama gives surjectivity, and equal R -ranks give injectivity.

Suppose a finite projective $\mathbb{Z}_5 G$ -module M has generic fiber obtained from a rational G -module V . In their identified $\mathbb{Q}_5$ -space, approximate a $\mathbb{Z}_5$ -basis of M by rational vectors modulo $5M$. They form both a $\mathbb{Z}_5$ -basis of M and a $\mathbb{Q}$ -basis of V . The G -matrices in this basis have entries in $\mathbb{Q} \cap \mathbb{Z}_5 = R$. Their R -span is a stable lattice completing to M , and is projective by the preceding reduction criterion.

The finite character input is the following part of the 5-modular decomposition data for $3.A_7$ [66, 17]. Use four Frobenius pairs of absolute Brauer characters, denoted A, B, C, D , of respective absolute degrees 3, 6, 18, 21. Their $\mathbb{F}_5$ -simple modules have dimensions 6, 12, 36, 42, and their projective covers have dimensions 90, 60, 90, 90. For one faithful central character, the relevant rows are

GAP ordinary index	degree	A	B	C	D
10	6	0	1	0	0
16	21	0	0	0	1
18	21	1	0	1	0
20	24	1	0	0	1
22	24	0	1	1	0

and the conjugate rows have indices 11, 17, 19, 23, 21. All other rows are zero in these columns. The ordinary Galois orbits are $\{10, 11\}$, $\{16, 17\}$, $\{18, 19\}$, $\{20, 21, 22, 23\}$. Indices refer to the retained GAP table, not a numbering across all covers in the printed tables.

Brauer reciprocity shows that a completed projective whose reduction has cover multiplicities (a, b, c, d) has generic ordinary multiplicities $b, d, a + c$ on the first three paired orbits, and $a + d, b + c$ on the two halves of the degree-24 orbit. Its character is rational-valued exactly when

$$a + d = b + c. \tag{3}$$

This is necessary for every rationally defined projective with this support.

Consider the four rays

$$r_1 = (1, 1, 0, 0), \quad r_2 = (1, 0, 1, 0), \quad r_3 = (0, 1, 0, 1), \quad r_4 = (0, 0, 1, 1).$$

Each gives a completed projective, since idempotents lift over the complete ring $\mathbb{Z}_5G$, and each satisfies (3). Multiplication of its generic character by $N = 168$ gives a rational form without assuming Schur index one. Indeed, for an ordinary Galois orbit $\mathcal{O}$, the sum of its primitive complex central idempotents lies in $\mathbb{Q}G$, and $\mathbb{Q}Ge_{\mathcal{O}}$ has character $\sum_{\chi \in \mathcal{O}} \chi(1)\chi$. Since 168 is divisible by 6, 21, 24, the desired multiple is a nonnegative sum of these rational characters. Semisimple $\mathbb{Q}_5G$ -modules are determined by their characters after scalar extension, so the descent argument gives projectives L_i over RG with reduction vectors $168r_i$.

Now $r_1 + r_4 = r_2 + r_3$, and reduction detects isomorphism of projectives. Hence

$$L_1 \oplus L_4 \cong L_2 \oplus L_3.$$

The common rank is $168(150 + 180) = 55,440$. Every nonzero direct summand of L_1 has reduction supported on A, B ; finite-dimensional Krull–Schmidt and (3) force its vector onto the positive ray through r_1 . The same holds for each of the other three rays. They meet only at zero. Thus no nonzero summand on one side can be isomorphic to a summand on the other. Splitting successively terminates by positive finite R -rank, giving incompatible indecomposable refinements.

The character data are a genuine dependency: the four-ray calculation alone would not construct modules. The archive reconstructs the ordinary table from the actual group and the faithful decomposition rows from modular representations, in addition to comparison with published tables. The L_i themselves are not asserted to be indecomposable. Source: `research/4.55-proof.md`.

C.17 A resolvable group without an expansive sequence: Problem 21.107

As usual, the open sets in the definition of an expansive sequence are understood to be nonempty: every such open set must meet all sufficiently late terms of a sequence of pairwise disjoint finite subsets. Take a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, obtained by extending the cofinite filter, and the countable Boolean group $B = [\mathbb{N}]^{<\omega}$ under symmetric difference. For $A \in \mathcal{U}$, put $H_A = [A]^{<\omega}$ and use these subgroups as an identity neighborhood base. Intersections are $H_A \cap H_C = H_{A \cap C}$, so this defines a group topology. It is Hausdorff by deleting a coordinate on which two elements differ, and nondiscrete because every $A \in \mathcal{U}$ is infinite.

The sets

$$D_k = \{s \in B : v_2(|s| + 1) = k\}, \quad k \geq 0,$$

partition B into dense subsets. Given $x + H_A$ and k , choose an arbitrarily large $r = 2^k(2q + 1) - 1 \geq |x|$, and add to x a subset of $A \setminus x$ of size $r - |x|$. This gives a point of the coset in D_k .

Suppose (F_n) is any sequence of pairwise disjoint finite subsets of B . Infinitely many empty terms already preclude expansiveness. Otherwise discard the finitely many empty terms and the at most one term containing the identity, leaving an infinite sequence of nonempty finite families of nonempty supports. Put $S_n = \{\max s : s \in F_n\}$. For every r , only finitely many S_n meet $\{1, \dots, r\}$, since there are just $2^r - 1$ nonempty supports with maximum at most r and the F_n are disjoint. Choose a subsequence with $\max S_{n_j} < \min S_{n_{j+1}}$. Let E be the union of its even-indexed S_{n_j} . Either $E \in \mathcal{U}$ or its complement is in $\mathcal{U}$. Whichever belongs to $\mathcal{U}$ misses all S_{n_j} of the opposite parity. The corresponding open subgroup H_A therefore misses infinitely many F_{n_j} , a contradiction.

Thus countable resolvability does not imply existence of an expansive sequence. Only an ordinary free ultrafilter is used, with no selectivity assumption. The underlying filter

topology is standard; the dense partition and obstruction are given explicitly. Source: `research/21.107-proof.md`.

C.18 An invertible adjacency operator on all functions: Problem 15.89

Let $G = C_3 * C_2 = \langle a, b : a^3 = b^2 = 1 \rangle$ and

$$S = \{aba, aba^2, a^2ba, a^2ba^2\}.$$

Its Cayley graph, with edges $g \sim gs$, is simple and four-regular: free-product normal form gives four distinct nonidentity words, and $S = S^{-1}$. It is connected because, for $s_1 = aba, s_2 = a^2ba, s_1s_2^{-1} = a^2$ and $a^2s_1a^2 = b$. The group is infinite by the distinct reduced words $(ab)^n$, and left translation is vertex-transitive.

On the space $\mathbb{C}^G$ of all functions, put $(R_gf)(x) = f(xg)$, $U = R_a + R_{a^2}$, and $B = R_b$. Then $R_gR_h = R_{gh}$, and

$$U^2 = U + 2I, \quad U^{-1} = (U - I)/2, \quad B^2 = I.$$

The adjacency operator is exactly $A = UBU$, hence

$$A^{-1} = \frac{1}{4}(U - I)B(U - I).$$

Both products with A equal the identity. All these expressions are finite sums of translations, so they act on arbitrary, even unbounded, functions. There is no support or convergence condition. In particular zero is not an eigenvalue, answering the question negatively.

The factorization has the familiar zig-zag form: a move within a triangle, across a matching edge, and within the next triangle. The general operation is due to [76]; the invertibility of finite factors also occurs in [20]. The proposed contribution is this particular infinite simple vertex-transitive example and its use for the printed question, not the graph-product operation. Source: `research/15.89-proof.md`.

C.19 Computing minimum palindromic length: Problem 16.9

In the free group F_n with specified basis $x_1, \dots, x_n$, let $\text{pl}(w)$ be the minimum number of palindromes whose product is w , with $\text{pl}(1) = 0$. The following algorithm uses $O(L^3)$ arithmetic operations and $O(L^2)$ integer storage for an input word of length L . Its stored integers have magnitude $O(L)$.

Let $\tau(x_i) = x_i^{-1}$, and form $H = F_n \rtimes \langle t \rangle$, with $t^2 = 1$ acting by τ . The substitutions $s_0 = t, s_i = x_it$ give the presentation

$$H = \langle s_0, \dots, s_n : s_i^2 = 1 \text{ for all } i \rangle.$$

The subgroup F_n is its even-parity subgroup. An element $p \in F_n$ is a palindrome exactly when $\tau(p) = p^{-1}$, hence when pt is an involution. Every nonidentity involution of this free product is conjugate to an s_i , by cyclic reduction, and has odd parity. Thus the involutions, or reflections, are precisely the elements pt for palindromes p .

If $r(h)$ is reflection length in H , then

$$\text{pl}(w) = \min\{r(w), r(wt)\}. \quad (4)$$

Indeed, a product $w = p_1 \cdots p_k$ gives

$$wt^k = (p_1t)(\tau(p_2)t) \cdots (\tau^{k-1}(p_k)t).$$

Conversely, write a reflection factorization of wt^j , $j = 0, 1$, as $(q_1t) \cdots (q_mt)$. Parity gives $m \equiv j \pmod{2}$, so $w = q_1\tau(q_2) \cdots \tau^{m-1}(q_m)$; each factor is a palindrome. This proves both inequalities in (4).

For a word V on the involution alphabet, let $d(V)$ be the minimum number of unmatched positions in a noncrossing partial pairing of equal letters. This is unchanged by insertion or deletion of two adjacent equal letters. Pairing the inserted letters gives one inequality. For the reverse, start with a pairing before deletion. If the deleted positions are paired together, remove that pair. If neither has a partner, remove them. If just one has another partner, leave that partner unmatched. If both have other partners, pair those partners to each other. Their labels agree, and noncrossing is preserved: the old arcs joined along the short deleted interval give a new arc disjoint from the others. None of these repairs increases the number unmatched.

It follows from free-product reduction that d depends only on the represented element. Moreover $d(h) = r(h)$. A conjugate vs_iv^{-1} has a pairing leaving only its central letter unmatched, so a product of k reflections has $d \leq k$. Conversely, delete the unmatched letters of a pairing. The remaining fully paired word reduces to the identity by successively removing innermost pairs. Deleting one letter uses the identity

$$Xs_iY = (Xs_iX^{-1})(XY).$$

Successive deletions express the original word as a product of that many reflections. This proves equality. The general Coxeter deletion principle is prior work of Dyer [33]; the special case needed here has been proved directly.

Replace each x_i in the input by $i0$ and each x_i^{-1} by $0i$, obtaining W of length at most $2L$. Then

$$\text{pl}(w) = \min\{d(W), d(W0)\}.$$

For half-open intervals of $V = W0$, set $D(i, i) = 0$ and compute

$$D(i, j) = \min \left\{ \begin{array}{l} 1 + D(i + 1, j), \\ \min_{\substack{i < k < j \\ V_i = V_k}} (D(i + 1, k) + D(k + 1, j)) \end{array} \right\}.$$

The minimum over an empty set is $+\infty$. The first option leaves position i unmatched. Pairing it with k separates the remaining positions into the inside and the following interval, by noncrossing. This exhausts all possibilities, proving the recurrence. There are $O(L^2)$ intervals with $O(L)$ choices each.

Stored minimizing choices recover the unmatched positions, whose successive deletions recover reflection factors r_i . Put $q_i = r_it$, rewrite their even-parity words in the x_i , and apply τ to every second factor as above. This gives a minimum palindrome factorization. The total output length is $O(L^2)$, since each conjugating prefix has length $O(L)$ and there are $O(L)$ factors. Bit complexity adds the usual $O(\log(L + 1))$ overhead for the stored integers; encoding generator indices is separate. This computes group palindromic length, allowing cancellation between factors, not merely a partition of the reduced word. Source: `research/16.9-proof.md`.

C.20 The cofinite theory of finite groups: Problem 18.18

Let T_{fin} consist of group-language sentences true in every finite group, and T_{cof} those true in all but finitely many finite groups up to isomorphism. The latter is equivalent to truth in all sufficiently large finite groups. Both T_{cof} and its complement fail to be computably enumerable. The input theorem is Mal'cev's undecidability of T_{fin} [61]: finite countermodels

can be enumerated, so T_{fin} is co-computably-enumerable and cannot also be computably enumerable.

For a sentence ϕ , effectively relativize every quantifier to the definable subgroup

$$C(x; a, b) : \quad xa = ax \wedge xb = bx,$$

renaming bound variables first. Write the resulting formula as $\phi^C(a, b)$. Formula induction gives its truth exactly when $C_G(a, b) \models \phi$. Then $P_\phi = \forall a \forall b \phi^C(a, b)$ satisfies

$$\phi \in T_{\text{fin}} \iff P_\phi \in T_{\text{cof}}.$$

The forward direction is immediate. For the reverse, if finite $H \not\models \phi$, use $H \times S_n$, $n \geq 3$, and two generators of the symmetric factor. Their common centralizer is $H \times 1$, giving countermodels of unbounded order. This proves the first non-enumerability assertion.

For the complement we need a fixed first-order sentence σ whose finite models are exactly the S_n , $n \geq 5$. The following explicit interpretation supplies it. Choose an involution $t \neq 1$, let C_t be its conjugacy class, and let $Q(\mathbf{u})$ say that four entries of $\mathbf{u}$ are distinct members of C_t and pairwise noncommuting. Put

$$\begin{aligned} M(c, \mathbf{u}) : \quad & c \in C_t \wedge \bigwedge_{i=1}^4 (c = u_i \vee cu_i \neq u_i c), \\ E(\mathbf{u}, \mathbf{v}) : \quad & \forall c \in C_t (M(c, \mathbf{u}) \leftrightarrow M(c, \mathbf{v})). \end{aligned}$$

On Q -tuples, E is equality of the incidence sets and hence an equivalence relation. Call these classes vertices. This is abbreviation only: vertex quantifiers use four group variables and vertex equality uses E . Require that there be at least five vertices; that each $c \in C_t$ be incident with exactly two vertices; that every pair of distinct vertices have exactly one common incident c ; that

$$E(cuc^{-1}, \mathbf{u}) \iff \neg M(c, \mathbf{u});$$

and that the common centralizer of C_t be trivial. These are finitely many fixed first-order assertions; their conjunction, existentially quantifying t , defines σ .

In a finite model, conjugation acts faithfully on the finite vertex set: an element fixing all vertices fixes their unique edges and therefore all C_t . Each c preserves its incident pair, moves both endpoints by the displayed assertion, and fixes every other vertex. Its action is that pair's transposition. Every pair occurs, so the faithful image is the full symmetric group. Conversely, in S_n , take t a transposition. Four distinct pairwise intersecting two-element subsets have a common point: the only alternative for three is a triangle, which admits no fourth distinct edge meeting all three. The sets defined by M are therefore exactly the stars at points. Every point supplies a four-tuple when $n \geq 5$. All the assertions hold, including trivial centralizer because transpositions generate centerless S_n . This proves the claim about σ .

We also need every finite H as a common centralizer in *every sufficiently large* S_n . Write $h = |H|$, enumerate $H = \{h_0 = 1, h_1, \dots, h_{h-1}\}$, and on $\Omega = H \times \{0, \dots, h-1\}$ define

$$a(g, j) = (g, j + 1 \bmod h), \quad b(g, j) = (gh_j, j).$$

The fixed points of b form precisely level zero. A commuting permutation must therefore have form $(g, j) \mapsto (u(g), j)$ by commutation with a ; commutation with b gives $u(gh_j) = u(g)h_j$, hence $u(g) = u(1)g$. The common centralizer is exactly the left regular copy of H . The action of $\langle a, b \rangle$ is transitive, by moving between levels to realize all right multiplications.

For $n \geq 2h^2 + 3$, add a disjoint orbit of $k = n - h^2 > h^2$ points, extending a, b by a k -cycle and a transposition of consecutive points. Their action there is S_k , with trivial centralizer.

The two orbit sizes differ, so no commuting permutation exchanges them. The total common centralizer in S_n is therefore H , as required.

Finally form the effective sentence

$$R_\phi = (\neg\sigma) \vee \exists a \exists b \neg\phi^C(a, b).$$

If $\phi \in T_{\text{fin}}$, the second disjunct is false in every finite group, and R_ϕ fails in every S_n , $n \geq 5$. If a finite H refutes ϕ , all nonsymmetric groups satisfy the first disjunct, and every sufficiently large symmetric group satisfies the second by the construction. Thus

$$\phi \in T_{\text{fin}} \iff R_\phi \notin T_{\text{cof}}.$$

Computable enumerability of the complement would give it for T_{fin} , a contradiction. The syntactic transformation does not search for H ; the witness is needed only for its correctness proof. No asymptotic-density interpretation of *almost all* is asserted. Source: `research/18.18-proof.md`.

C.21 Solubility from restricted coprime commutators: Problem 19.56

For finite G , let

$$X(G) = \{[x, y] : |x|, |y| \text{ are powers of distinct primes}\}.$$

Suppose $|ab| \geq |a||b|$ for $a, b \in X(G)$ of coprime orders. Then G is soluble. In fact it suffices to impose the inequality when the outputs a, b also have prime-power orders. The prime-power restrictions on the *inputs* are the additional issue here, compared with nearby commutator criteria [67, 11].

First, a primary element $a \in X(G)$ normalizing a subgroup R of different prime-power order must centralize R . For $z \in R$, the element $b = [a, z]$ lies in $R \cap X(G)$, while $ab = a^z$. If $b \neq 1$, the hypothesis gives $|a| = |ab| \geq |a||b| > |a|$, a contradiction. This adapts the normalizer argument in [67], with the required input orders checked explicitly.

For $P \in \text{Syl}_p(G)$, let $O^p(G)$ be the p -residual, the intersection of normal subgroups with p -group quotient. The hyperfocal theorem in the finite-group form of [18, Lemma 2.2] gives

$$P \cap O^p(G) = \langle [x, g] : x \in Q \leq P, g \in N_G(Q), (|g|, p) = 1 \rangle.$$

Decompose g into its commuting primary components inside $\langle g \rangle$. They still normalize Q , and $[x, uv] = [x, v][x, u]^v$ expresses each generator as a product of elements of $P \cap X(G)$. Conversely, every $X(G)$ -value vanishes in a p -group quotient because at least one input does. Hence

$$P \cap O^p(G) = \langle P \cap X(G) \rangle.$$

For perfect G , its p -group quotient is trivial, so these primary $X(G)$ -values generate every Sylow subgroup.

Take a nonsoluble counterexample of least order. The condition passes to subgroups, so every proper subgroup is soluble and $G = G'$. Its soluble radical R is contained in every maximal subgroup: otherwise $G = RM$ makes G/R soluble. Thus $R \leq \Phi(G)$. The Frattini subgroup of a finite group is nilpotent: the Frattini argument for any of its Sylow subgroups gives $G = \Phi(G)N_G(P)$, forcing $N_G(P) = G$. Thus $R = \Phi(G)$, and G/R is minimal nonabelian simple.

For a Sylow subgroup R_p of $\Phi(G)$, the normalizer observation makes every primary $X(G)$ -value of another prime centralize R_p . The generation result makes its centralizer contain all

the corresponding Sylow subgroups of G . That normal centralizer has p -power index, and perfection forces it to be G . It follows that

$$\Phi(G) = Z(G), \quad S = G/Z(G) \text{ is minimal simple.}$$

The order condition has not been passed to a quotient at any point.

We use the standard local consequence of Thompson's minimal-simple classification recorded in [11, Proposition 18]: a finite minimal simple group contains a nontrivial elementary abelian 2-subgroup V with a nontrivial action by an odd-order subgroup normalizing it. Choose an element of that subgroup acting nontrivially and decompose it into primary components; at least one component $\bar{y}$, of odd prime-power order, acts nontrivially. This weaker primary-order choice suffices here. The archive also supplies explicit local matrices for all five minimal-simple families, giving an odd-prime-order choice.

The preimage of V in G has central derived subgroup, hence is nilpotent of class at most two. Its unique Sylow 2-subgroup P maps onto V . Choose a primary lift y of $\bar{y}$ by taking the appropriate primary component of any lift; it normalizes P . For some $u \in P$,

$$a = [u, y] \in P \cap X(G)$$

has nonidentity image $\bar{a} \in V$, an involution.

The Baer–Suzuki theorem, in its usual p -element form recalled in [40], gives a conjugate $\bar{a}^{\bar{g}}$ such that $\langle \bar{a}, \bar{a}^{\bar{g}} \rangle$ is not a 2-group, since $O_2(S) = 1$. Two involutions generate a dihedral group. Their product has order divisible by an odd prime r , and its appropriate power c has order r and is inverted by $\bar{a}$.

The preimage of $\langle c \rangle$ is abelian, as its quotient modulo the central kernel is cyclic. Its unique Sylow r -subgroup R_r maps onto $\langle c \rangle$ and is normalized by a , in the preimage of the dihedral group. The normalizer observation therefore makes a centralize R_r . In the quotient this says that $\bar{a}$ centralizes c , contradicting its inversion action on an element of odd order. This proves the theorem.

The hyperfocal theorem, minimal-simple local structure, and Baer–Suzuki theorem are imported dependencies. The normalizer and central-extension strategy has precedents in the cited works; the required replacement is generation and witness construction with coprime primary inputs. Source: `research/19.56-proof.md`.

C.22 20.92(a): the printed brace average for large primes

For a left brace write $a * b = a \circ b - a - b$. The question asks whether the specific operation

$$a \cdot b = - \sum_{i=0}^{p-2} 2^{-i} ((2^i a) * b)$$

is pre-Lie on every k -dimensional $\mathbb{F}_p$ -brace when p is sufficiently large relative to k . The following bound is coarse but uniform.

Proposition C.4. *For $k \geq 2$, put $E = k^{2k-2}$ and $D = (k-1)E$. The displayed operation is bilinear and pre-Lie if $p > k$ and $\text{ord}_p(2) > D$. In particular $p > \max\{k, 2^D\}$ suffices. For $k = 1$ every odd prime works, with zero product.*

Proof. Let $V = (A, +)$, and set $\lambda_a(b) = -a + a \circ b$. The brace identities make

$$\Gamma = \left\{ G_a = \begin{pmatrix} \lambda_a & a \\ 0 & 1 \end{pmatrix} : a \in A \right\}$$

a regular affine group of order p^k . The p -group $\lambda(A)$ has a full invariant flag with trivial action on successive quotients: the orbit equation gives a nonzero fixed vector in every nonzero $\mathbb{F}_p$ -module of a finite p -group, and one repeats on the quotient. In a suitable basis $\Gamma \leq \text{UT}_{k+1}(\mathbb{F}_p)$. Write J for the strictly upper triangular matrix algebra. Then $J^{k+1} = 0$, and truncated exponential and logarithm are inverse when $p > k$.

The subgroup assertion of Lazard's correspondence [58, pp. 178–180] implies that $\mathfrak{g} = \log \Gamma$ is a Lie subring of J , hence an $\mathbb{F}_p$ -linear Lie subalgebra of dimension k . Projection π onto the translation column is injective on $\mathfrak{g}$: zero translation in X gives zero translation in $\exp X$, and regularity forces $\exp X = 1$. Thus π is an isomorphism, and

$$\mathfrak{g} = \left\{ X_x = \begin{pmatrix} L_x & x \\ 0 & 0 \end{pmatrix} : x \in V \right\}$$

for a linear map $x \mapsto L_x$, all L_x strictly upper triangular. Lie closure gives

$$[L_x, L_y] = L_{L_x y - L_y x}.$$

Therefore $x \triangleright y = L_x y$ is a bilinear pre-Lie product; the equation is precisely the symmetry of its associator.

The translation of $\exp X_x$ is the polynomial

$$W(x) = \sum_{j=0}^{k-1} \frac{L_x^j x}{(j+1)!}.$$

It has degree at most k , constant term zero, and linear term x . Regularity makes $W : V \rightarrow V$ bijective. If Ω denotes its inverse as a function, then $\lambda_a = \exp L_{\Omega(a)}$. We need a polynomial representative of Ω of degree independent of p ; finite-set interpolation would not provide that.

Define the polynomial

$$C(x, y) = \pi \log(\exp X_x \exp X_y).$$

Every surviving associative monomial has length at most k , so $\deg C \leq k$. Truncated BCH is a Lie polynomial over $\mathbb{F}_p$ in this class. Consequently, after scalar extension to the polynomial ring, Lie closure gives the formal identities

$$X_{C(x, y)} = \log(\exp X_x \exp X_y), \quad W(C(x, y)) = W(x) + \exp(L_x)W(y). \quad (*)$$

This step uses a polynomial identity, not an inference from equality on finite-field points.

Put $V_i = \langle e_1, \dots, e_{k+1-i} \rangle$ for $1 \leq i \leq k+1$. For every x , $L_x V_i \subseteq V_{i+1}$. Thus for $r \in V_i$,

$$W(r) - r \in V_{i+1}, \quad (\exp L_x - I)W(r) \in V_{i+1}.$$

These inclusions hold after any scalar extension. Starting with $x_0(a) = 0$, set

$$r_{i-1}(a) = a - W(x_{i-1}(a)), \quad x_i(a) = C(x_{i-1}(a), r_{i-1}(a)).$$

Inductively $r_{i-1} \in V_i$, because $(*)$ changes the residual to $r_{i-1} - \exp(L_{x_{i-1}})W(r_{i-1}) \in V_{i+1}$. Hence $W(x_k(a)) = a$ formally. Since $x_1(a) = a$ and every subsequent step multiplies the previous degree bound by at most k^2 , $\deg x_k \leq E$. On V , $x_k = \Omega$. It has zero constant term and linear term a , the latter by comparison in $W(x_k(a)) = a$.

It follows that

$$P(a) = \exp(L_{\Omega(a)}) - I = \sum_{j=1}^D P_j(a)$$

has a polynomial representative of degree at most D , with homogeneous linear part $P_1(a) = L_a$. If $\text{ord}_p(2) > D$, then

$$\sum_{i=0}^{p-2} 2^{i(j-1)} = \begin{cases} -1 & j = 1, \\ 0 & 2 \leq j \leq D. \end{cases}$$

The stated average is therefore $L_a b = a \triangleright b$ in the original additive coordinates. The scalar 2 need not be a primitive root. Finally, if $p > 2^D$, an order $d \leq D$ would force $p \mid 2^d - 1 < p$, a contradiction. In dimension one, $\lambda(A) \leq \mathbb{F}_p^\times$ is a p -group and is trivial. $\square$

The affine correspondence, residual correction, and weighted coefficient extraction have prior antecedents, especially Trappeniers [90, Proposition 5.5 and Section 6]. The additional step here is the dimension-only polynomial bound using the common triangular flag. It does not assume the stronger condition $L_{V_i} V_j \subseteq V_{i+j}$, or strong nilpotence of the brace. Part (b), the correspondence itself, is recorded as prior work in the Notebook. The retained proof and exact controls are `research/20.92a-proof.md` and `results/20.92a-summary.json`; the controls include actual braces which are not right nilpotent.

C.23 9.47: a scattered compact space excluded by the E-topology

The E-topology on the closed subgroups $L(G)$ of a locally compact Hausdorff group is the full Vietoris topology, with subbasic sets

$$D_1(O) = \{H : H \subseteq O\}, \quad D_2(O) = \{H : H \cap O \neq \emptyset\}$$

for arbitrary open $O \subseteq G$. In particular its upper subbasis is more extensive than that of the Chabauty topology. A *clopen selector* on X assigns a clopen neighborhood U_x to every x , with

$$x \in U_y, y \in U_x \implies x = y, \quad y \in U_x \implies U_y \subseteq U_x.$$

We prove that a compact Hausdorff space embeds in the E-space of closed noncompact subgroups of some locally compact group exactly when it has such a selector. A known height-four compact space fails this test. The ingredients are Protasov's neighborhood lemma [75, 54] and the Dow–Watson example [32]; their use here does not make either ingredient new.

Lemma C.5. *If $\mathcal{F} \subseteq L(G)$ is compact and $H \in \mathcal{F}$ is noncompact, then $\{K \in \mathcal{F} : K \subseteq H\}$ is a neighborhood of H in $\mathcal{F}$.*

Proof. The space $L(G)$ is Hausdorff: if $g \in H \setminus K$, choose an open $V \ni g$ with $\overline{V} \cap K = \emptyset$, and use $D_2(V)$ and $D_1(G \setminus \overline{V})$. Suppose the conclusion fails, so H is in the closure of $\mathcal{A} = \{K \in \mathcal{F} : K \not\subseteq H\}$. For each neighborhood W of H in $\mathcal{F}$, let

$$E_W = \bigcup_{K \in \mathcal{A} \cap W} (K \setminus H).$$

This set is not relatively compact. Otherwise its closure C is compact; choose $h \in H \setminus (C \cup C^{-1}C)$ and an identity neighborhood V so that Vh misses both sets. Some $K \in \mathcal{A} \cap W$ meets Vh , at x . Since $Vh \cap C = \emptyset$, $x \in H$. For $z \in K \setminus H$, both z, zx belong to C , contradicting $CVh \cap C = \emptyset$.

Fix a compact identity neighborhood U . Inductively choose $K_i \in \mathcal{A}$, $x_i \in K_i \setminus H$, and a compact neighborhood V_i of x_i disjoint from H , with

$$K_i \cap V_j = \emptyset \ (j < i), \quad x_i U \cap x_j U = \emptyset \ (j < i).$$

The first condition restricts K_i to an open neighborhood W_i of H ; the second follows by choosing $x_i \in E_{W_i}$ outside the compact union $\bigcup_{j < i} x_j U U^{-1}$. Local compactness supplies V_i . The set $\mathcal{B} = \{K_i\}$ is discrete, because $D_2(\text{int } V_i)$ meets it in finitely many points, which Hausdorff separation reduces to K_i .

Also $D = \{x_i\}$ is closed discrete in G : choose open $Q \ni 1$ with $Q^{-1}Q \subseteq U U^{-1}$; every translate gQ contains at most one x_i . If $K \in \overline{\mathcal{B}} \setminus \mathcal{B}$, it cannot meet $\text{int } V_i$, whose associated lower neighborhood meets $\mathcal{B}$ only finitely. Thus $K \cap D = \emptyset$, but the open upper neighborhood $D_1(G \setminus D)$ contains no K_i , a contradiction. Hence $\mathcal{B}$ is closed as well as discrete, impossible inside compact $\mathcal{F}$. $\square$

If every member of $\mathcal{F}$ is noncompact, set $U_H = \{K \in \mathcal{F} : K \subseteq H\}$. It is closed, since its complement is $D_2(G \setminus H)$. At each $K \in U_H$ the lemma gives a neighborhood contained in $U_K \subseteq U_H$, so it is also open. Subgroup inclusion supplies the selector axioms.

Here is the counterexample, including the obstruction. For every $f : \omega \rightarrow \omega$, choose an infinite $A_f \subseteq \omega$ so that distinct A_f 's meet finitely. Such a family is obtained by coding finite initial segments of the distinct binary branches $0^{f(0)}10^{f(1)}1 \dots$. Start with $Y = (\omega + 1) \times \omega$, a disjoint union of convergent sequences. Write $u_n = (\omega, n)$, $v_{m,n} = (m, n)$, and set

$$B_f = \{u_n : n \in A_f\} \cup \{v_{m,n} : n \in A_f, m > f(n)\}.$$

Adjoin a point a_f with basic neighborhoods consisting of a_f and the part of B_f in rows $n \geq N$. Call the resulting space Z . These neighborhoods are compact: after covering a_f , only finitely many compact row tails remain. Almost disjointness separates different a_f 's; omission of a row separates them from points of Y . Thus Z is locally compact Hausdorff, and these compact neighborhoods are clopen. The infinite set $\{a_f\}$ is closed discrete, so Z is noncompact. Let $X = Z \cup \{\infty\}$ be its one-point compactification. Its successive derived sets are

$$X' = \{u_n\} \cup \{a_f\} \cup \{\infty\}, \quad X'' = \{a_f\} \cup \{\infty\}, \quad X''' = \{\infty\}, \quad X'''' = \emptyset.$$

It is consequently scattered of height four.

Suppose X had a selector. Choose $t(n)$ so that $v_{m,n} \in U_{u_n}$ whenever $m > t(n)$. For any f with $f(n) > t(n)$ for all n , the neighborhood U_{a_f} contains u_n for all sufficiently large $n \in A_f$. Transitivity makes it contain the infinite set

$$D_f = \{v_{t(n)+1,n} : n \in A_f, n \geq N\}.$$

This is closed discrete in Z : each row meets it at most once, the neighborhood based on B_f misses it, and every other a_g has a neighborhood omitting the finitely many common rows. Thus no compact subset of Z contains D_f , so $\infty \in \overline{D_f} \subseteq U_{a_f}$. There are infinitely many such f , for example $f_j(n) = t(n) + j$, $j \geq 1$. Every neighborhood of ∞ , including U_∞ , contains all but finitely many a_{f_j} , since compact subsets of Z meet the closed discrete set $\{a_f\}$ finitely. For some j , both $a_{f_j} \in U_\infty$ and $\infty \in U_{a_{f_j}}$, contradicting antisymmetry. The necessary condition above now answers 9.47 negatively.

For completeness, the condition is also sufficient. Given a compact X with selector, take the discrete abelian group

$$G = \mathbb{Z} \oplus \bigoplus_{y \in X} C_2, \quad H_x = \mathbb{Z} \oplus \bigoplus_{\{y : x \notin U_y\}} C_2.$$

Each H_x is closed and noncompact, and antisymmetry makes $x \mapsto H_x$ injective. Membership of any fixed element in H_x is a finite intersection of clopen conditions, proving continuity for the lower subbasis. If $z \in U_x$, transitivity implies $H_z \subseteq H_x$; this proves continuity for the upper subbasis. A continuous injection from compact X to Hausdorff $L(G)$ is an embedding. The full archive proof is [research/9.47-proof.md](#).

C.24 9.45: a coordinate test for orthogonal lattice bases

For $a \in \mathbb{Q}^n$, let $S(a) = \mathbb{Z}^n + \mathbb{Z}a$. Subtracting an integer vector, write

$$a = r/m, \quad 0 \leq r_j < m, \quad \gcd(m, r_1, \dots, r_n) = 1.$$

The following finite test uses $O(mn)$ integer arithmetic operations; no polynomial bound in the binary length of m is asserted.

For each $1 \leq k < m$, center every coordinate of $kr \bmod m$ in $[-m/2, m/2]$. At a half-coordinate retain both signs; discard a coset with more than two such ties. Among the resulting at most four vectors w , retain those with first nonzero coordinate positive which satisfy, with $Q(w) = \sum_j w_j^2$,

$$Q(w) \mid mw_j \quad \text{for every } j, \quad Q(w) \mid \sum_j r_j w_j. \quad (**)$$

Call their list $\mathcal{W}$, and let z count the zero coordinates of r .

Proposition C.6. *The lattice $S(a)$ has an orthogonal basis if and only if $z + |\mathcal{W}| = n$. In that case a basis is $\{e_j : r_j = 0\} \cup \{w/m : w \in \mathcal{W}\}$.*

Proof. For a Euclidean lattice L , a vector $b \neq 0$ spans an orthogonal rank-one direct summand exactly when $c_b = b/\|b\|^2 \in L^*$: subtract the integer multiple $(x \cdot c_b)b$ from each $x \in L$. Call such a vector splitting. For splitting vectors b, v , both

$$\frac{b \cdot v}{\|b\|^2}, \quad \frac{b \cdot v}{\|v\|^2}$$

are integers. Their product is the squared cosine. If the lines differ, it is less than one and hence zero; if they coincide, the ratio and its reciprocal are integers, so $v = \pm b$. All splitting lines are therefore mutually orthogonal. There are at most n , and if there are n , the orthogonal expansion of every $x \in L$ has integer coefficients, proving that their generators form a basis.

Now $L = S(r/m)$ contains $\mathbb{Z}^n$, so $L^* \subseteq \mathbb{Z}^n$. For a splitting vector write $c = c_b$, $d = \|c\|^2$. Then $c \in \mathbb{Z}^n$, d is a positive integer, and $b = c/d$. If $d = 1$, $b = \pm e_j$, which splits precisely when $r_j = 0$. If $d \geq 2$, then $2|c_j| \leq d$ in every coordinate, so $|b_j| \leq 1/2$. If h coordinates attain equality, then $hd^2/4 \leq d$, giving $h \leq 2$. Moreover $\|b\|^2 = 1/d < 1$, so $b \notin \mathbb{Z}^n$. The gcd condition shows that r/m has order exactly m modulo $\mathbb{Z}^n$. Thus every remaining splitting vector, up to its unique first-positive sign, occurs in the centered enumeration.

For a listed w , $b = w/m$ lies in L , and $c_b = mw/Q(w)$. Membership in L^* says that its coordinates and its scalar product with r/m are integers, which is exactly (**). Different cosets or tie choices give different vectors. The test therefore lists all splitting lines exactly once, proving the result. $\square$

Half-coordinate signs matter: for $a = (1/6, 1/6)$, the vector $(3, -3)$ passes (**) while $(3, 3)$ fails. For $a = (1/2, 1/2, 1/2)$, there are no splitting lines. A three-dimensional positive example is $a = (1, 5, 2)/6$, with basis

$$(1, -1, 2)/6, \quad (1, -1, -1)/3, \quad (1, 1, 0)/2.$$

The archive also proves that when $m = p^k$, at most two coordinates of r can be nonzero in an orthogonal example. Indeed an integral orthogonal basis matrix C of L^* has determinant p^k , Smith form $\text{diag}(1, \dots, 1, p^k)$, and diagonal CC^t whose entries are powers of p . The rank inequality modulo p gives $\text{rank}(CC^t) \geq n - 2$. All but at most two row norms are therefore one, forcing the other residue coordinates to vanish.

Bardakov already solved the two-dimensional case and gave a higher-dimensional sufficient condition [8]. Normalized-dual arguments and orthogonal decomposition are prior methods; compare [22, 42]. The candidate contribution is this particular cyclic-coset criterion in arbitrary dimension. The archive `research/9.45-proof.md` contains the prime-power classification; `results/9.45-summary.json` binds independent finite controls.

C.25 10.62: a periodic perfect group generated by involutions

Proposition C.7. *For every sufficiently large odd prime p , there is an infinite three-generated perfect group G of exponent $2p$, with no subgroup of order four, whose involutions form one conjugacy class and generate G . Any two distinct involutions have product of order p .*

The deep existence input is Amelio’s geometric quotient construction [3, Proposition 6.1, Theorem 6.7, Remark 6.8]. We use its construction as well as its final theorem: the first quotient may impose a prescribed eligible primitive odd-power relation. This feature is needed to eliminate the quotient of order two. The cone-off and small-cancellation estimates underlying that result are imported, not reproved by the experiment.

Proof. Start with

$$F = \langle a, b, c \mid a^2 = b^2 = c^2 = 1 \rangle = C_2 * C_2 * C_2$$

acting by left multiplication on its trivalent Cayley tree with unit-length edges. Vertex stabilizers are trivial; a nontrivial point stabilizer is the order-two group interchanging an edge’s endpoints. A finite subgroup fixes the center of a finite invariant subtree and therefore has order at most two. The action is acylindrical: the number of elements moving one point at most R is bounded by the size of a uniform vertex ball of radius $R + 1$. This also shows that elliptic subgroups are finite. The translations ab, ac have different endpoint pairs, so the action is non-elementary. Every nonelliptic element translates by at least two, and the injectivity radius is exactly two.

Amelio calls an action tame when there is no subgroup of order four and each maximal loxodromic subgroup has finite normal subgroup of order at most two. Here an axis stabilizer acts faithfully on a simplicial line, since it fixes no vertex nontrivially; it is cyclic or infinite dihedral and has trivial finite normal subgroup. Thus the action is tame. The additional numerical invariants in Theorem 6.7 are finite. More precisely, its invariant ν is at most one: if h is loxodromic and $\langle g, h^{-1}gh \rangle$ is elliptic, then $g \neq 1$ would force h to fix the unique midpoint fixed by g , a contradiction. Thus $\tau = \max\{\nu, 3\} = 3$. Regard the tree as δ -hyperbolic for $0 < \delta < \min\{1/L_S, 1/100\}$, with L_S the source’s constant. Every short element used in its overlap invariant Ω is then elliptic. The 13δ -enlargement of its almost-fixed set lies within 17δ of its unique midpoint. Distinct midpoints are at least one apart, while a non-elementary tuple contains two distinct nontrivial elliptic elements. Hence the relevant intersections are empty and $\Omega = 0$. Using positive δ respects the source’s strict displacement inequalities. Take Q to be all loxodromic elements, explicitly permitted as a strongly stable family by Remark 6.6.

The word $w = abc$ is primitive in its maximal loxodromic subgroup. Its axis has periodic label pattern $abcabc\cdots$, whose least positive period is three; free vertex action makes its translation stabilizer exactly $\langle w \rangle$. An orientation reversal would identify abc with cba up to cyclic rotation, which is impossible. Thus the full axis stabilizer is $\langle w \rangle$.

Choose an initial scale $\lambda > 0$ sufficiently small that $3\lambda \leq L_S\delta_1$, together with the initialization conditions in Theorem 6.7. Remark 6.8 permits this smaller scale at the cost of increasing the critical exponent. The new injectivity radius is still $2\lambda > 0$; the required lower bound tends to zero as the critical exponent grows. All choices are made before choosing p .

For a sufficiently large odd prime p , use the exponent family $\{p\}$ and include w in the first maximal admissible family of primitive loxodromic elements. Proposition 6.1 then kills w^p .

The theorem constructs epimorphisms

$$F = F_0 \twoheadrightarrow F_1 \twoheadrightarrow F_2 \twoheadrightarrow \cdots \twoheadrightarrow G$$

with infinite limit, embedding the initial elliptic subgroups, and with every initial loxodromic element mapping to an element of order dividing $2p$ or to an initial elliptic image. Consequently a, b, c survive as involutions generating G , every element of G has order dividing $2p$, and

$$(abc)^p = 1. \quad (****)$$

Each F_i is tame and has no subgroup of order four. This property passes to the limit here: a finite subgroup of G has finitely many multiplication-table relations; choose lifts of its elements, and a stage at which all those relations hold. They define an embedding into that stage, because composition to G is the original inclusion. Thus G also has no subgroup of order four.

Let x, y be distinct involutions of G , including any not identified beforehand with original involutions. Put $t = xy$. If t had even order d , then $z = t^{d/2}$ would be an involution centralized by x . Also $x \notin \langle t \rangle$, since otherwise y would lie there too and a cyclic group has only one involution. Thus $\langle x, z \rangle \cong C_2^2$, a contradiction. Hence $|t| = p$. With $k = (p+1)/2$,

$$t^{-k}xt^k = xt^{2k} = y,$$

so all involutions are conjugate. There are at least two, since otherwise G would be cyclic of order two. The exponent is exactly $2p$.

In the abelianization all three generating involutions have the same image ϵ , of order dividing two. Relation (****) gives $3p\epsilon = 0$, and oddness implies $\epsilon = 0$. Thus G is perfect, and in particular has no subgroup of index two. $\square$

No numerical threshold for p , finite presentation, or simplicity claim follows here. The new candidate is the application with the selected odd relation; the geometric construction is Amelio's. Finite dihedral and abelianization controls in `results/10.62-summary.json` check only the elementary deductions. They do not construct this infinite group. The complete parameter and source audit is `research/10.62-review.md`.

C.26 10.32: a logarithmic bound for symmetric-group power words

Proposition C.8. *Let $r, s \neq 0$, with at least one odd, and $m = \prod_{\ell|rs} \ell$, the product of the distinct prime divisors of $|rs|$. The word $x^r y^s$ is universal on S_n whenever*

$$n \geq \max\{1126, 16 \log m\}.$$

More generally, every real $c > 8$ admits $N_0(c)$ for which $n \geq \max\{N_0(c), c \log m\}$ suffices.

Proof. We use two permutation-covering results. First, if two classes in S_d have h, k cycles, counting fixed points, and

$$h + k + 1 \leq L \leq d, \quad L \equiv h + k - 1 \pmod{2},$$

some representatives have product an L -cycle. This is Boccara's positive-genus three-partition result in the form reproved by Song–Xu–Ye [86, Propositions 1.4 and 2.1]. Second, if $u, v \geq 4$ and $\lfloor 3n/4 \rfloor + 1 \leq u + v \leq n$, every even permutation is a product of two permutations each consisting of a disjoint u -cycle and v -cycle [16, Lemma 5].

Suppose an odd q satisfies $3n/8 + 1 < q \leq n/2$. Every odd $g \in S_n$ is a product of a $2q$ -cycle and two disjoint q -cycles. For a transposition g , choose the two q -cycles with its letters in different supports; multiplying their inverse by g merges them. Otherwise let s_0 be the support size, t its number of nontrivial cycles, and $e = s_0 - t$, an odd integer at least three. Restrict to an invariant subset of size $d = \max\{s_0, 2q\}$. The class of g there has $h = t + d - s_0$ cycles; the class of two q -cycles has $k = d - 2q + 2$. If $s_0 \leq 2q$, then $h + k + 1 = 2q - e + 3 \leq 2q$. Otherwise, using $t \leq s_0/2$,

$$h + k + 1 = s_0 + t - 2q + 3 \leq 3n/2 - 2q + 3 < 2q.$$

Also $h + k - 1 = 2d - e - 2q + 1$ is even. The first covering result, followed by inversion of its second factor and simultaneous conjugation, gives the assertion. Removing the unneeded fixed points before applying it is essential.

Now let $q \geq 5$ be prime in that interval with $q \nmid rs$, and suppose r is odd. For odd targets the factors just obtained have orders $2q, q$; for even targets the second covering result with $u = v = q$ supplies two factors of order q . Their orders are coprime to r, s , respectively, so taking inverse exponents modulo those orders gives $g = x^r y^s$. If s is the odd exponent, reverse the argument's factor order using $U^s V^r = V^r (V^{-r} U V^r)^s$. Negative exponents cause no change.

It remains to find a prime in the interval missing from m . Let $\vartheta(x) = \sum_{\ell \leq x} \log \ell$. Rosser–Schoenfeld [77, Theorems 4 and 9] give

$$\vartheta(x) > x \left(1 - \frac{1}{2 \log x}\right) \quad (x \geq 563), \quad \vartheta(y) < 1.01624y \quad (y > 0).$$

For $n \geq 1126$, $\log(n/2) > 6$, and $1.01624 < 49/48$. Consequently

$$\vartheta(n/2) - \vartheta(3n/8 + 1) > \frac{29n}{384} - \frac{49}{48} > \frac{n}{16}.$$

If all primes in the interval divided m , this would contradict $\log m \leq n/16$. The explicit bound follows. The same source gives $\vartheta(x) = x + o(x)$, so the interval's prime-log sum is $n/8 + o(n)$, eventually exceeding n/c for any $c > 8$. This proves the second statement. $\square$

If both exponents are odd, every degree works: every permutation is a product of two involutions, by reversing each cycle. An exponent 1 or -1 also makes universality immediate. The stated constants are sufficient, not optimal. The candidate contribution is this symmetric-group deduction from the credited covering and prime estimates, rather than new versions of those imported theorems. See `research/10.32-proof.md` and `results/10.32-summary.json`.

C.27 15.65: nonrational classical-group limits

For the full families $U(n, q)$, $\mathrm{Sp}(2m, q)$, $O(2m + 1, q)$, $O^+(2m, q)$, and $O^-(2m, q)$, both limiting proportions asked about are nonrational functions of q . For symplectic and orthogonal groups the assertion holds separately on even and on odd prime powers. The Notebook calls a matrix separable when its *minimal* polynomial is squarefree; this is semisimplicity. The enumeration source uses separable for squarefree *characteristic* polynomial. We distinguish the two and prove nonrationality for both, as well as for cyclic matrices. This is a statement about functions of field size, not about irrationality of an individual probability.

All enumeration and limit-existence formulas below are those of Fulman–Neumann–Praeger (FNP) [36]. The proposed additional deduction is their analytic continuation in

inverse field size and the obstruction to rational interpolation. Let $t = 1/q$. The count polynomials for unitary factors are

$$\begin{aligned} N_d(q) &= \frac{1}{d} \sum_{a|d} \mu(a)(q^{d/a} + 1) & (d \text{ odd}), \\ M_1(q) &= (q^2 - q - 2)/2, \\ M_d(q) &= \frac{1}{2d} \sum_{a|d} \mu(a)(q^{2d/a} - q^{d/a}) & (d > 1 \text{ odd}), \\ M_d(q) &= \frac{1}{2d} \sum_{a|d} \mu(a)q^{2d/a} & (d \text{ even}). \end{aligned}$$

For symplectic and orthogonal factors fix $e = 1$ for even q or $e = 2$ for odd q , and use

$$\begin{aligned} J_d(q) &= \frac{1}{2d} \sum_{\substack{a|d \\ a \text{ odd}}} \mu(a)(q^{d/a} + 1 - e), \\ K_1(q) &= (q - e - 1)/2, \\ K_d(q) &= \frac{1}{2}N(q; d) & (d > 1 \text{ odd}), \\ K_d(q) &= \frac{1}{2}(N(q; d) - J_{d/2}(q)) & (d \text{ even}), \\ N(q; d) &= \frac{1}{d} \sum_{a|d} \mu(a)q^{d/a} & (d > 1). \end{aligned}$$

These are FNP Lemmas 1.3.12 and 1.3.16. In particular

$$N_1(1/t) = 1 + 1/t, \quad J_1(1/t) = (1/t + 1 - e)/2.$$

For $0 < |t| < 1$, their explicit formulas give bounds by a constant times $|t|^{-d}$ for N_d, J_d, K_d , and $|t|^{-2d}$ for M_d . For instance, bounding the number of divisors by d suffices. These estimates will justify the infinite products as analytic functions.

The analytic obstruction. Suppose a holomorphic function $a(t)$ has a zero of multiplicity $h > 0$ at $b \neq 0$, $P(t)$ is holomorphic and nonconstant near b , and $K(t)$ is a nonzero single-valued meromorphic germ there. Then a branch of $K(t) \exp(P(t) \log a(t))$ cannot continue a rational germ. One circuit around b multiplies it by $\exp(2\pi i h P(t))$. Rational single-valuedness would make this identically one on an open set, forcing $P' = 0$. An integer-order zero or pole of K cannot cancel this monodromy.

Each germ constructed below has a finite nonzero value at $t = 0$ and agrees with the indicated probability at sufficiently small positive $t = 1/q$. If a rational function of q agreed on all powers of a fixed prime, its transform in t would have no pole at zero and would agree with the germ by the identity theorem. The analytic obstruction therefore excludes even that restricted form of rational interpolation.

Cyclic limits. Put

$$A(u) = \frac{1 + u - u^2}{1 + u}, \quad B(u) = \frac{1 - u + u^2}{1 - u} = \frac{1 + u^3}{1 - u^2}.$$

FNP Theorems 2.1.9 and 2.2.9 give

$$\begin{aligned} C(U; q) &= (1 + t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} A(t^d)^{N_d(1/t)} \prod_{d \geq 1} B(t^{2d})^{M_d(1/t)}, \\ C(\text{Sp}; q) &= \prod_{d \geq 1} A(t^d)^{J_d(1/t)} B(t^d)^{K_d(1/t)}. \end{aligned}$$

Powers mean exponentials of logarithms initially vanishing at zero. The first A -factor has a simple zero at $\alpha = (1 - \sqrt{5})/2$. Remove it. On $|t| < 3/4$, every remaining A -factor has degree $d \geq 2$, and

$$\left| \frac{t^{2d}}{1 + t^d} \right| \leq \frac{(3/4)^4}{1 - (3/4)^2} = \frac{81}{112} < 1.$$

The logarithms of all B -factors are single-valued on $|t| < 1$ by the second expression for B . Their logarithms vanish to order at least $2d$, or $4d$ after substitution t^{2d} . Multiplication by the count polynomials therefore leaves summable bounds by constant multiples of $|t|^d$ or $|t|^{2d}$, uniformly on compact subdiscs. All apparent singularities at zero are removable. The remaining product is consequently holomorphic and nonvanishing on $|t| < 3/4$. The first exponents $N_1(1/t)$ and $J_1(1/t)$ are nonconstant, so the obstruction at α proves nonrationality. In the unitary case the real local exponent is $1 + 1/\alpha = \alpha$, also giving a direct noninteger-order proof. FNP Theorem 2.3.11 gives

$$C(O_{\text{odd}}; q) = (1 - t)C(\text{Sp}; q), \quad C(O^{\pm}; q) = \frac{(1 - t)^e + 1}{2} C(\text{Sp}; q).$$

These nonzero rational multipliers transfer the conclusion to all full orthogonal families.

A zero needed for the literal separability question. Define holomorphic functions on $|u| < 1$ by

$$\mathcal{U}(u) = \sum_{m \geq 0} \frac{u^{m^2}}{\prod_{i=1}^m (1 - (-u)^i)}, \quad \mathcal{L}(u) = \sum_{m \geq 0} \frac{u^{m^2}}{\prod_{i=1}^m (1 - u^i)}.$$

On $|u| \leq r < 1$, denominator products are bounded below by $\prod_{i \geq 1} (1 - r^i) > 0$, proving normal convergence. Set $a_m(r) = r^{m^2} / \prod_{i=1}^m (1 - r^i)$. For $m \geq 1$,

$$\frac{a_{m+1}(r)}{a_m(r)} \leq \frac{r^3}{1 - r^2}, \quad \sum_{m \geq 1} a_m(3/7) \leq \frac{210}{253} < 1.$$

Hence both functions are nonzero on $|u| \leq 3/7$. On the negative real axis, $\mathcal{U}(-r) = 1 - a_1(r) + a_2(r) - \dots$. For $r \leq 3/5$ these positive terms decrease. Alternating bounds give positivity for $r \leq 1/2$, a lower bound $9/56$ at $1/2$, and the following negative upper bound at $3/5$:

$$1 - a_1 + a_2 - a_3 + a_4 = -\frac{1128797279}{26656000000} < 0.$$

Thus there is a first negative zero $b \in (-3/5, -1/2)$, of finite positive multiplicity, with $\mathcal{U}(t) > 0$ for $b < t \leq 0$. Also $\mathcal{L}(-r) > 1 - r/(1 + r) > 0$ for $0 < r \leq 13/20$: its alternating term ratios are bounded by $r^3/(1 - r^2) < 1$. No simplicity or exact location of b is needed.

Put

$$E(u) = (1 - u)\mathcal{U}(u), \quad D(u) = (1 - u)\mathcal{L}(u).$$

They are nonzero on $|u| \leq 3/7$, and both are $1 + O(u^2)$. Let SS denote semisimplicity, namely squarefree minimal polynomial. FNP Theorems 3.1.13 and 3.1.15 give

$$SS(U; q) = (1 + t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} E(t^d)^{N_d(1/t)} \prod_{d \geq 1} D(t^{2d})^{M_d(1/t)},$$

$$SS(\text{Sp}; q) = (1 - t)^e \mathcal{P}(t)^e T_e(t),$$

$$T_e(t) = \prod_{d \geq 1} E(t^d)^{J_d(1/t)} D(t^d)^{K_d(1/t)},$$

$$\mathcal{P}(t) = 1 + \sum_{m \geq 1} \frac{t^{m(2m+1)}}{\prod_{i=1}^m (1 - t^{2i})}.$$

The function $\mathcal{P}$ is single-valued holomorphic on the unit disc, with value one at zero. Take $R = 13/20$. All arguments of factors of degree at least two lie in $|u| \leq R^2 = 169/400 < 3/7$. Their logarithms and the count-polynomial bounds thus give normally convergent, nonvanishing remainders on $|t| < R$. For T_e , separate also the degree-one $D(t)$ -factor. It has a holomorphic logarithm near $[b, 0]$, by the positivity just proved; multiplication by $K_1(1/t)$ is removable at zero since $\log D(t) = O(t^2)$. The only required branching at b comes from $E(t)$, with nonconstant exponent $N_1(1/t)$ or $J_1(1/t)$. The prefactors can have only integer-order zeros there. The obstruction proves both nonrationality claims.

For the orthogonal case put

$$\mathcal{Q}(t) = 1 + \sum_{m \geq 1} \frac{t^{m(2m-1)}}{\prod_{i=1}^m (1 - t^{2i})}.$$

It is likewise holomorphic on the unit disc, with value one at zero. The reciprocal orders of the two even-dimensional full orthogonal groups sum to the terms of $\mathcal{Q}(1/q) - 1$. FNP Theorem 3.1.18 gives

$$\begin{aligned} SS(O_{\text{odd}}; q) &= (1 - t)^e \mathcal{Q}(t)^{e-1} \mathcal{P}(t) T_e(t), \\ SS(O^{\pm}; q) &= \frac{(1 - t)^e}{2} (\mathcal{Q}(t)^e + (e - 1) \mathcal{P}(t)^2) T_e(t). \end{aligned}$$

The extra factors are single-valued and not identically zero: at zero they equal 1 and $e/2$. Their possible finite-order zeros cannot cancel the monodromy of $E(t)^{J_1(1/t)}$. This proves the literal Notebook assertion for all the stated families.

The other meaning of separability. For completeness let RS mean squarefree characteristic polynomial, and set $V(u) = (1 - u)(1 + 2u)/(1 + u)$. FNP Theorems 2.1.3, 2.2.3 and 2.3.4 give

$$\begin{aligned} RS(U; q) &= (1 + t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} V(t^d)^{N_d(1/t)}, \\ RS(\text{Sp}; q) &= (1 - t)^e \prod_{d \geq 1} V(t^d)^{J_d(1/t)}, \\ RS(O_{\text{odd}}; q) &= RS(\text{Sp}; q), \\ RS(O^{\pm}; q) &= 2^{e-2} RS(\text{Sp}; q). \end{aligned}$$

The first V -factor vanishes at $-1/2$. All higher factors have zero-free logarithms on $|t| < 13/20$, and the same estimates prove normal convergence. The first exponents are nonconstant, so the branching obstruction applies even when their values at $-1/2$ happen to be integers. This proves nonrationality for the source's convention too.

The symplectic semisimple formula uses $M^*(q; d)$, as in FNP Theorem 3.1.15 and its underlying product; an intermediate display in Lemma 3.1.9(e) prints $M^*(q; 2d)$. The archive records that index discrepancy and a separate coefficient discrepancy. No numerical coefficient table is a premise here. The distinction between the dimension variable of FNP's generating functions and our field-size variable t is essential. See `research/15.65-proof.md`, `research/15.65-classical-proof.md`, and their source audits.

C.28 15.92: continuum many alternating-residual triangle quotients

For every $r > 6$, the triangle group

$$\Gamma_r = \langle x, y \mid x^2 = y^3 = (xy)^r = 1 \rangle$$

has continuum many pairwise nonisomorphic infinite quotients in which every nonidentity element survives an epimorphism onto a finite alternating group. The proof combines Conder's finite permutation constructions with the following separating-kernel argument.

Lemma C.9. *Suppose a finitely generated group Γ has epimorphisms $\theta_i : \Gamma \rightarrow A_{n_i}$, with $n_i \rightarrow \infty$, and one element v whose image is always nontrivial and moves at most a fixed number S of points. Then Γ has continuum many pairwise nonisomorphic infinite alternating-residual quotients.*

Proof. By diagonal subsequence selection, arrange that each element of the countable group Γ is eventually always trivial or eventually always nontrivial under θ_i . The eventually trivial elements form a normal subgroup R ; let $G = \Gamma/R$, with $1 \neq \bar{v} \in G$. The action on a nonprincipal ultraproduct of the underlying finite sets is faithful: a word eventually nontrivial has a sequence of moved points. The support of $\bar{v}$ has at most S elements, since $S+1$ distinct moved ultraproduct points would contradict the coordinate support bound on an ultrafilter-large set.

Let $N = \langle \bar{v}^G \rangle$. On the union Ω_* of all translates of $\text{supp}(\bar{v})$, the group N acts faithfully by finitary permutations. There are at most S G -orbits. Because $N \trianglelefteq G$, its orbits within each G -orbit have equal size. Thus a finite M bounds all finite N -orbit sizes. Let η be the action of N on the union of its finite orbits, and put $L = \ker \eta$. The exponent of $\eta(N)$ divides $M!$.

Suppose $\phi : G \rightarrow A_m$ is onto, $m \geq 5$, and $\phi(\bar{v}) \neq 1$. Then $\phi(N) = A_m$. Set $H = \ker(\phi|_N)$, a finite-index normal subgroup of N . Every H -orbit inside an infinite N -orbit is infinite, since the latter is a finite union of translates of the former. For $x \in L$ and $y \in N$, there is $h \in H$ moving $\text{supp}(x)$ disjoint from $\text{supp}(y)$. Indeed H is locally finite as a finitary permutation group. For finite sets $A = \text{supp}(x)$, $B = \text{supp}(y)$, choose finitely many elements of H so that the finite subgroup they generate has every orbit of $a \in A$ larger than $|A||B|$. A uniform element of this finite subgroup avoids all equalities $h(a) = b$ by the union bound. Empty supports are immediate. Thus $h x h^{-1}$ commutes with y , and $\phi(h) = 1$ gives $\phi(x) \in Z(\phi(N)) = 1$. Therefore A_m is a quotient of $\eta(N)$, of exponent dividing $M!$. For m at least an odd prime greater than M , this is impossible because A_m contains a cycle of that prime length. All sufficiently large alternating quotients of G consequently kill $\bar{v}$.

For sufficiently large i , θ_i cannot factor through G . Hence $\theta_i(R) \neq 1$, and normality and simplicity give $\theta_i(R) = A_{n_i}$. Inductively select indices i_j and elements $r_j \in R$ so that

$$\theta_{i_k}(r_j) \neq 1 \iff k = j.$$

At step j , choose i_j beyond the cutoffs for all preceding r_k , with $|A_{n_{i_j}}| > \prod_{k < j} |A_{n_{i_k}}|$. The subgroup $L_j = R \cap \bigcap_{k < j} \ker \theta_{i_k}$ has index in R at most this product, so it cannot be killed by the surjection $\theta_{i_j}|_{L_j}$. Choose $r_j \in L_j$ with nontrivial image. Since $r_j \in R$, all sufficiently late maps kill it, supplying the next cutoff.

For each infinite $I \subseteq \mathbb{N}$, put $K_I = \bigcap_{j \in I} \ker \theta_{i_j}$. The witnesses give $r_j \notin K_I$ exactly when $j \in I$; hence there are continuum many different kernels. Each quotient embeds in the corresponding product of alternating groups with onto coordinates, and is infinite because their orders are unbounded. For each fixed countable quotient isomorphism type, there are only countably many epimorphisms from a finitely generated Γ , as images of a finite generating tuple determine a map. The continuum many kernels therefore give continuum many types. $\square$

We specify the finite input, which is stronger than mere existence of alternating quotients.

Write $r = h + 6d$, $h \in \{7, 8, 9, 10, 11, 12\}$, $d \geq 0$, and choose

$$p = \begin{cases} 11 & h = 7, 8, 11, \\ 13 & h = 9, 10, 12. \end{cases}$$

Conder's thesis [25, Chapter 5, pp. 117–120] constructs arbitrarily large actions of the extended triangle group with generators x_n, y_n, t_n satisfying

$$x_n^2 = y_n^3 = (x_n y_n)^r = t_n^2 = (x_n t_n)^2 = (y_n t_n)^2 = 1.$$

The parity choice makes $\langle x_n, y_n \rangle = A_n$, including when r is even. In these same actions $w_n = x_n y_n t_n$ has one p -cycle, no other cycle length divisible by p , and all cycle lengths divide $720720 = 2^4 3^2 5 7 11 13$. These cycle and parity facts are read from the construction, not inferred from its headline generation theorem.

The relations give $w_n^2 = x_n y_n x_n y_n^{-1}$. The even integer $E = 720720/p$ is coprime to p and divisible by every other occurring cycle length, so w_n^E is a single nontrivial p -cycle. Thus the one fixed word

$$v_r = (xyxy^{-1})^{32760} \quad (p = 11), \quad v_r = (xyxy^{-1})^{27720} \quad (p = 13)$$

has nontrivial support at most 13 in all the selected alternating images of the ordinary triangle group. The lemma proves the result. The presentation printed in the Notebook omits the final $= 1$; that literal group also surjects onto Γ_r , so the same quotients answer it. See `research/15.92-proof.md` and its source-data controls. Neither residual alternatingness of Γ_r itself under the chosen family nor independence of arbitrary quotient kernels is assumed.

C.29 4.75: noncentral involutions modulo the odd-order radical

Use the same group G as in Section C.25; this is a second consequence of one geometric construction. Every 2-subgroup has exponent two because G has exponent $2p$. It is therefore abelian, and the absence of a subgroup of order four forces its order to be at most two. The maximal 2-subgroups are exactly the cyclic groups generated by involutions, and are locally cyclic. No infinite-group Sylow theorem is required.

For a periodic group, $O_{2'}(G)$ denotes the largest normal subgroup all of whose elements have odd order. It exists: if A, B are normal odd-order-element subgroups, any involution of AB has trivial image in AB/A and would lie in A , a contradiction. Periodicity then makes all elements of AB odd-order. The directed union of finite products gives the largest such subgroup. Write it as O . Every involution i has nontrivial image in G/O .

If iO were central, all conjugates of i would have the same image. They generate G , so $G/O = \langle iO \rangle$ would be cyclic. It is also a quotient of the perfect group G , and hence perfect, forcing it to be trivial. This contradicts $i \notin O$. Thus every involution has noncentral image in $G/O_{2'}(G)$, answering 4.75 negatively even with Sylow 2-subgroups of order two. Determination of the radical itself is unnecessary. See `research/4.75-proof.md`.

C.30 9.4: finite generators of group-isotope varieties

Use ordinary quasigroups in the three-operation, one-sorted signature $(*, \backslash, /)$. If a nontrivial finite group generates a variety $\mathcal{M}$, the class IM of all isotopes of groups in $\mathcal{M}$ cannot be generated by any finite quasigroup, or by any finite family of finite quasigroups. The same obstruction applies to quasivarieties and pseudovarieties.

Define terms $t_0(x, y) = y$, $t_{j+1}(x, y) = x * t_j(x, y)$. Every quasigroup of size at most n satisfies

$$t_{n!}(x, y) = y, \tag{***}$$

because each left translation is a permutation whose cycle lengths divide $n!$. Identities persist under each of the three generation operations just listed.

Choose a subgroup C_p of the nontrivial finite generating group. For any $N \geq 1$, let $m = N + 1$, $A = \mathbb{F}_p^m$, and let T cycle its standard basis $e_0, \dots, e_{m-1}$. The operation

$$a * b = a + T(b), \quad a \setminus c = T^{-1}(c - a), \quad c/b = c - T(b)$$

defines a quasigroup isotopic to A ; the isotopy is $(\text{id}, T, \text{id})$. Only a subgroup and finite direct powers of the original group were used, so this finite quasigroup belongs to IM in all three interpretations. At $x = 0, y = e_0$, however,

$$t_N(0, e_0) = T^N(e_0) = e_N \neq e_0.$$

Taking $N = n!$ contradicts (***) for every proposed finite family of finite generators.

There is also a local-finiteness obstruction. On $A = \bigoplus_{i \in \mathbb{Z}} \mathbb{F}_p e_i$, for the same prime p chosen above, use the bilateral shift and the same operation. This infinite quasigroup is generated by e_0 : $e_0 \setminus e_0 = 0$, the terms $0 * e_i$ and $0 \setminus e_i$ move in both directions, and $a + b = a * (0 \setminus b)$ forms all finite sums. A variety generated by a finite algebra Q is locally finite, since its free algebra on k generators has at most $|Q|^{|Q|^k}$ term functions.

The interpretation is essential. In a three-sorted reversible automaton, $x * (x * y)$ need not be well typed. Gvaramiya distinguishes these objects from ordinary quasigroups, and states the ordinary isotope variety theorem in [41, Section 3, Theorem 8]; compare also [56]. The argument concerns ordinary algebraic generation, as in the supplied statement, and makes no assertion about generation defined to include closure under isotopy. Its simplicity makes independent checking of the intended convention and priority particularly valuable. See `research/9.4-proof.md`.

C.31 12.40: unbounded Brauer-degree parts at fixed Sylow order

Proposition C.10. *For every $k \geq 3$, some odd positive integer a gives an irreducible 2-Brauer character φ of $G = \text{PSL}_6(5^a)$ with*

$$|G|_2 = 8192, \quad \varphi(1)_2 = 2^k.$$

Thus no finite-valued bound $\varphi(1)_p \leq f(|G|_p)$ exists, already for the fixed prime $p = 2$.

Proof. Bray's projective-point module [15, Main Theorem] has an absolutely irreducible heart in characteristic two of dimension

$$F(q) = \frac{q^6 - 1}{q - 1} - 2 = q^5 + q^4 + q^3 + q^2 + q - 1.$$

Here $q = 5^a$. To recall why the assertion applies, let P be the projective permutation module over an algebraically closed field of characteristic two, U its constant line, and W its augmentation kernel. Its even permutation degree puts $U \subseteq W$. Perfectness of $\text{SL}_6(q)$ excludes a trivial submodule of W/U : its preimage would be an extension of two trivial modules, which splits, contradicting the one-dimensional invariant space of P . The permutation form makes W/U self-dual, so there is no trivial quotient either. Each nontrivial simple factor has dimension at least $q^5 - 1$: on the elementary abelian unipotent subgroup $\mathbb{F}_q^5$ it has a nontrivial weight, and the Levi subgroup $\text{SL}_5(q)$ permutes all nontrivial weights transitively. The unipotent subgroup normally generates $\text{SL}_6(q)$, so it cannot act trivially on a nontrivial simple factor. Since $2(q^5 - 1) > F(q)$, there is only one nontrivial factor. Absence of trivial submodules and quotients now makes the heart simple. Scalars fix projective points, so it descends to $\text{PSL}_6(q)$.

For odd a , $q \equiv 5 \pmod{8}$ and $q \equiv -1 \pmod{3}$; hence $\gcd(6, q-1) = 2$. The order formula is

$$|G| = \frac{q^{15}}{2} \prod_{i=2}^6 (q^i - 1).$$

The five factors have 2-adic valuations 3, 2, 4, 2, 3. Their sum, less the central factor of two, is 13, independently of a .

It remains to obtain arbitrarily prescribed degree valuations at actual prime powers. Repeated squaring gives $v_2(5^{2^j} - 1) = j + 2$, so for $k \geq 3$,

$$5^{2^{k-2}} \equiv 1 + 2^k \pmod{2^{k+1}}.$$

For odd x , $F'(x)$ is odd, and the binomial theorem gives

$$F(x + 2^k) \equiv F(x) + 2^k \pmod{2^{k+1}}.$$

Therefore, whenever odd a satisfies $2^k \mid F(5^a)$, exactly one of a and $a + 2^{k-2}$ gives divisibility by 2^{k+1} ; the other gives valuation exactly k . Start with $a = 1$, since $F(5) = 3904$ is divisible by eight, and continue using the divisible choice. The unused choice at each step supplies the required odd exponent and exact valuation. $\square$

The representation family and its irreducibility are prior mathematics; the fixed group-order valuation and unbounded degree valuation are the deduction proposed here. No assertion about ordinary character degrees is being made. See `research/12.40-proof.md`; finite module and integer controls corroborate the formulas without replacing the arbitrary- k argument.

C.32 13.42: a tensor completion that is not locally nilpotent

Let $A = \mathbb{Z}[t]$ and $N = F(x, y)/\gamma_4 F(x, y)$. Its unrestricted tensor A -completion is not locally nilpotent. Here an A -group satisfies the power, conjugation, and commuting-product axioms in Myasnikov–Remeslennikov [68, Definitions 1–2]. The universal category allows scalar torsion and does not restrict targets to nilpotent groups. This distinction is part of the claim.

Let L be the rational BCH group of the Lie algebra with basis X, Y, U, V, W , brackets

$$[X, Y] = U, \quad [X, U] = V, \quad [Y, U] = W,$$

and all other basic brackets zero. Coordinates (a, b, c, d, e) denote $\exp(aX + bY + cU + dV + eW)$, and $\beta(a, b, c, d, e) = b$ is a homomorphism $L \rightarrow (\mathbb{Q}, +)$. These are finite polynomial operations: realize the Lie algebra in the free associative algebra on X, Y truncated above degree three and use $\exp Z = 1 + Z + Z^2/2 + Z^3/6$. For $l = (a, b, c, d, e)$, $h = (A, B, C, D, E)$, BCH gives the first three coordinates of $l^h = h^{-1}lh$ as

$$(a, b, c + aB - bA).$$

It also gives the commuting criterion

$$[l, h] = 1 \iff aB - bA = aC - cA = bC - cB = 0.$$

For example, equality of the degree-two coordinates in the two products first gives $aB - bA = 0$; then equality in degree three gives the other two minors. Thus commuting elements have proportional triples (a, b, c) , (A, B, C) , including the zero cases.

Let $M = \mathbb{Q}[\mathbb{Q}]$ additively, with basis T^q , $q \in \mathbb{Q}$. Give it the right L -action $v^l = vT^{\beta(l)}$, and define

$$F(a, b, c, d, e) = \begin{cases} aT^{c/a} & b = 0, a \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

The conjugation formula, coordinate powers, and proportional-triple criterion prove, respectively,

$$\begin{aligned} F(l^h) &= F(l)T^{\beta(h)}, & F(l^n) &= nF(l), \\ F(lh) &= F(l) + F(h) \quad ([l, h] = 1), & F(l)T^{\beta(l)} &= F(l). \end{aligned}$$

For the third assertion, the defining function of a triple is homogeneous over $\mathbb{Q}$, hence additive on any line. It is not asserted to be additive on arbitrary pairs.

Form the right-action semidirect product with multiplication

$$H = L \ltimes M, \quad (l, v)(h, w) = (lh, vT^{\beta(h)} + w),$$

write $i(v) = (1, v)$, and put $f(l, v) = i(F(l))$. Then

$$f(g^h) = f(g)^h, \quad [g, f(g)] = 1, \quad f(gh) = f(g)f(h) \text{ if } [g, h] = 1,$$

and, for all integers n, m ,

$$f(g^n f(g)^m) = f(g)^n.$$

The last identity follows by projecting to L . If g, h commute, equivariance also gives $[f(g), h] = [g, f(h)] = 1$.

For $P(t) \in \mathbb{Z}[t]$, define

$$g^P = g^{P(0)} f(g)^{P'(0)}.$$

This supplies all the A -group axioms. Unit, zero and addition follow from $[g, f(g)] = 1$. If $P(0) = a, P'(0) = b, Q(0) = c, Q'(0) = d$, then

$$(g^P)^Q = g^{ac} f(g)^{bc+ad} = g^{PQ},$$

using $f(g^P) = f(g)^a$. Equivariance proves the conjugation axiom. For commuting g, h , the cross-commutations just established allow rearrangement to give $(gh)^P = g^P h^P$. Thus H is an A -group in the required unrestricted category. It is torsion-free as an ordinary group, being an extension of torsion-free groups, although its scalar action factors through $A/(t^2)$.

Set $x = (\exp X, 0)$, $y = (\exp Y, 0)$. They generate a group of class at most three, so the map from the free nilpotent group N extends by the tensor-completion universal property to an A -homomorphism $N^A \rightarrow H$. But $x^t = i(T^0)$, and direct multiplication gives

$$[i(v), y] = i(v(T - 1)).$$

For every $r \geq 1$, the image of $[x^t, y, \dots, y]$, with r copies of y , is $i((T - 1)^r) \neq 1$: its coefficient at T^r is one. The fixed pair x^t, y therefore generates a nonnilpotent subgroup of N^A , proving the claim. The map need not be onto H , and injectivity of $N \rightarrow H$ is unnecessary. Completion within nilpotent A -groups would be a different construction. See `research/13.42-proof.md`.

C.33 14.26: restricted wreath products in a nilpotent quasivariety

Let $\mathcal{C}$ be all torsion-free nilpotent groups, with no fixed class bound. If $G \in \mathcal{C}$, then $G \wr \mathbb{Z} \in \mathcal{C}$ for the restricted wreath product. This is the direct wreath product of the question: Budkin's definition uses finite-support functions [19, Section 1].

A finite set $B \ni 1$ locally embeds in a group if an injective identity-preserving map preserves every product staying in B . Since $\mathcal{C}$ is closed under finite direct products,

$$G \in Q(\mathcal{C}) \iff \text{every finite } B \subseteq G, 1 \in B, \text{ locally embeds in some } C \in \mathcal{C}.$$

Indeed, write down the finite multiplication table of B . For each unequal pair, some realization in $\mathcal{C}$ must separate it, or the corresponding quasiidentity would be violated by G . The product of these finitely many realizations separates all pairs. Conversely, a quasiidentity violation involves only finitely many intermediate word values, which a local embedding would preserve.

We construct the needed local models. Fix $c, R \geq 1$ and a finite nonempty set $I \subseteq \mathbb{Z}$ of size M , with diameter at most R . Choose $m \geq 2R(c-1) + M-1$, a rational vector space with basis $a_0, \dots, a_m$, and a nilpotent derivation determined by $D(a_j) = a_{j+1}$, $D(a_m) = 0$. Put

$$f(z) = \sum_{j=0}^m z^j a_j / j!.$$

In its symmetric algebra let J be the ideal generated by all coefficients in z of $f(z)f(z+d)$, $1 \leq d \leq R$, and form

$$\mathcal{A} = \text{Sym}\langle a_0, \dots, a_m \rangle / (J + \text{Sym}^{>c}).$$

This is finite-dimensional, commutative, and graded by ordinary degree. The ideal is D -stable because $D(f(z)f(z+d)) = \frac{d}{dz}(f(z)f(z+d))$. On the degree-truncated algebra, D raises the total index of each monomial, bounded by cm , so $D^{cm+1} = 0$, also after quotienting. Thus $\alpha = \exp D$ is an automorphism, with

$$\alpha^s f(i) = f(i+s), \quad f(i)f(j) = 0 \quad (0 < |i-j| \leq R). \quad (\text{a})$$

For each $1 \leq k \leq c$, the vectors $f(i)^k$, $i \in I$, are linearly independent in $\mathcal{A}$. To prove it, a linear functional on Sym^k corresponds to a symmetric polynomial $Q(z_1, \dots, z_k) = \ell(f(z_1) \cdots f(z_k))$ of degree at most m in each variable. Conversely the coefficients of any such polynomial prescribe ℓ , with factorial denominators. If Q vanishes when $z_2 = z_1 + d$, $1 \leq d \leq R$, the functional kills the degree- k part of J . Coefficient extraction and Vandermonde spanning by the other $f(z_j)$'s justify this for every quadratic generator multiplied by an arbitrary degree- $k-2$ term. Define

$$\Delta_k = \prod_{a < b} \prod_{d=1}^R ((z_a - z_b)^2 - d^2), \quad C_k = \prod_{a < b} \prod_{d=1}^R (-d^2).$$

If L_i is the degree- $(M-1)$ Lagrange polynomial on I , then

$$Q_i = \frac{\Delta_k}{C_k} L_i \left(\frac{z_1 + \cdots + z_k}{k} \right)$$

has the required degree bound in each variable, kills those relations, and has $Q_i(j, \dots, j) = \delta_{ij}$. Empty products handle $k = 1$; the cutoff adds no relation in these degrees. This proves independence.

Let $\mathcal{A}_+$ be the positive-degree ideal, U the upper unitriangular group of matrix size $(c+1) \times (c+1)$ with strict entries in $\mathcal{A}_+$, and

$$H = U \rtimes \langle t \rangle, \quad tut^{-1} = \alpha(u).$$

For $g \in \text{UT}_{c+1}(\mathbb{Q})$, define $\rho_i(g)$ by the off-diagonal entries

$$\rho_i(g)_{ab} = g_{ab} f(i)^{b-a} \quad (a < b).$$

Superdiagonal distances add, so this is a homomorphism, and the independence above makes it injective for $i \in I$. Equation (a) gives $\alpha\rho_i = \rho_{i+1}$. For distinct positions in I , the corresponding strictly upper matrices multiply to zero in either order. Thus

$$\prod_{i \in I} \rho_i(g_i) = 1 + \sum_{i \in I} (\rho_i(g_i) - 1)$$

and the independence of each $f(i)^k$ makes this direct-product map injective.

The group H is torsion-free nilpotent. On each successive superdiagonal quotient of U , conjugation by U is trivial and t acts as α . Writing $\delta = \alpha - 1$, we have $\delta^{cm+1} = 0$. Insert the images of successive powers of δ in every superdiagonal quotient. Their inverse images give a central series, of length at most $c(cm + 1) + 1$ including H/U . Torsion projects trivially to $H/U = \mathbb{Z}$; in U , a positive power multiplies the first nonzero superdiagonal by a positive integer, so cannot kill it over $\mathbb{Q}$.

Now take a finite subset $S \ni 1$ of $N \wr \mathbb{Z}$, with $N \in \mathcal{C}$. The finitely many base coordinates generate a finitely generated torsion-free nilpotent group, which embeds in $\text{UT}_{c+1}(\mathbb{Q})$ for some c . This classical embedding input follows from Jennings' augmentation-separation theorem [47, Theorem 5.2]: a finite-dimensional quotient $\mathbb{Q}[N]/\Delta^s$ is faithful, and the Δ -filtration makes the action unitriangular. Choose I to contain all supports in S and all shifted supports needed in products $uv = w$ within S ; take $I = \{0\}$ if empty. With R bounding its diameter, the map

$$(f, n) \mapsto \left(\prod_i \rho_i(f(i)) \right) t^n$$

preserves those products: conjugation by t^n shifts the indices, different positions commute, and equal positions multiply correctly. It is injective by the quotient H/U and the direct-product independence above. Hence every such S locally embeds in $\mathcal{C}$.

Finally, if $G \in Q(\mathcal{C})$, collect from a finite $S \subseteq G \wr \mathbb{Z}$ all base coordinates and all coordinate products needed by its finite multiplication table. Locally embed this finite set in a group $N \in \mathcal{C}$. Coordinate replacement, leaving the integer exponents unchanged, locally embeds S into $N \wr \mathbb{Z}$, where the preceding construction applies. The local criterion proves $G \wr \mathbb{Z} \in Q(\mathcal{C})$. The model and its nilpotency class depend on S ; no global embedding of the wreath product in a nilpotent group is asserted. See `research/14.26-proof.md`.

C.34 15.76(b): automorphisms of categories of free soluble groups

Let $\mathcal{S}_d$ be the full variety of groups of derived length at most $d \geq 1$. Every automorphism of its category of finite-rank free groups is naturally isomorphic to the identity. This includes the full metabelian variety. It does not assert the same for every proper subvariety of soluble groups: the proper metabelian variety studied by Fernandes–Tsurkov provides a counterexample [7]. The reduction of category automorphisms to verbal operations is an established method, used in the nilpotent predecessor [93]. The additional step here is the affine coefficient induction for the full soluble varieties.

We first recall the affine coefficient model. For an absolutely free group $F = F(x_1, \dots, x_n)$, a normal subgroup N , and $Q = F/N$, map x_i to $(\bar{x}_i, e_i)$ in

$$Q \ltimes \mathbb{Z}[Q]^n, \quad (q, U)(r, V) = (qr, U + qV).$$

Its kernel is $[N, N]$. Indeed the covering graph of the rose corresponding to N has vertices Q and edges $q \rightarrow q\bar{x}_i$. The two affine coordinates of a word are its endpoint and signed integral edge chain. A loop has zero chain exactly when its class in the first homology of this graph

is zero, which is its image in $N/[N, N]$. Finite support makes the argument valid for infinite covering graphs.

For a word $w(x, y)$, write its affine value as

$$w((a, U), (b, V)) = (w(a, b), f(a, b)U + g(a, b)V).$$

The boundary of its edge chain gives the ordered group-ring identity

$$f(a, b)(a - 1) + g(a, b)(b - 1) = w(a, b) - 1. \quad (\text{b})$$

We prove by induction on d that a normalized associative word in $\mathcal{S}_d$, meaning

$$w(x, 1) = x, \quad w(1, y) = y, \quad w(w(x, y), z) = w(x, w(y, z)),$$

must equal xy or yx as an identity. For $d = 1$, exponent sums suffice. For $d \geq 2$, induction and, if necessary, reversal of the two arguments allow us to assume $w(a, b) = ab$ in $\mathcal{S}_{d-1}$. Let Q be free in that variety on a, b, c . The affine group $Q \ltimes \mathbb{Z}[Q]^3$ belongs to $\mathcal{S}_d$. Associativity, evaluated at its three module basis vectors, gives

$$f(ab, c)f(a, b) = f(a, bc), \quad g(ab, c) = g(a, bc)g(b, c).$$

Set $h(t) = f(1, t) \in \mathbb{Z}[t, t^{-1}]$. The first equation gives $f(b, c)h(b) = h(bc)$; putting $c = b^{-1}$ shows that h is a Laurent unit with $h(1) = 1$. Hence $h = t^k$ for an integer k , and

$$f(a, b) = (ab)^k a^{-k}.$$

Only the units of the ordinary cyclic Laurent ring are used: highest and lowest powers in a product show that a unit is $\pm t^k$. Likewise, using $g(t) = g(t, 1)$, the second equation gives

$$g(a, b) = (ab)^l b^{-l}$$

for an integer l . Map (b) to the free abelian quotient, with variables X, Y . It becomes

$$Y^k(X - 1) + X^l(Y - 1) = XY - 1.$$

Differentiate in X and set $X = 1$, obtaining $Y^k = l + (1 - l)Y$. Coefficient comparison gives exactly $(k, l) = (0, 1)$ or $(1, 0)$.

In the first case $f = 1, g = a$, the affine coefficients of xy . Apply the kernel model to $F(x, y)$ with $N = F^{(d-1)}$: equality of affine values proves $w = xy$ modulo $F^{(d)}$. In the second case $f = aba^{-1}, g = 1$; equation (b) forces $aba^{-1} = b$ in Q . For $d \geq 3$, Q maps onto the nonabelian metabelian group S_3 , so this is impossible. For $d = 2$, Q is abelian and these are exactly the affine coefficients and scalar coordinate of yx . The same kernel model gives $w = yx$. Undoing any initial reversal proves the induction.

We finish the categorical reduction explicitly. Rank one is recognized by its infinite commutative endomorphism monoid: rank zero has a one-element monoid, and ranks at least two have noncommuting basis projection and interchange endomorphisms. Coproducts of rank-one objects then recognize every finite rank. Conjugate the given functor by chosen object isomorphisms to fix objects, and by basis automorphisms to fix the maps $\iota_i : F_1 \rightarrow F_n$ selecting the basis. Their images form a coproduct diagram and hence a free basis, justifying the second step. At F_1 this normalization is the identity.

For $a \in F_n$, let $f_a : F_1 \rightarrow F_n$ send the generator to a , and set $s_n(a) = \Psi(f_a)(x)$. These are bijections fixing the basis and identity. Functoriality gives, for every homomorphism $u : F_n \rightarrow F_m$,

$$\Psi(u) = s_m u s_n^{-1}.$$

With $w = s_2(xy)$, it also gives $s_n(ab) = w(s_n(a), s_n(b))$. Thus w transports a group law and is normalized associative on all free objects, hence throughout $\mathcal{S}_d$. The word result applies. If $w = xy$, every s_n is a basis-fixing homomorphism and is the identity. If $w = yx$, let η_n invert each basis generator as an ordinary automorphism, and let j_n be the inversion set map. Then $s_n = j_n \eta_n$. Every group homomorphism commutes with the inversion maps, so $\Psi(u) = \eta_m u \eta_n^{-1}$, an inner functor. Undoing the earlier object and basis conjugations supplies a natural isomorphism for the original functor on every object and morphism. The argument works equally with all named free objects or a skeleton; rank zero contributes its unique maps. The affine kernel model is the classical Magnus method, proved here from graph homology. No general group-ring unit theorem or classification of automorphisms of free soluble groups is imported. See `research/15.76-proof.md`.

C.35 16.38: periodic subgroups in a set product

Proposition C.11. *If G is soluble and $A, B \leq G$ are periodic, every subgroup contained in the set AB is periodic.*

Proof. First, periodic soluble groups are locally finite. A finitely generated one has finite quotient by its last nontrivial derived subgroup, by induction on derived length. That subgroup is finitely generated by Schreier's theorem, and abelian periodic, hence finite.

We need a cocycle observation. Let a locally finite group C act on a torsion-free abelian group D , and let $\delta(cd) = \delta(c) + c\delta(d)$. The image of δ cannot contain both t and $2t$ for $t \neq 0$. Indeed, choose preimages c_1, c_2 , and put $H = \langle c_1, c_2 \rangle$, which is finite. In $\mathbb{Q} \otimes D$, averaging gives

$$v = |H|^{-1} \sum_{h \in H} \delta(h), \quad \delta(h) = v - hv.$$

Average an inner product on the finite-dimensional real span of Hv . The vectors $v, v-t, v-2t$ then have equal norm. Expanding the last two equalities yields $2\langle v, t \rangle = \|t\|^2$ and $4\langle v, t \rangle = 4\|t\|^2$, so $t = 0$. This averaging is over the finite group H , not over all of C .

Suppose now that $D \leq E$ is torsion-free abelian and $A, B \leq E$ are locally finite. For $\pi : E \rightarrow E/D$, let $C = \pi(A) \cap \pi(B)$. The preimages of C in A, B map isomorphically to C , since their intersections with D are trivial. Write their inverse sections as a, b . In $\pi^{-1}(C) = D \rtimes a(C)$, put $b(c) = \delta(c)a(c)$; then δ is a cocycle and C is locally finite. A factorization lying in D must have inverse quotient coordinates, and hence

$$D \cap AB = \{a(c)b(c)^{-1} : c \in C\} = -\delta(C).$$

The observation shows that this set cannot contain both g and g^2 for any $1 \neq g \in D$.

It suffices to prove that $\langle g \rangle \subseteq AB$ implies that g has finite order. Induct on the derived length of G . The abelian case is immediate. Otherwise let D be its last nontrivial derived subgroup. Induction in G/D gives $g^r \in D$ for some $r > 0$. The torsion subgroup T of D is normal in G , and D/T is torsion-free abelian. Apply the preceding observation in G/T to $g^r T$ and $g^{2r} T$, both of which belong to the appropriate set product. It gives $g^r \in T$, proving the claim. $\square$

The set AB itself can contain elements of infinite order. For example, let $D = \bigoplus_{i \geq 1} \mathbb{Z}$, let $C = \bigoplus_{i \geq 1} C_2$ act by coordinate sign changes, and set $\delta(c)_i = c_i$ for $c_i \in \{0, 1\}$. This is a cocycle. In $D \rtimes C$, the two complements

$$A = \{(0, c)\}, \quad B = \{(\delta(c), c)\}$$

are elementary abelian, whereas $D \cap AB = -\delta(C)$ contains nonzero infinite-order elements and excludes their doubles. The theorem concerns subgroups contained in AB . It also does not assert the locally soluble version of Problem 9.65. See `research/16.38-proof.md`.

C.36 19.93: unbounded lower p -central width without a wreath quotient

Write $P_1(Q) = Q$, $P_{i+1}(Q) = P_i(Q)^p[P_i(Q), Q]$. For every fixed prime $p \geq 7$ and integer K , we construct a finite two-generator p -group with no quotient $C_p \wr C_p$ and some factor P_i/P_{i+1} of order greater than p^K .

First write $C_p \wr C_p = \mathbb{F}_p^p \rtimes \langle t \rangle$ and let $s(v)$ be the sum of the coordinates of v . The map $(v, t^a) \mapsto (s(v), a)$ is onto $\mathbb{F}_p^2$. The element (v, t^a) has p -th power one exactly when $s(v)a = 0$: for $a \neq 0$ its p -th power has constant vector $s(v)$, and for $a = 0$ it is always one. If u, v generate the wreath product and $u^p = v^p = 1$, their images must lie on different coordinate axes, so $(uv)^p \neq 1$. Consequently the three relations

$$x^p = y^p = (xy)^p = 1 \quad (\dagger)$$

exclude this quotient for a group generated by x, y .

Consider the complete filtered algebra

$$A = \mathbb{F}_p \langle\langle X, Y \rangle\rangle / \overline{(X^p, Y^p, (X + Y + XY)^p)}.$$

Let A_n be its degree filtration, $a_n = \dim A_n/A_{n+1}$, and $H(t) = \sum a_n t^n$. All three relations have initial degree p ; in the last relation the degree- p part $(X + Y)^p$ is the sum of all distinct length- p words and is nonzero. The complete filtered Golod–Shafarevich inequality [34, Theorem 2.6] states coefficientwise that

$$(1 - 2t + 3t^p) \frac{H(t)}{1 - t} \geq \frac{1}{1 - t}.$$

At $t = 2/3$ the polynomial factor is at most $-115/729 < 0$, so $H(2/3)$ diverges. The denominator $1 - t$ is retained: these relations are not homogeneous.

For $N \geq 2$, let $R_N = A/A_N$ and $M_N = A_1/A_N$. The finite group $1 + M_N$ is a p -group, because M_N is nilpotent and $|1 + M_N| = |M_N|$. Set

$$Q_N = \langle 1 + X, 1 + Y \rangle \leq 1 + M_N.$$

The defining relations give $(\dagger)$, excluding the wreath quotient. Reduction to $A_1/A_2 \cong \mathbb{F}_p^2$ shows that Q_N requires exactly two generators. The natural map $\mathbb{F}_p[Q_N] \rightarrow R_N$ is onto: its image contains X, Y , and every truncated series is a polynomial. For its augmentation ideal I_N , the map kills I_N^N ; hence

$$\dim R_N \leq \dim \mathbb{F}_p[Q_N]/I_N^N. \quad (\ddagger)$$

Here is an elementary bound on the right side if all lower p -central widths were bounded. For any finite p -group Q , put $b_i = \dim_{\mathbb{F}_p} P_i/P_{i+1}$, and choose basis lifts g_{ij} , ordered by increasing i . The series terminates, and successive quotient normal forms give unique products $\prod g_{ij}^{e_{ij}}$, $0 \leq e_{ij} < p$. Expanding in this fixed order shows that $\prod (g_{ij} - 1)^{e_{ij}}$ span $\mathbb{F}_p[Q]$. For the augmentation ideal I ,

$$g - 1 \in I^i \quad (g \in P_i).$$

This follows inductively from $h^p - 1 = (h - 1)^p$ and

$$[h, g] - 1 = h^{-1}g^{-1}((h - 1)(g - 1) - (g - 1)(h - 1)).$$

Thus only products with $\sum i e_{ij} < N$ are needed modulo I^N . If every $b_i \leq K$, and c_n are the coefficients of

$$P_K(t) = \prod_{i \geq 1} (1 + t^i + \cdots + t^{(p-1)i})^K,$$

then $\dim \mathbb{F}_p[Q]/I^N \leq \sum_{n < N} c_n$. This product converges for $0 < t < 1$, being bounded by $\prod_{i \geq 1} (1 - t^i)^{-K}$; the logarithms are bounded by $K \sum_i t^i / (1 - t)$.

If every Q_N had every $b_i \leq K$, (§) would give $\sum_{n < N} a_n \leq \sum_{n < N} c_n$ for all N . In particular $a_n \leq \sum_{j \leq n} c_j$, and therefore

$$H(t) \leq \frac{P_K(t)}{1 - t} < \infty \quad (0 < t < 1),$$

contradicting divergence at $2/3$. This proves the claimed unboundedness, already for $(p, d) = (7, 2)$. The finite groups are obtainable from algebra truncations, but the proof gives no practical size bound. It excludes wreath quotients of Q_N itself, exactly as asked, not quotients of every subgroup. See `research/19.93-proof.md`.

C.37 20.1: a totally 2-closed insoluble group with nontrivial Fitting subgroup

The example is $J_1 \times C_{13}$. The imported input is that J_1 is totally 2-closed [4, Theorem 1.1 and Proposition 4.8]. We give the product argument for arbitrary faithful actions, including intransitive ones.

For $X \leq \text{Sym}(\Omega)$, its 2-closure consists exactly of permutations s such that each ordered pair (α, β) has an $x \in X$ agreeing with s at both points. In particular s preserves every point orbit. For a finite group, total closure for finite faithful sets implies the same for arbitrary sets: choose a finite invariant faithful subset Ω_0 , using one moved point for each nonidentity group element. The restriction of s agrees with a unique group element there. Applying finite closure to $\Omega_0 \cup \alpha^X$ for every α forces the same element everywhere.

Lemma C.12. *If finite groups A, B are totally 2-closed and have coprime orders, then $A \times B$ is totally 2-closed.*

Proof. In any faithful action, $\alpha^A \cap \alpha^B = \{\alpha\}$. Indeed, $\alpha^a = \alpha^b$ makes ab^{-1} fix α . Since a, b commute and have coprime orders, the Chinese remainder theorem expresses a as a power of ab^{-1} , so a fixes α .

Let Δ_B be the set of B -orbits. The induced A -action is faithful: an element fixing every such orbit moves each α within $\alpha^A \cap \alpha^B$, so fixes every point. Similarly B acts faithfully on Δ_A . If s is in the 2-closure of $A \times B$, it preserves both orbit equivalence relations, since each is a union of ordered-pair orbits. For two B -orbits, choose representatives and apply the pair criterion; the witnessing element ab induces the A -element a on the pair of blocks. Thus s induces an element of the 2-closure of A on Δ_B , and total closure gives one $a_0 \in A$ for all blocks. Similarly one $b_0 \in B$ gives its action on Δ_A . The permutation $s(a_0 b_0)^{-1}$ fixes every orbit of both factors setwise, hence every singleton intersection, and is the identity. $\square$

A prime cyclic group is totally 2-closed: in a faithful action choose a point with trivial stabilizer; pairing it with every other point forces one common group element to realize a 2-closure permutation. Now

$$|J_1| = 175560 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$$

is coprime to 13, so the lemma proves the claim about closure. The group is insoluble because it has the simple quotient J_1 . Every normal nilpotent subgroup projects trivially to J_1 , whereas the central C_{13} is nilpotent. Thus

$$F(J_1 \times C_{13}) = C_{13} \neq 1.$$

The supplied question imposes no direct-indecomposability restriction. Prior related results include the nilpotent product case of Chen–Ren and coprime transitive-action decompositions [23, 1]; the lemma above handles all faithful actions of the stated factors. The J_1 theorem retains its classification and computational dependencies. See `research/20.1-proof.md`.

C.38 19.62: the derived carpet is closed

Let Φ be a reduced crystallographic root system and K a commutative ring with identity. A carpet $A = (A_r)_{r \in \Phi}$ is a family of additive subgroups of K satisfying all Chevalley conditions

$$c_{ij,rs}a^ib^j \in A_{ir+js} \quad (a \in A_r, b \in A_s, i, j > 0, ir + js \in \Phi).$$

Here r, s are distinct and nonopposite, and $c_{ij,rs}$ is the absolute value of the corresponding integral commutator constant. Signs do not affect additive subgroup membership. Write $\partial A = B$ for the family additively generated by these parameters at each destination root. Then $B \subseteq A$ and B is itself a carpet: a rule applied to elements of B is also a rule applied to elements of A , so its value belongs to the appropriate subgroup of B .

The candidate proves the stronger inclusion

$$A_p^2 B_{-p} \subseteq B_p \quad (p \in \Phi), \tag{5}$$

where $A_p^2 = \{a^2 : a \in A_p\}$. Consequently $B_p^2 B_{-p} \subseteq B_p$. Nuzhin’s already published square-closedness criterion then says that B is closed:

$$\langle x_r(B_r) : r \in \Phi \rangle \cap x_p(K) = x_p(B_p).$$

The criterion, and hence the answer to Problem 19.63, predates the experiment [70, Theorem 2]. The ledger’s grouping of 19.62 with 19.63 must not be read as a new solution of the latter.

Substantial cases of Problem 19.62 were also known. Nuzhin’s 2023 discussion records the answer for type A_ℓ ; the earlier triple-product inclusion quoted there, together with its Theorem 1, also gives the simply laced types D_ℓ, E_ℓ . Its Corollary 2 treats B_ℓ, C_ℓ, F_4 when $\gcd(\text{char } K, 2) = 1$, and G_2 when $\gcd(\text{char } K, 6) = 1$ [69, Section 3]. The integer derivations below apply without these characteristic restrictions; the already covered cases are not new answers.

Reduction to finite integer derivations. Fix $x \in A_p$. It suffices to prove

$$cx^2u^iv^j \in B_p \quad \text{whenever} \quad ir + js = -p, \quad u \in A_r, \quad v \in A_s, \tag{*}$$

since multiplication by x^2 distributes over every additive expression for an element of B_{-p} . The intersection of Φ with the plane spanned by r, s is an irreducible rank-two root system, of type A_2, B_2 , or G_2 . Every root string between its roots stays in that plane. Thus the restricted carpet has precisely the required absolute Chevalley constants, and any derived parameter in the restriction belongs to the ambient B . No commutation of restriction with the entire operation ∂ is assumed.

The following finite proof calculus establishes (*). Initial nodes are independent variables with specified membership in A_r . A Chevalley rule applied to two earlier nodes gives $cf^ig^j \in B_{ir+js} \subseteq A_{ir+js}$. If two applications give integer multiples m_1w, m_2w of the same monomial at the same root, their integer linear combinations give $\gcd(m_1, m_2)w \in B$. Each final coefficient must divide the requested coefficient. These operations use no division in K and remain valid in characteristics two and three.

There are 6, 20, and 78 distinct requests in types A_2, B_2, G_2 , respectively, after coincident requests are combined by taking the gcd of their coefficients. Short certificates contain 655 nodes in total, at most nine per request, including initial nodes. Relabeling the carpet by Weyl transformations reduces the display to 19 requests, one destination root of each length. This is transport of a universal implication; it does not assert that the particular carpet is Weyl invariant. The ancillary checker verifies the absolute-rule invariance, the coverage by those representatives, and all 104 requests directly. The 19 representative derivations are printed in Appendix F.

For illustration, take G_2 with short simple root a , long simple root b , and $p = 2a + b$. For $x \in A_p, u \in A_a, v \in A_{-3a-b}$, the rules give

$$3xu \in B_{3a+b}, \quad xv \in B_{-a}, \quad x^2v \in B_{a+b}.$$

Applying the rules again gives $3x^2uv \in B_p$ from the first two expressions and $2x^2uv \in B_p$ from the third and u . Their difference gives $x^2uv \in B_p$. This illustrates why retaining both small integer coefficients can prove an assertion that an argument dividing by a structure constant would miss.

A direct proof of the known closedness criterion. We retain the following alternative argument as part of the experiment's mathematical record. Assume $A_r^2 A_{-r} \subseteq A_r$ for every root and work in the integral Chevalley Lie algebra tensored with K . Put

$$M = \sum_r A_r e_r + \sum_r \langle A_r A_{-r} \rangle_{\mathbb{Z}} h_r.$$

Root coordinates in this additive group are independent; in particular $M \cap K e_r = A_r e_r$. The group M is a Lie subring. Root-root brackets use the $(1, 1)$ carpet condition or, for opposite roots, the displayed Cartan summands. For a Cartan generator uvh_r and a root $s \neq \pm r$, Jacobi expresses their bracket as two successive root brackets, each allowed by the carpet rules. For $s = r$ or $-r$, polarization supplies the required coefficient:

$$2uvw = (u + w)^2 v - u^2 v - w^2 v \in A_r \quad (u, w \in A_r, v \in A_{-r}).$$

The Cartan part commutes with itself. This also shows that M is exactly the Lie subring generated by the root summands.

Every $x_s(t)$, $t \in A_s$, preserves M . On ue_r with $r \neq \pm s$, its root-string terms have coefficients $C_{i1, sr} t^i u$, covered by the carpet rules. It fixes e_s , while

$$ue_{-s} \mapsto ue_{-s} + tuh_s - t^2 ue_s$$

uses the square assumption. The inverse generator gives equality. If a root p admits a root q with $\langle p, q^\vee \rangle = \pm 1$, all generating root elements send h_q into $h_q + M$. Thus $x_p(t)$ in the carpet group forces $\pm t e_p \in M$, and hence $t \in A_p$. Such a q exists in simply laced systems of rank at least two by adjacency; in B_n , $n \geq 3$, use $e_1 + e_2$ for a short root e_1 , and $e_2 + e_3$ for a long root $e_1 + e_2$. The same choices cover F_4 , and the same-length hexagons cover G_2 .

For C_n , including $B_2 = C_2$ and rank one, use natural symplectic matrices indexed by $1, \dots, n, -1, \dots, -n$. The root vectors are

$$E_{ij} - E_{-j, -i}, \quad E_{i, -j} + E_{j, -i}, \quad E_{i, -i},$$

and their negative-root counterparts. Together with $H_i = E_{ii} - E_{-i, -i}$ they form a free Chevalley basis over K . Every root vector N satisfies $N^2 = 0$. Choose a diagonal matrix D with entries $d_i \in \{0, 1\}$ at i and $1 - d_i$ at $-i$. Then

$$[D, e_r] = \lambda_r e_r, \quad \lambda_{e_i - e_j} = d_i - d_j, \quad \lambda_{e_i + e_j} = d_i + d_j - 1, \quad \lambda_{2e_i} = 2d_i - 1.$$

For any prescribed p , the d_i can make $\lambda_p = \pm 1$. Since $NDN = 0$,

$$(I + tN)D(I - tN) = D - t\lambda_r N.$$

Each initial root generator sends D into $D + M$, and invariance of M extends this to every word. The formula detects $t \in A_p$ for a new root element $I + te_p$. This also works in every standard Chevalley isogeny form: an equality of root words gives equality of actions on the abstract Chevalley Lie algebra. The quotient of their natural symplectic lifts centralizes that algebra. Commuting with $E_{i,-i}$ and $E_{-i,i}$ forces a matrix to be scalar on each two-coordinate block and zero between blocks; the short-root matrices force the scalars to agree. Such a quotient is therefore scalar over any commutative ring and commutes with D . The same detection argument applies. This proves the criterion without dividing by two.

The retained proof and audits are `research/19.62-rank-two-proof.md`, `research/19.62-closedness-criterion.md`, and `results/19.62-rank-two-packet.json`. The portable ancillary files include the proof calculus, target coverage checks, and integer certificates.

C.39 19.61: root closure preserves the derived carpet

Use the notation of Section C.38. For a carpet A , put

$$H = \langle x_r(A_r) : r \in \Phi \rangle, \quad D_r = \{t \in K : x_r(t) \in H\}.$$

The candidate asserts that D is a carpet and $\partial D = \partial A$, over every commutative ring. Troyanskaya previously proved carpet closedness of this root closure in odd characteristic, with characteristic divisible by three also excluded in type G_2 [91]. The argument here uses Nuzhin's known square criterion and a universally valid finite adjoining lemma. Priority of the stronger statement is not established by those source comparisons.

Lemma C.13. *Let $B = \partial A$, fix $a \in A_r, b \in A_{-r}$, and replace A_r by $A_r + \mathbb{Z}a^2b$, leaving all other entries unchanged. The resulting family A' is a carpet and $\partial A' = B$.*

Proof. Consider an ordered Chevalley rule whose first input changes from x to $x + na^2b$. Its parameter expands as

$$c(x + na^2b)^i u^j = \sum_{k=0}^i c \binom{i}{k} n^k x^{i-k} a^{2k} b^k u^j.$$

The $k = 0$ term lies in B_{ir+js} by definition. For every other term, the required universal implication is

$$c \binom{i}{k} x^{i-k} a^{2k} b^k u^j \in B_{ir+js}, \quad x, a \in A_r, \quad b \in A_{-r}, \quad u \in A_s. \quad (6)$$

The variables x, a are independent even though their root labels coincide. Multiplication by any integer n^k is allowed in the additive subgroup B_{ir+js} . The analogous substitution into the second input is checked separately, retaining the constant of the original ordered rule; reversing an ordered source pair in G_2 can change its absolute constant. A nontrivial rule has distinct nonopposite source roots, so only one input subgroup changes.

The same root-plane argument as in Section C.38 reduces (6) to types A_2, B_2, G_2 . The ancillary certificates apply exactly the integer proof calculus explained there. Their complete

sizes are

type	targets	nodes	largest derivation
A_2	12	72	6
B_2	48	380	12
G_2	240	2739	19
total	300	3191	19

Every noninitial node lies in B by a carpet application and, where needed, a Bezout combination of two such applications. The last coefficient divides the coefficient in (6). Thus each certificate is a finite integer algebraic proof valid under every substitution in K . They prove all 300 implications. For example, in type A_2 with $p = r + s$, the only changed first-input term is a^2bu . The rules give $au \in B_p$, then $abu \in B_s$, and then $a^2bu \in B_p$. The longer certificates perform the same operation where coefficients two and three intervene.

The coverage checker reconstructs the requests by repeated sparse polynomial multiplication, independently of the binomial formula used by the target generator. It covers both inputs of all ordered rules: 416 substitutions and 600 nonconstant terms, giving the 300 requests after coincident requests are combined by gcd. The checked rule tables and the complete derivations accompany this paper; their search procedure is not a mathematical premise.

Expansion now gives $\partial A' \subseteq B$; monotonicity gives the reverse inclusion. Since $B \subseteq A \subseteq A'$, every defining parameter of A' lies in its designated entry. This is precisely the carpet condition. $\square$

Starting with $C^{(0)} = A$, define

$$C_r^{(n+1)} = C_r^{(n)} + \langle u^2v : u \in C_r^{(n)}, v \in C_{-r}^{(n)} \rangle_{\mathbb{Z}}.$$

Each $C^{(n)}$ is a carpet with derived family B . Indeed, adjoin any finite collection of the displayed generators one at a time. Its parameters remain available after earlier adjoinings, so the lemma applies throughout. These finite extensions form a directed family whose union is $C^{(n+1)}$. A carpet condition involves finitely many elements, and a derived parameter uses only two input elements. Hence directed union preserves both carpet membership and equality of the derived family.

Put $C_r = \bigcup_n C_r^{(n)}$. Again C is a carpet and $\partial C = B$. Any $u \in C_r, v \in C_{-r}$ lie in a common stage, so u^2v lies in the next stage. Thus $C_r^2 C_{-r} \subseteq C_r$, and the known square criterion makes C closed. Since $A \subseteq C$, one has $H \subseteq E(\Phi, C)$; closedness yields

$$A \subseteq D \subseteq C.$$

The root-group addition law makes each D_r additive. Every commutator parameter formed from D belongs to $\partial C = B \subseteq A \subseteq D$, proving that D is a carpet. Finally monotonicity gives

$$B = \partial A \subseteq \partial D \subseteq \partial C = B.$$

The constructed square completion C need not equal D ; the proof only requires the displayed sandwich. It does not assert that the adjoined root elements already belong to H . In rank one there are no nonopposite commutator conditions and all derived families are zero, so the conclusion is immediate. See `research/19.61-proof.md` and `results/19.61-square-completion-packet.json`.

C.40 21.76: closed irreducible nets without a diagonal completion

For every field k and every $n \geq 3$, set

$$R = k[x, y], \quad K = k(x, y), \quad J = yR, \quad \sigma_{12} = \sigma_{21} = kx + J, \quad \sigma_{ij} = J \text{ otherwise } (i \neq j).$$

This gives an irreducible closed elementary net over K which cannot be completed by diagonal additive groups. In particular it supplies every odd characteristic requested.

Recall that an elementary net requires $\sigma_{ik}\sigma_{kj} \subseteq \sigma_{ij}$ for pairwise distinct indices; irreducibility here means every level is nonzero. All levels contain J , and at least one of $(i, k), (k, j)$ is outside the two exceptional positions, so every required product lies in $RJ = J$. Thus it is an irreducible elementary net.

We first check the rank-one pair. In $\mathrm{SL}_2(k(x))$, let

$$H = \left\langle U(a) = \begin{pmatrix} 1 & ax \\ 0 & 1 \end{pmatrix}, V(b) = \begin{pmatrix} 1 & 0 \\ bx & 1 \end{pmatrix} : a, b \in k \right\rangle.$$

Its intersection with either elementary root subgroup has parameters exactly kx . To see this, combine adjacent factors of the same kind and remove zero parameters. Any remaining word is alternating with all parameters nonzero. If its length $m \geq 2$ is even and it starts with U , choosing all m off-diagonal terms gives the unique degree- m contribution in entry $(1, 1)$, nonzero because $E_{12}E_{21} \cdots E_{21} = E_{11}$. If m is odd, a degree- $(m-1)$ contribution omits one factor. Omitting the last gives E_{11} , omitting the first gives E_{22} , and omitting any interior factor gives zero by adjacent equal matrix units. Thus entry $(1, 1)$ has nonzero leading term of degree $m-1$. For a word starting with V , interchange the indices. No such word is an elementary transvection, whose diagonal entries are both one. Length one and zero give exactly the stated intersections. This is a special case of Koibaev's prior closed-pair theorem [53]; its needed special case has just been proved.

Let $E(\sigma)$ be generated by $I + aE_{ij}$ with $a \in \sigma_{ij}$. All these matrices and their inverses lie in $\mathrm{SL}_n(R)$. If $I + aE_{ij} \in E(\sigma)$, initially with $a \in K$, its matrix entry therefore gives $a \in R$. Reduction at $y = 0$, now a legitimate ring map on R , sends $E(\sigma)$ to $\mathrm{diag}(H, I_{n-2})$. At an ordinary position this forces $a \bmod y = 0$; at either exceptional position the rank-one result gives $a \bmod y \in kx$. In each case $a \in \sigma_{ij}$, proving closedness. No evaluation homomorphism on the entire field K was used.

A diagonal completion would force $x^2 \in \sigma_{11}$ from the two exceptional levels, and then $x^3 \in \sigma_{11}\sigma_{12} \subseteq \sigma_{12}$. Reduction modulo y would give $x^3 \in kx$, impossible for transcendental x . Thus no completion exists.

The archive also gives a one-variable characteristic-three example. In $\mathbb{F}_9 = \mathbb{F}_3[j]$, $j^2 = -1$, the two elementary matrices with upper parameter 1 and lower parameter j generate a group of order 120, established by an explicit finite coset table. Its intersections with the order-nine root groups therefore have parameters $\mathbb{F}_3$ and $j\mathbb{F}_3$. Using exceptional levels $\mathbb{F}_3 + t\mathbb{F}_9[t]$, $j\mathbb{F}_3 + t\mathbb{F}_9[t]$ and all other levels $t\mathbb{F}_9[t]$ repeats the reduction argument over $\mathbb{F}_9(t)$; a completion would force j into the first exceptional level. The full table is `research/21.76-finite-table.md`. This supplementary finite certificate is not needed for the two-variable proof, which is independent of the carpet candidates in Problems 19.61–19.62. See `research/21.76-proof.md`.

C.41 11.115: equal commutator subgroups from a proper power

Let $X = F(a, b)$, fix $m \geq 2$, and put

$$N = \langle\langle a^m \rangle\rangle_X, \quad T = \ker(X \longrightarrow C_m), \quad a \mapsto 1, \quad b \mapsto 0.$$

Then $1 < N < T < X$: a^m is nontrivial, $a \notin T$, and $b \in T \setminus N$, as the exponent-sum map to $\mathbb{Z}$ in b shows. In particular N is not maximal in X . Nevertheless

$$[T, N] = [X, N].$$

To prove this, set $K = [T, N]$. Both T, N are normal in X , so $K \trianglelefteq X$. In X/K , the image of a^m commutes with $b \in T$, and it already commutes with a . It is therefore central. Its normal closure, the image of N , is central too, giving $[X, N] \leq K$. The reverse inclusion is immediate. This disproves the proposed strict containment.

More explicitly,

$$K = \langle\langle [a^m, b] \rangle\rangle_X, \quad X/K = \langle a, b \mid [a^m, b] = 1 \rangle.$$

This quotient is $F(s_0, \dots, s_{m-1}) \rtimes \mathbb{Z}$, where a generator of $\mathbb{Z}$ cyclically permutes the s_i 's: send a to that generator and b to s_0 . The inverse map uses $s_i = a^i b a^{-i}$; the final cyclic step is just $[a^m, b] = 1$. Thus $N/K \cong \mathbb{Z}$ is central and $X/N \cong C_m * \mathbb{Z}$. The common subgroup K is nontrivial because $[a^m, b]$ is a nonempty reduced word, and $K < N$ by exponent sum in a . For the optional stronger condition that T itself not be maximal, take $m = 4$ and insert the kernel of the map to C_2 .

The supplied question imposes no torsion-free quotient or primitive-element assumption. Those extra conditions would exclude this example. The short proof is self-contained; its intended scope and historical priority still warrant outside scrutiny. The full packet is `rese arch/11.115-proof.md`.

C.42 20.122(a)–(c): minimum intersections outside the Fitting subgroup

For nilpotent $A, B, C \leq G$, the question defines $\text{Min}_G(A, B, C)$ using all inclusion-minimal intersections $A \cap B^x \cap C^y$, and $\text{min}_G(A, B, C)$ using those of minimum order. Both can lie outside $F(G)$, even for soluble G . The distinction matters: the experiment's earlier smaller example addressed only inclusion-minimal intersections. Zenkov's earlier examples separate the analogous notions for pairs of subgroups [98]. The argument below computes the full triple-intersection family for a different group.

Let $\Omega = \mathbb{F}_3^2$, $o = 0$, and let $H \cong D_8$ be the eight signed coordinate permutations. It is generated by $d(x, y) = (-x, y)$ and $s(x, y) = (y, x)$. Put

$$Q = \mathbb{F}_3^2 \rtimes H, \quad E = C_2^\Omega = \langle z_w : w \in \Omega \rangle, \quad G = E \rtimes Q.$$

In its natural action on $\Omega \times \mathbb{F}_2$, a base element flips selected fibers and Q permutes their indices. This soluble group has order $512 \cdot 72 = 36864$. Take

$$Z = \langle z_o \rangle, \quad A = Z \times H, \quad B = C = G_{(o,0)}.$$

The orders of A, B, C are 16, 2048, 2048, so all are nilpotent. The definition permits $B = C$.

The two points in each fiber have the same stabilizer,

$$B_w = \langle z_t : t \neq w \rangle \rtimes Q_w.$$

Transitivity shows that these nine distinct groups are exactly the conjugates of B , accounting for base-group conjugation as well. For each nonzero w , H_w is an order-two reflection group R_L , where $L = \langle w \rangle$. The four lines give four distinct reflections: sign change of either coordinate, interchange, and negative interchange. Stabilizers of nonzero vectors on distinct

lines intersect trivially. It follows that all 81 ordered pairs of conjugates give precisely:

u, v	$A \cap B_u \cap B_v$	order	pairs
$u = v = o$	H	8	1
exactly one is o , other on L	R_L	2	16
both nonzero on L	$Z \times R_L$	4	16
nonzero on different lines	Z	2	48

Membership in B_o kills Z ; all other B_w 's retain it. The remaining condition is to fix the specified vectors in H , which proves every table entry. There are ten distinct intersections, none trivial. The five groups Z, R_L are exactly both the minimum-order and the inclusion-minimal members. The reflections generate H , so

$$\min_G(A, B, B) = \text{Min}_G(A, B, B) = A.$$

Finally $F(G) = E$. First $O_2(Q) = 1$: a normal 2-subgroup centralizes the normal translation subgroup $\mathbb{F}_3^2$, whose centralizer in Q is itself. Also $C_G(E) = E$, because an affine part centralizing all independent base flips fixes all nine indices, and the action is faithful. For any normal nilpotent subgroup of G , its characteristic odd Sylow subgroups centralize E , and hence are trivial. It is therefore a 2-group whose image in Q is trivial, so lies in E . Conversely E is normal abelian. The reflection $d \in A$ is outside E , proving failure of all assertions (a)–(c). In fact $A \cap B_o \cap B_{(0,1)} = \langle d \rangle$ is already a minimum-order intersection with an outside witness.

The argument is elementary and determines the entire conjugacy and intersection family; it is not an extrapolation from a sample. The related affine stabilizer pattern has prior antecedents recorded in `research/20.122-minimum-source-audit.md`. The proof and independent enumeration packet are `research/20.122-minimum-proof.md` and `results/20.122-minimum-packet.json`.

C.43 17.39: no uniform number of system normalizers

For every odd prime p and $d \geq 1$, there is a supersoluble metabelian group G of order $2p^{2d+1}$, with trivial hypercenter, for which the least number of system normalizers having trivial intersection is exactly $d+1$. Its Frattini subgroup has order p . Fixing $p = 3$ disproves a uniform bound even at fixed Frattini order.

Write $V = \mathbb{F}_p^d$, and define

$$P = V \times V \times \mathbb{F}_p, \quad (u, v, z)(a, b, c) = (u + a, v + b, z + c + u \cdot b).$$

These are upper unitriangular block matrices of block sizes $1, d, 1$. The automorphism $\alpha(u, v, z) = (u, -v, -z)$ has order two. Set $G = P \rtimes \langle t \rangle$, with $x^t = \alpha(x)$. The commutator formula is

$$[(u, v, z), (a, b, c)] = (0, 0, u \cdot b - a \cdot v),$$

so $Z(P) = P' = \{(0, 0, z)\} =: Z$. The element t inverts Z , and every element outside P does too. Thus $Z(G) = 1$, and its entire hypercenter is trivial. The subgroup $\{(0, v, z)\}$ is abelian normal with abelian quotient, making G metabelian. Coordinate flags in the two copies of V in G/Z , preceded by Z , give a normal series of prime-order cyclic factors, so G is supersoluble.

A Sylow system consists here of the normal Sylow p -subgroup P and a Sylow 2-subgroup. Its system normalizer is a conjugate of

$$D = N_G(\langle t \rangle) = C_G(t) = \{(u, 0, 0) : u \in V\} \times \langle t \rangle.$$

All such conjugates occur, by Sylow conjugacy, and $G = DP$ allows conjugators in P . Represent an element of G as (u, v, z, e) , meaning $(u, v, z)t^e$, $e = 0, 1$. For $g = (a, b, c)$ and $q = c - a \cdot b$, direct multiplication gives

$$D^g = D_{b,q} = \{(u, 0, u \cdot b, 0) : u \in V\} \\ \cup \{(u, -2b, -2q - u \cdot b, 1) : u \in V\}.$$

Every pair b, q occurs, and different pairs give different subgroups. Thus there are exactly p^{d+1} system normalizers.

For k arbitrary normalizers D_{b_i, q_i} , their intersection with P consists of

$$\{(u, 0, u \cdot b_1, 0) : u \cdot (b_i - b_1) = 0, \ 2 \leq i \leq k\}.$$

At most $k - 1$ independent equations are imposed on V ; for $k \leq d$ this intersection is nontrivial. Conversely take $D_{0,0}$ and $D_{e_i,0}$ for all d standard basis vectors. Their common part in P is trivial. Already the first two have no common element outside P , whose second coordinates would have to be both 0 and $-2e_1$. Their full intersection is therefore trivial, proving the exact minimum.

To locate the example relative to the known Frattini-free positive case [6], note that $\Phi(G) = Z$. Indeed $P^p = 1$, by

$$(u, v, z)^m = (mu, mv, mz + \binom{m}{2}u \cdot v),$$

so $\Phi(P) = P'P^p = Z$. If a maximal subgroup M of G missed this normal subgroup, then $G = M\Phi(P)$ and $P = (P \cap M)\Phi(P)$, contrary to the nongenerator property of $\Phi(P)$. Thus $Z \leq \Phi(G)$. On the other hand $G/Z \cong V \times (V \rtimes C_2)$, with inversion on the second copy. The coordinate-hyperplane subgroups in either copy, extended by the other factors, and the index-two subgroup $V \times V$, are maximal with trivial total intersection. Their preimages give $\Phi(G) \leq Z$. The fixed nontrivial Frattini subgroup therefore explains why the prior positive result does not apply. See `research/17.39-proof.md`.

D Partial results and bounded computations

None of the results in this appendix contributes an entry to the 46-candidate count. Their hypotheses and remaining cases are part of their statements. Several are substantial theorems even though they do not resolve a complete Notebook question.

D.1 20.100: distinct powers, reduction, and four finite proofs

For a positive integer n , assume G has no element of order $2, \dots, n + 1$. The question asks whether every n -element set can be ordered as $a_1, \dots, a_n$ with $a_1, a_2^2, \dots, a_n^n$ pairwise distinct. The identity is allowed. Sun's Conjecture 4.1 is this assertion; his Theorem 1.4 covers torsion-free abelian groups and $n \leq 3$ [87]. The experiment obtained a torsion-free theorem without commutativity, a reduction to finite abelian groups, an explicit finite counterexample bound for fixed n , and complete certificates for $n = 4, 5, 6, 7$.

Proposition D.1. *Any finite list of infinite-order elements, with repetitions allowed, can be assigned prescribed distinct positive integer exponents so that the resulting powers are distinct.*

Proof. On the cyclic subgroups generated by the list, nontrivial intersection is an equivalence relation: intersect the two finite-index subgroups of the middle infinite cyclic group to prove

transitivity. In each of the finitely many classes choose a nonidentity common power c . Write $c = a^{k_a}$, $k_a \neq 0$, for each entry in that class and give it length $1/|k_a|$. If $a^i = b^j$, then the entries lie in the same class, and raising to the integer $k_a k_b$ gives $c^{i k_b} = c^{j k_a}$. Infinite order implies $i/|k_a| = j/|k_b|$. Sort the entire list by nondecreasing positive length and assign the exponents in increasing order. The products of exponent and length strictly increase, so no equality is possible. One additional identity element can be assigned any unused exponent. $\square$

For fixed n , the question for all groups is equivalent to the question for finite abelian groups. To see the nontrivial direction, partition the finite-order nonidentity entries according to the cyclic subgroups they generate. All prime divisors of their orders exceed $n + 1$; otherwise G has an element of forbidden prime order. Thus $\langle a^i \rangle = \langle a \rangle$ for $1 \leq i \leq n$. Distinct cyclic groups cannot contribute equal powers in this exponent range. Represent the entries in each cyclic group in a separate abstract cyclic coordinate. This preserves exactly their relevant power equalities. Replace all infinite-order entries by distinct nonidentity elements in one additional cyclic coordinate of prime order greater than $n + 1$, and represent the identity by the identity. An affirmative finite abelian result assigns the exponents. Keep the assignments on finite-order entries; reassign the remaining exponents to the infinite-order entries using the proposition. Finite- and infinite-order powers cannot coincide. This proves the equivalence, without embedding G in an abelian group.

There is also a finite size bound. Let p_n be the least prime greater than $n + 1$. A counterexample at $n \geq 2$, if one exists, exists in a finite abelian group of order at most

$$p_n n^{n-1}.$$

Indeed, from a finite abelian counterexample choose a collision for every exponent permutation and impose those relations $ix_a = jx_b$ on a free module over $S = \mathbb{Z}[1/P]$, where P is the product of primes at most $n + 1$. Its distinguished elements remain distinct by mapping to the counterexample, and every permutation fails. In the multigraph of relations, a spanning tree in a component of k vertices expresses each variable as an S -unit multiple of one generator. A nontree edge either gives the zero relation or gives an annihilator, up to an S -unit, equal to the difference of the two coefficient products around a simple cycle. A nonzero difference has absolute value less than n^k . The component is therefore free of rank one over S , or cyclic finite of order less than n^k , after stripping inverted primes. If every component is finite, the total order is at most $n^n < p_n n^{n-1}$. Otherwise the finite components involve at most $n - 1$ variables; replace all infinite-order variables by a separate C_{p_n} as above. A good assignment would lift back to a contradiction, and the resulting order is at most $p_n n^{n-1}$. This is decidability for each fixed n , not a uniform proof in n .

The certificate is a proof of an all-group assertion. A node consists of an integer relation matrix R on n distinguished variables, interpreted over S . It asserts that no counterexample tuple satisfies those relations. There are two rules. A leaf proves that a coordinate difference belongs to the S -row span of R , forcing two entries equal. A branch chooses one exponent permutation and treats every one of its $\binom{n}{2}$ possible collisions. For each collision it points to an earlier proved node, with a bijective variable relabeling, and proves that every relation of that child follows from the parent relations and the collision. The children need not describe exact quotients; the indicated implications suffice. The final node has no relations. Induction along earlier-node references proves that no counterexample exists in any S -module, hence in any eligible finite abelian group and, by the reduction, in any eligible group.

The producer represents connected components by rational weights and annihilators and uses gcd operations to merge collisions. The GAP checker instead uses integer Smith normal

form: it verifies $UMV = D$, unimodularity of U, V , and diagonality of D , then tests divisibility after removing exactly the inverted prime factors and requires zero in all free coordinates. This checks the mathematical implications without trusting the producer’s normalization. A separate data-only parser checks the certificate grammar, excluding executable instructions that could alter the checker.

n	Nodes	Collision cases	Relation implications
4	227	1,056	3,953
5	10,493	87,570	426,345
6	265,104	3,245,190	19,094,383
7	42,891,332	786,577,344	5,447,182,055

For $n = 7$, six GAP workers partitioned identifiers by residue modulo six. Each read all literal states but checked its assigned nodes; together their disjoint classes cover the proof, with every child identifier smaller than its parent. All six recorded actual process exit zero. The completed aggregate was written at 13:08:05 UTC on 12 September, about 26 hours after launch. The complete compressed certificate occupies 1,271,256,410 bytes. The six 14 GiB workspace ceilings are configuration limits, not a measurement of peak memory. These historical completion records have been inspected in the manuscript phase; the 26-hour verification has not been represented as a fresh paper-phase replay.

As a hypothesis control, allowing order five at $n = 4$ yields the four nonidentity elements of C_5 . In additive coordinates modulo five, distinct nonzero outputs would have product -1 , whereas the input product times the product of exponents is 1. This is a counterexample to the weakened assumption and is stored as such. Sun’s higher- n small cyclic searches are a separate kind of evidence from these universal fixed- n certificates. The unrestricted conjecture remains open in this investigation. Detailed proofs, grammar and receipts are indexed by `research/20.100-reduction.md` and `research/20.100-n7-report.md`.

D.2 20.89/19.83: almost Engel closure under additional hypotheses

An element a is almost Engel in G if one finite set $E(a)$ contains $[x, {}_n a]$ for every sufficiently large n , for each $x \in G$. The entry time may depend on x . The question asks whether these elements form a subgroup of an arbitrary linear group. A finite sink inside $\langle a_1, \dots, a_r \rangle$ does not by itself provide a sink for a product of the a_i relative to all of G . The following deductions supply that ambient quantifier in specified cases; the general positive-characteristic case remains unresolved.

The local reduction. In every characteristic, a subgroup generated by finitely many elements almost Engel in G is finite-by-nilpotent. Here is the reduction and its imported inputs. Restrict to the finitely generated linear group $X = \langle a_1, \dots, a_r \rangle$, which is residually finite. Choose a finite-index normal subgroup H avoiding the nonidentity members of their finite sinks. Then each a_i is Engel in $H\langle a_i \rangle$: for $h \in H$ eventual commutators lie both in H and in its avoided sink, and $[ha_i^k, {}_n a_i] = [h, {}_n a_i]^{a_i^k}$. Gruenberg’s theorem identifies the Engel elements of a linear group with its Hirsch–Plotkin radical. Hence $[H, a_i] \subseteq R = \text{HP}(H)$. In X/R , all a_i centralize H/R ; its centralizer has a central subgroup of finite index, namely its intersection with H/R . The locally nilpotent linear group R is soluble. Consequently X is virtually soluble. Shumyatsky’s Theorem 3.3 now makes X finite-by-nilpotent [85]; the residual-finiteness and linear-radical inputs are recorded also in [95]. No product closure has been assumed in this deduction.

Let S be the set in question and $L = \langle S \rangle$. Every finitely generated subgroup of L is finite-by-nilpotent, by collecting the finitely many members of S used in its words. Call this

property $\mathcal{P}$. In any group with $\mathcal{P}$, the torsion elements form a locally finite characteristic subgroup: finitely many torsion elements generate a finite-by-nilpotent group whose nilpotent quotient is generated by finitely many torsion elements, hence finite. This proves closure as well as local finiteness.

The unique minimal finite sink consists of the periodic points of $x \mapsto [x, a]$. Each such point belongs to $\langle a^G \rangle$. Express its finitely many points using finitely many conjugates of a , including a itself. Their subgroup is finite-by-nilpotent by the local reduction. In its nilpotent quotient a periodic repeated commutator must be trivial, so the entire minimal sink lies in the finite kernel. Thus its generated subgroup is finite. It need not be normal in the ambient group. In particular, for a torsion-free linear group every almost Engel element is Engel, proving closure there in every characteristic.

Two lifting facts. If X/D is locally nilpotent for some finite normal D , one can choose D characteristic: take one of minimum order. For an automorphism σ , the quotient by $D \cap \sigma(D)$ embeds in the product of the two locally nilpotent quotients. That product is locally nilpotent, as are its subgroups. Minimality forces $D = \sigma(D)$.

Suppose X has $\mathcal{P}$, $N \triangleleft X$ is torsion-free, and X/N is finite-by-(locally nilpotent). Choose a finite characteristic kernel in X/N and let R be its preimage. The torsion subgroup $K = T(R)$ is locally finite and injects into the finite group R/N , so is finite. For a finitely generated $Y \leq X$, a sufficiently late lower central term is finite by $\mathcal{P}$, and a sufficiently late term lies in R since X/R is locally nilpotent. A common later term therefore lies in K . Thus X/K is locally nilpotent. The construction preserves any group of automorphisms preserving N .

Characteristic zero. The group L contains no nonabelian free subgroup, since such a group cannot have $\mathcal{P}$. The characteristic-zero form of the Tits alternative applies without finite generation and makes L virtually soluble [89, Theorem 1]. Over an algebraic closure of the field let A be its Zariski closure and U the unipotent radical of A° . The connected component is soluble, and A°/U is a torus; the quotient A/U is linear algebraic [65]. Since $L \triangleleft G$, the subgroup $N = L \cap U$ is normal in G . In characteristic zero it is torsion-free: the finite matrix logarithm gives $\log(u^m) = m \log(u)$. The group L/N is therefore linear and virtually abelian.

Shumyatsky's Lemma 3.2 gives the following prior consequence: in a virtually abelian group, almost Engel elements form a subgroup. Indeed it makes the normal closure of each such element finite-by-(locally nilpotent); choose its finite kernel characteristic as above and use the Hirsch–Plotkin radical after factoring out the finitely many kernels needed for a product. The images of S generate L/N , so every element of L/N is almost Engel there. Only now apply Shumyatsky's Theorem 4.2: an almost Engel linear group is finite-by-hypercentral, hence finite-by-(locally nilpotent). The torsion-free lifting fact gives a finite G -normal subgroup $K \leq L$ with L/K locally nilpotent. For $l \in L, x \in G$, normality gives $[x, l] \in L$. The two elements $[x, l]K, lK$ generate a nilpotent subgroup, so $[x, {}_n l] \in K$ eventually. Thus K is one ambient sink for every l , proving $S = L$. The minimum-kernel argument makes K characteristic in L , and hence in G .

This proves closure for every characteristic-zero linear group, without generation or solubility assumptions on G . It does not make the full torsion subgroup of L finite; an infinite central torsion subgroup is possible. Both the unrestricted Tits step and the torsion-free unipotent kernel require characteristic zero in this argument.

Two further characteristic-free cases. If all finite subgroups of G have order at most b , the locally finite torsion subgroup $K = T(L)$ has order at most b : $b + 1$ of its elements

would generate a larger finite subgroup. Property $\mathcal{P}$ then gives L/K locally nilpotent, and the same commutator argument proves $S = L$, with K a characteristic common sink. If G has a torsion-free subgroup of index d , every finite subgroup acts freely on its d cosets, so its order divides d ; in particular $|K| \mid d$. This is not automatic in positive characteristic: the group generated by $\text{diag}(t, 1)$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ over $\mathbb{F}_p(t)$ contains elementary abelian subgroups of orders p^r for all r .

For an upper triangular group, order matrix entries by decreasing superdiagonal distance and refine its unipotent radical into one-entry additive factors B_s . Conjugation acts on the factor at (i, j) by the character $\chi_s(g) = g_{jj}/g_{ii}$. Define

$$P = \bigcap_{B_s \text{ infinite}} \ker \chi_s.$$

If multiplication by $\chi - 1 \neq 0$ on an additive subgroup of a field has one finite eventual sink, a nonzero orbit first shows that $\chi - 1$ has finite multiplicative order. All points are then periodic and must lie in the sink, so the additive subgroup is finite. This proves necessity of membership in P . Conversely P has a finite normal series whose factors are finite or P -central, including its abelian diagonal quotient. The Hall–Baer finite-central-series lemma makes it finite-by-nilpotent [95]. A finite lower-central term $K = \gamma_{c+1}(P)$ is normal in G ; since $[G, P] \subseteq P$, $[x, {}_{c+1}g] \in K$ for $x \in G, g \in P$. This proves that P is exactly the almost Engel subgroup, with a uniform commutator-length bound in this triangular case.

Normal subgroups of algebra unit groups. Let A be a finite-dimensional associative unital algebra over an infinite field F , and $G \triangleleft A^\times$. Every almost Engel element of G is Engel. For $a \in G$, all but finitely many $c_t = 1 + ta$ are units: $\det(I + tL_a)$, in the faithful left regular representation, is a nonzero polynomial with constant term one. Each such c_t commutes with a and normalizes G , hence permutes its minimal sink $E(a)$. If $c_s^{-1}ec_s = c_t^{-1}ec_t = f$ for $s \neq t$, then

$$e + sea = f + saf, \quad e + tea = f + taf.$$

Subtracting gives $ea = af$, and substitution gives $e = f$; thus e commutes with a . Finiteness of $E(a)$ and infinitely many parameters give such a repetition for every e . The commutator map consequently sends the whole sink to 1; since it permutes the minimal sink, that sink is $\{1\}$. Gruenberg’s theorem now proves the subgroup assertion. This includes normal subgroups of $\text{GL}_n(F)$ and $\text{SL}_n(F)$. Over $\mathbb{F}_q$, the same argument gives $|E(a)| \geq q - \dim_F A + 1$ for a nontrivial minimal sink; in a matrix subalgebra it sharpens to $q - s(a) + 1$, where $s(a)$ counts the distinct eigenvalues in $\mathbb{F}_q$.

Normality supplies the parameter conjugations. An arbitrary finite $S_3 \leq \text{GL}_2(\mathbb{Q})$ has non-Engel almost Engel elements, so this hypothesis cannot simply be dropped. Proofs, exact finite controls, and the precise classical imports are indexed by `research/20.89-character0-proof.md`, `research/20.89-residual-reduction.md`, `research/20.89-triangular-proof.md`, `research/20.89-bounded-torsion-proof.md`, and `research/20.89-normal-algebra-proof.md`.

D.3 21.121(b): sharp soluble bounds and permanence

Use the definition of admissible p -Jordan exponents from Section C.1. The unresolved part asks for a universal upper bound three. The following arguments prove exponent two under a soluble-subgroup hypothesis, give a nonsoluble example with exact exponent strictly between two and three, and preserve the entire set of admissible exponents under specified operations. The definition and prior general framework are those in [45, 24].

A characteristic abelian subgroup lemma. If a finite group X has an abelian subgroup of index at most j , it has a characteristic abelian subgroup of index at most j^2 . Recall the Chermak–Delgado argument in this form. For $H \leq X$ put $m(H) = |H||C_X(H)|$. The product-size formula and centralizer inclusions give

$$m(H)m(K) \leq m(H \cap K)m(\langle H, K \rangle).$$

Thus the intersection D of all subgroups of maximum measure also has maximum measure and is characteristic. Its centralizer has maximum measure as well, since $D \subseteq C_X(C_X(D))$. Minimality of the intersection implies $D \subseteq C_X(D)$, so D is abelian. For the given abelian A ,

$$|X||D| \geq m(D) \geq m(A) \geq |A|^2 \geq |X|^2/j^2,$$

proving the bound. The lemma does not force $D \subseteq A$. It will be applied inside a p' -group when prime-to- p order is required.

An abelian p' -subgroup $C \leq \text{Aut}(P)$, for a nontrivial finite p -group P , satisfies

$$|C| \leq |P/\Phi(P)| - 1.$$

Indeed the kernel of $\text{Aut}(P) \rightarrow \text{Aut}(P/\Phi(P))$ is a p -group: it acts freely on the $|\Phi(P)|^d$ tuples lifting a fixed basis of $P/\Phi(P)$, because each lift tuple generates P . Thus C acts faithfully on $\mathbb{F}_p^d$. Complete reducibility and commutativity embed it in a product of multiplicative groups of finite fields of orders p^{d_i} , with $\sum d_i = d$. Their orders have product at most $p^d - 1$.

The soluble theorem. Suppose every finite p' -subgroup of Γ has a normal abelian subgroup of index at most J , and every finite subgroup of Γ is soluble. For each finite $G \leq \Gamma$ there is a characteristic abelian p' -subgroup B with

$$[G : B] \leq J^3 |G_{(p)}|^2.$$

First let $R \neq 1$ be finite soluble with $O_{p'}(R) = 1$. Its Fitting subgroup $F(R) \neq 1$ is a p -group and $C_R(F(R)) \subseteq F(R)$. For the latter standard fact, if the centralizer modulo $Z(F(R))$ were nontrivial, choose a minimal normal elementary abelian subgroup there. Its preimage is a normal class-two nilpotent subgroup strictly exceeding the centralizer's intersection with $F(R)$, contradicting the Fitting theorem. A Hall p' -subgroup H therefore acts faithfully on $F(R)$. Choose $A \triangleleft H$ abelian of index at most J and apply the coprime-action bound:

$$|R| = |R_{(p)}||H| \leq J|R_{(p)}|(|R_{(p)}| - 1).$$

Hall-subgroup existence here is the soluble Hall theorem, obtainable by induction with Schur–Zassenhaus.

For general G , set $N = O_{p'}(G)$ and $R = G/N$. Then $O_{p'}(R) = 1$; the same J works for its p' -subgroups, whose full preimages are p' -groups. The characteristic lemma gives $B \text{ char } N$ abelian with $[N : B] \leq J^2$. Since $N \text{ char } G$, also $B \text{ char } G$, and the preceding estimate proves the assertion, with $R = 1$ handled directly. If every finite subgroup instead has a soluble subgroup of index at most d , take its core, of index at most $d!$. The bound becomes $d!J^3 |G_{(p)}|^2$, with B normal in G .

When every finite subgroup has normal Sylow p -subgroup P , a sharper bound is available without solubility. Write $G = P \rtimes H$ by Schur–Zassenhaus, choose $A \triangleleft H$ abelian with $[H : A] \leq J$, and put $B = C_A(P)$. Both P and H normalize B , so $B \triangleleft G$. For $P \neq 1$,

$$[G : B] \leq J|P|(|P/\Phi(P)| - 1) < J|P|^2;$$

for $P = 1$, use A . Exponent two is sharp for every p : the locally finite affine group $\text{AGL}_1(\overline{\mathbb{F}}_p)$ has these hypotheses with $J = 1$. Its subgroups $\text{AGL}_1(\mathbb{F}_q)$, $q = p^n$, have order $q(q - 1)$ and no nontrivial normal p' -subgroup. Such a subgroup would centralize the normal translations, whose centralizer is the translation group itself. Hence their least normal abelian p' -indices are $q(q - 1)$, excluding every exponent less than two.

An exact nonsoluble exponent. Let $\Gamma = \bigoplus_{\mathbb{N}} A_5$, the finite-support direct product, and $\rho = \log(60)/\log(4)$. Then its 2-Jordan exponent is exactly ρ , attained with constant one. For a subgroup $H \leq A_5^n$, separate coordinates with full projection from those with proper projection. The first projection T is a product of diagonal simple factors, hence A_5^r ; induction proves this by projecting off the last coordinate, whose kernel is either A_5 or trivial. In the latter case a surjection $A_5^r \rightarrow A_5$ is supported on exactly one factor because its factor images commute and are normal. The second projection B is soluble, since all proper subgroups of A_5 are soluble. Goursat's argument gives a common quotient of T and B ; it is perfect and soluble, hence trivial. Thus $H = T \times B$. Every odd-order subgroup of B is abelian, since its coordinate projections are cyclic subgroups of orders 1, 3, 5. The soluble theorem with $J = 1$ gives a normal abelian odd-order $D \leq B$ of index at most $|B_{(2)}|^2$. Therefore

$$[H : 1 \times D] \leq 60^r |B_{(2)}|^2 \leq (4^r |B_{(2)}|)^{\rho} = |H_{(2)}|^{\rho}.$$

Every finite subgroup of Γ has finite support, so this is a uniform bound. Full powers A_5^r have no nontrivial abelian normal subgroup and exclude every smaller exponent. Since $16 < 60 < 64$, one has $2 < \rho < 3$. This example is not virtually soluble: a soluble subgroup of A_5^r has index at least 5^r .

Preserving exponents. Suppose e is admissible for Γ with constant J . For any finite $G \leq \Gamma$, apply the characteristic lemma inside $N = O_{p'}(G)$. The original abelian normal subgroup lies in N , giving $[G : N] \leq J|G_{(p)}|^e$, and the p' -subgroup bound gives a characteristic abelian $D \leq N$ of index at most J^2 . Thus

$$[G : D] \leq J^3 |G_{(p)}|^e.$$

Replacing normal by characteristic in the definition does not change its admissible exponents or attainment. If $\Lambda \triangleleft \Gamma$ and finite subgroups of Γ/Λ have order at most b , apply this form to $G \cap \Lambda$; its characteristic subgroup is normal in G . For a real e , the constant $bJ^3 \max(1, b^{-e})$ works in Γ . Restriction gives the reverse implication. Taking a core proves the same invariance for any finite-index subgroup.

More generally, assume a fixed generalized p -Jordan constant J_0 for Γ , and let $N \triangleleft \Gamma$ be finite. Then Γ and Γ/N have exactly the same admissible exponents. Downward passage uses finite full preimages. For upward passage, take finite $G \leq \Gamma$, $K = G \cap N$, and the preimage M of a normal abelian p' -subgroup in G/K . Put $C = C_M(K)$. It is normal in G , $[M : C] \leq |N|!$, and $C' \subseteq K \subseteq C_G(C)$; so C has class at most two. The soluble theorem supplies $D \text{ char } C$, abelian of p' -order, with $[C : D] \leq J_0^3 |K_{(p)}|^2$. If the quotient bound has constant J , then

$$[G : D] \leq JJ_0^3 |N|! \max(1, |N|^{2-e}) |G_{(p)}|^e.$$

This preserves the exponent and its attainment. For a finite normal p -kernel the generalized hypothesis follows from the quotient bound, since p' -subgroups inject into the quotient. For other finite kernels it is necessary: the infinite extraspecial group at a prime $q \neq p$ has central kernel C_q and abelian quotient, but contains finite extraspecial groups with unbounded least abelian index.

These results leave the unrestricted bound three unresolved. The full arguments and constants are in `research/21.121b-soluble-proof.md`, `research/21.121b-normal-sylow-proof.md`, `research/21.121b-a5-proof.md`, `research/21.121-permanence-proof.md`, and `research/21.121-finite-kernels.md`.

D.4 20.124: a Rota–Baxter operator in every infinite free rank

The question asks for a weight-one Rota–Baxter operator on a nonabelian free group with image its derived subgroup. The experiment constructs one in every infinite rank. The bijective-cocycle framework is prior work [9, 39]; the construction and identity can be given directly.

For an infinite cardinal κ , partition a free basis into levels $X_0, X_1, \dots$, each of size κ . Let F_m be free on levels through m . Both F_m and F'_m have cardinality κ ; for the lower bound on F'_m , use the distinct commutators with one fixed basis element. Choose a surjection $X_{m+1} \rightarrow F'_m$ and define an endomorphism ϕ by those maps, killing X_0 . It lowers levels, so each element is killed by an iterate. Moreover $\phi(F) = \bigcup_m F'_m = F'$.

For any locally terminating endomorphism of any group put

$$D(g) = g\phi(g)^{-1}, \quad T(g) = g\phi(g)\phi^2(g) \cdots$$

Each product is finite for its particular g . Cancellation gives $D(T(g)) = g = T(D(g))$, so these are inverse set bijections. Define $B(g) = \phi(T(g))$, whence $B(D(u)) = \phi(u)$. For $g = D(u), h = D(v)$,

$$gB(g)hB(g)^{-1} = uv\phi(uv)^{-1} = D(uv).$$

Therefore

$$B(gB(g)hB(g)^{-1}) = \phi(uv) = B(g)B(h),$$

the required identity, and $\text{im } B = \text{im } \phi = F'$. If $\phi(x) = w$, the explicit preimage is $B(xw^{-1}) = w$. The descendant multiplication is carried from the original free group by D . In finite rank an endomorphism image is finitely generated whereas F' has infinite rank. This limits this particular method; it does not rule out another Rota–Baxter map. See `research/20.124-infinite-rank-proof.md`.

D.5 17.118: characteristic exponent subgroups

Let a finite p -group G have an exponent- p subgroup H of index p . The question asks for a characteristic exponent- p subgroup of index bounded solely in terms of p . The experiment proves bounds p^{c+1} if H has class at most c , and p^c if G has class at most c . The optimal universal bounds for $p = 2, 3$ are 4, 27.

Here is the class argument. Intersect all automorphic images of H , obtaining characteristic exponent- p subgroup K with $G/K \cong \mathbb{F}_p^r$. Choose r images H_i defining independent coordinate hyperplanes and lifts x_i of the dual coordinate basis. Together with generators of K , they generate G , and omitting x_i leaves a subgroup inside H_i . For a class- c group with this deletion property, $r > c$ forces exponent p . To justify deletion without assuming it is an endomorphism of G , form the free class- c group and impose $w^p = 1$ for every word omitting a distinguished x_i . The deletion maps descend as commuting idempotents e_i . Every p th power lies in $\bigcap_i \ker e_i$. On the weight- k lower-central factor, $k < r$, the product $\prod_i (1 - e_i)$ kills each generating bracket, since it omits some x_i . It is the identity on the common kernel. Induction thus places that kernel in $\gamma_r = 1$. This proves the deletion assertion and the bound for G . If only H has class c and $r > c + 1$, every generating commutator of weight $c + 1$ omits one x_i , hence vanishes in its H_i . So G has class at most c , and the same argument applies.

For $p = 2$, H is abelian, giving 4. For $p = 3$, Laffey’s theorem says that a finite 3-group not of exponent three has at most a $7/9$ proportion of cube roots of one [57]. The union of the independent hyperplanes H_i has proportion $1 - (2/3)^r$. At $r \geq 4$ this exceeds $7/9$. Thus $r \leq 3$, proving 27.

For every prime p , a uniform bound must be at least p^p . Take the associative $\mathbb{F}_p$ -algebra with basis all words without repeated letters in p letters $x_0, \dots, x_{p-1}$, including the empty word, and multiply by concatenation when the supports are disjoint and by zero otherwise. Let J be its positive-degree ideal and $G = 1 + J$. Then $J^{p+1} = 0$. If u has linear part $\sum a_i x_i$,

$$(1 + u)^p = 1 + \left(\prod_i a_i \right) \Omega,$$

where $\Omega \neq 0$ is the sum of the distinct length- p basis words. The preimage of any coordinate hyperplane has index p and exponent p , with class exactly $p - 1$. The subgroup $K = 1 + J^2$ has exponent p and index p^p . It is intrinsically characterized by

$$K = \{g : (gh)^p = h^p \text{ for every } h \in G\}.$$

For a nonzero linear part a , choose $b_j = 0$ where $a_j \neq 0$ and $b_i \neq -a_i$ for all other coordinates; then $\prod b_i = 0 \neq \prod (a_i + b_i)$, proving the characterization. Cycling the letters induces a p -cycle on $\mathbb{F}_p^p$. Any nonzero invariant subspace contains a nonzero constant vector, by taking the last nonzero iterate of $\sigma - 1$. That vector has a lift with nontrivial p th power. The image of a characteristic exponent- p subgroup must therefore be zero, forcing it inside K . This proves sharpness for $p = 2, 3$ and the general lower bound. It does not give unbounded index at a fixed p . See `research/17.118-partial-proof.md`; the bounded-class parameter is not known here to depend only on $p \geq 5$.

D.6 Other proved restrictions and a lower-bound construction

For Problem 21.114, write $a(X) = |X/X'|$; a finite group is weakly ab-maximal if $a(H) \leq a(X)$ for all $H \leq X$. The prior general framework is due to Lisi–Sabatini [60]. A central-product construction in the experiment yields order 8192 and derived length three, so a universal derived-length bound, if it exists, is at least three. Let P have cyclic central subgroup Z of order p^k with $Q = P/Z$ weakly ab-maximal. Form the Heisenberg group E on $(\mathbb{Z}/p^k\mathbb{Z})^r \times (\mathbb{Z}/p^k\mathbb{Z})^r \times \mathbb{Z}/p^k\mathbb{Z}$, where $r \geq 1$, with central cross term $u \cdot v'$, and identify $E' = Z(E)$ with Z in the central product G . Then $|G| = |P|p^{2rk}$, $a(G) = a(Q)p^{2rk}$, and its derived length is $\max(2, \text{dl } P)$. For $H \leq G$, enlarge to HZ , which can only increase $a(H)$. In $H/Z \leq Q \times V$, let $U = (H/Z) \cap V$ and let T be the projection on Q . Write $|Z/(H' \cap Z)| = p^t$. Then

$$a(H) \leq p^t |U| a(T) \leq p^t |U| a(Q).$$

The reduction of U modulo p^t is isotropic in the nondegenerate alternating pairing on $(\mathbb{Z}/p^t\mathbb{Z})^{2r}$. Character orthogonality gives its size at most p^{rt} , even when it is not a free module. Hence $|U| \leq p^{2rk-rt}$, proving $a(H) \leq a(G)$. Literal finite permutation models supply P of order 512, derived-series orders 512, 32, 2, 1, and central Z of order four, with Q of order 128 and $a(Q) = 16$. Taking $r = 1$ gives the stated group and $a(G) = 256$. The exact input checks enumerate every subgroup of the two retained Q -models; they are recorded in `research/21.114-central-products.md`. That file also proves that central extensions of these particular outputs still have derived length at most three. Iteration of this construction does not establish unboundedness.

For Problem 19.108, let p be odd and $|P| = p^{10}$, with abelian normal subgroup A of order at least p^6 . Then for every irreducible complex χ and every $\chi(x) \neq 0$, one has $|x| \mid |P|/\chi(1)^2$. Wilde’s prior results cover degree at most p^3 , groups of order at most p^9 , and give, for a faithful violating character, a conjugate of $\langle x \rangle$ intersecting it trivially [96]. If χ has nontrivial kernel, the order-at-most- p^9 theorem in the quotient lifts the bound: the order of x is at most its quotient order times the kernel order. Thus a violation requires faithful χ of degree p^4 .

The bound $\chi(1) \leq [P : A]$ forces $|A| = p^6$, and equality in Clifford theory gives $\chi = \lambda^P$. Nonvanishing puts x in the normal subgroup A . The disjoint conjugate cyclic subgroups then force $|x| = p^3$ and $A = \langle x \rangle \times \langle x \rangle^g \cong C_{p^3}^2$. The dual orbit of λ has size p^4 . Such an orbit of a p -subgroup of $\mathrm{GL}_2(\mathbb{Z}/p^3\mathbb{Z})$ is invariant under translation by $p^2\mathbb{F}_p^2$. To see this, its image modulo p has size at most p . If its image modulo p^2 had size p^3 , the congruence kernel would supply $I + pM_1, I + pM_2$ with independent directions at a point. Their p th powers are $I + p^2M_i$ modulo p^3 , forcing full p^2 -fibers and an orbit of size p^5 , a contradiction. The image modulo p^2 thus has size at most p^2 ; the size p^4 forces every fiber full. Translation in a coordinate where x is primitive multiplies the orbit character sum by a nontrivial p th root of unity, so it vanishes, the final contradiction. Nonsplit extensions are included. Groups with no such large abelian normal subgroup remain uncovered; see `research/19.108-partial.md`.

For Problem 18.120, suppose $G = AB$, A abelian and B of class at most two. Put $C = (B')^G = \langle (B')^a : a \in A \rangle$ and $K = \langle A, B' \rangle = AC$. If C is abelian, $B_0 = K \cap B$ is abelian and $K = AB_0$. Indeed an element $ac \in K \cap B$ centralizes B' ; as c does too, a centralizes B' , and commutativity of A makes it centralize all its A -conjugates. Two such elements commute. The product assertion follows by writing an element of K in AB . If G has class at most four, conjugates of elements of $B' \subseteq \gamma_2(G)$ differ from them by elements of $\gamma_3(G)$; all cross brackets have weight at least five, while B' is abelian. Thus C is abelian. This proves the class-four case without finiteness or $A \cap B = 1$. The defect-two subnormal case is also a direct deduction from McCann's earlier lemma [63]; see `research/18.120-partial.md`.

For Problem 6.47, if a binary word defines another group operation on a class-two group G , collection and the two identity laws give $x * y = xy[x, y]^k$. This is associative, with unchanged inverses and powers and with new commutator $[x, y]_* = [x, y]^{2k+1}$. Collect an identity in the free class-two group as a product of unary powers and binary commutator powers. Setting all but one, and then all but two, variables equal to one shows that each collected factor is separately an identity of G . Unchanged powers and the commutator formula transfer them all to the new group, which therefore lies in $\mathrm{var}(G)$. This is a direct proof of a restricted case within the subject already studied by Cooper [27], without a novelty claim; see `research/6.47-class-two.md`.

D.7 Bounded exclusions with no general conclusion

The following completed searches are useful negative evidence, with the stated coverage boundaries.

- For Problem 20.21, structural reductions and catalogue exclusions through order 1536 force any counterexample to have order at least 3072. All 26,760 retained kernel pairs at order 1536 were separated from their multiplication tables; catalogue and subgroup-enumeration coverage remain dependencies.
- For Problem 19.20, all 91,774 nonabelian groups through order 511 gave no equality case and 469 reverse inequalities. This does not decide the arbitrary-group equality question.
- For Problem 21.113, 2,750 ordinary character tables provided 9,850 table/prime cases. Of the corresponding modular cases, 7,573 were available and 2,277 were unavailable. No counterexample appeared in the available cases.
- For Problem 21.99, the retained coverage includes all 4,722 transitive groups of degrees 2–20; direct word-certificate checks cover the 86 groups of degrees 2–8. These are different scopes, not interchangeable verification counts.
- For Problem 18.43, exact matrix-trace fingerprints screened all 3,933,931,043 positive

binary necklaces of lengths 1–36, with no identity candidate. A future collision would require symbolic certification; larger lengths remain outside the search.

The corresponding reports and scripts are indexed in `reports/FINAL_REPORT.md` and `reports/STATUS.md`. The absence of a witness in these finite domains is not a proof for unbounded orders, dimensions, or word lengths.

E Rediscoveries and deductions from prior work

These results were excluded from the deadline candidate count. The reasons differ: an earlier paper may state the answer explicitly, a prior theorem may imply it by a further argument, or the literal formulation may be inconsistent with its own accompanying comment. We retain selected proofs because the experiment’s output includes mathematical exposition and source reconciliation as well as possible new results. A deduction from existing theorems is identified as such; we do not assert that its source explicitly named the Notebook question.

E.1 13.19: a small subdirect-quotient counterexample

The question asks whether Q/H must be a regular p -group when $H \triangleleft Q$ and both are subdirect in the same finite direct product. An earlier construction of Kearnes–Mayr–Ruškuc already implies a negative answer [50]. Specifically, their Section 6 has a finite perfect group $G = HM$, with M a 2-group and normal $N \leq M$, satisfying $[N, N] > [G, N, N]$. In their construction with $d = 2$, Theorem 4.2 gives $S = \Gamma_1(G, N) \leq G^4$ and $T = \Gamma_3(G, N) = \gamma_3(S)$, with S/T nonabelian of class two. Both contain the diagonal G , hence are subdirect. All nondiagonal generators lie in M^4 ; since the diagonal is killed in S/T , this quotient is a finite 2-group. A nonabelian 2-group is irregular. This is the prior implication used to remove 13.19 from the count.

The experiment’s smaller example is transparent. Let $D = \text{UT}_3(\mathbb{F}_2)$, writing its elements as

$$d(a, b, c)d(a', b', c') = d(a + a', b + b', c + c' + ab').$$

It is dihedral of order eight. In D^4 put

$$Q = \{(d(a_i, b_j, c_{ij}))_{1 \leq i, j \leq 2} : a, b \in \mathbb{F}_2^2, C \in M_2(\mathbb{F}_2)\}.$$

Its multiplication on the eight coordinates is

$$(a, b, C)(a', b', C') = (a + a', b + b', C + C' + a(b')^T),$$

so $|Q| = 256$. The map

$$\phi(a, b, C) = d(a_1 + a_2, b_1 + b_2, \sum_{i,j} c_{ij})$$

is onto and multiplicative, since the sum of the entries of $a(b')^T$ is $(a_1 + a_2)(b'_1 + b'_2)$. Its kernel H has order 32 and contains the entire diagonal D . Both H and Q thus project onto all four factors, with $Q/H \cong D$. For $s = d(1, 0, 0)$, $t = d(0, 1, 0)$, one has $s^2 = t^2 = 1$, but $(st)^2 = d(0, 0, 1) \neq 1$. Every derived-subgroup square is trivial, so no correction allowed in the regularity identity can repair this failure. The original question permits $p = 2$.

There is also a direct universal construction: every finite p -group K is such a quotient when the factors may vary. Let $m = |K|$, choose generators $g_1, \dots, g_d$, and let I be the augmentation ideal in $\mathbb{F}_p[K]$. Induction on $|K|$ gives $I^m = 0$: choose central z of order p ; the kernel of the map to $\mathbb{F}_p[K/\langle z \rangle]$ is $(z - 1)\mathbb{F}_p[K]$, whose p th power is zero, and use induction in the quotient. Let A be the nonunital free associative algebra on symbols x_{ij} ,

$1 \leq i \leq m, 1 \leq j \leq d$, truncated at degree m . In $1 + A$ take $Q = \langle 1 + x_{ij} \rangle$. Substitution $x_{ij} \mapsto g_j - 1$ maps Q onto K . For each i , delete the i th block of symbols and let G_i be the image of Q . These maps jointly embed Q : each surviving monomial has length less than m , omits some block, and retains its coefficient when that block is deleted. The kernel H of the map onto K projects onto every G_i , since $(1 + x_{kj})(1 + x_{ij})^{-1} \in H$ projects to $1 + x_{kj}$ for $k \neq i$. This proves the assertion, including simultaneous subdirectness. The trivial K is handled by $Q = H = C_p$. The construction is not claimed to be small or new. See `research/13.19-proof.md`, `research/13.19-universality.md`, and the source comparison in `research/13.19-review.md`.

E.2 17.34: a bounded-class separation proof using PBW

Let $\mathcal{N}_c$ be the full quasivariety of torsion-free nilpotent groups of class at most c . If $H \leq G \in \mathcal{N}_c$ is divisible, then its dominion is exactly H . The fixed-class strong amalgamation theorem of d’Elbée–Müller–Ramsey–Siniora already implies this [30, Theorem 4.35]. Use the lower-central filtration on the rational Lie algebra of the Malcev completion of G , and its intersection filtration on the subalgebra corresponding to H . These are compatible Lazard series even if the latter is not the intrinsic lower-central series of the subalgebra. Strong amalgamation separates the two copies outside H . Finite-dimensional amalgamation suffices in arbitrary rank: every finite set of constraints lies in a finitely generated, hence finite-dimensional, nilpotent Lie subalgebra; compactness then supplies the full pair of embeddings.

Here is the retained alternative proof. For a rational Lie algebra L of class at most c and subalgebra $\mathfrak{h}$, let $U = U(L)$, I its augmentation ideal, and $J = U\mathfrak{h}$, a left ideal. Set

$$M = I/(J + I^{c+1}), \quad d(x) = x + J + I^{c+1}.$$

This is a left U -module and $d([x, y]) = xd(y) - yd(x)$. Choose a basis of L adapted both to the lower-central filtration and to $\mathfrak{h}$, weighting a basis element by its largest filtration index. Order the basis with the complementary vectors before the $\mathfrak{h}$ -vectors. PBW monomials of weight at least n span exactly I^n . One inclusion follows since a vector of weight w lies in I^w ; the other follows by reordering, because brackets do not decrease total weight. Also J is spanned exactly by ordered monomials containing a nonempty terminal $\mathfrak{h}$ -block: multiplication on the right by an $\mathfrak{h}$ -vector only reorders that block. Thus the complementary single-letter vectors all survive independently in M , while the $\mathfrak{h}$ -vectors vanish. This proves $\ker d = \mathfrak{h}$, in any dimension.

In $P = L \ltimes M$, with M abelian, the two maps

$$f(x) = (x, 0), \quad g(x) = (x, d(x))$$

are injective Lie homomorphisms agreeing exactly on $\mathfrak{h}$. Induction gives $\gamma_r(P) \subseteq \gamma_r(L) + I^{r-1}M$. Since $I^c M = 0$, the class remains at most c . The numerator I is essential: replacing it by U would generally allow an unwanted extra nilpotency step.

Embed G in its rational Malcev completion and use the BCH correspondence, which preserves the class bound and works in arbitrary rank [5]. Unique roots make the divisible subgroup H a rational Lie subalgebra there. The BCH group of P lies in $\mathcal{N}_c$, and the restrictions of f, g to G agree exactly on H . This is precisely the required dominion separation. The result concerns the full $\mathcal{N}_c$, with no assertion for every proper subquasivariety. The class-two predecessor is Shakhova’s explicit 2015 result [84]. PBW separation in unrestricted Lie algebras is also prior work of Bergman [13]. See `research/17.34-proof.md` and `research/17.34-known-consequence.md`.

E.3 21.115: Fourier support and class-two linearization

Sambale’s preprint of 8 September 2026, two days before the experiment, proves the general complement bound for unions of cosets, including mixed left and right cosets [80]. The experiment first developed the following restricted arguments, then found that resolution and excluded the problem from its candidate count. The earlier Sambale–Tărnăuceanu paper treated several special cases [81]; the character-product method below also appears in Szegedy’s work [88].

If cosets $g_i H_i$, $1 \leq i \leq n$, do not cover a finite abelian group A , choose x_0 outside their union. Characters separate points in A/H_i , so choose χ_i trivial on H_i with $\chi_i(x_0) \neq \chi_i(g_i)$. The nonzero function

$$f(x) = \prod_{i=1}^n (\chi_i(x) - \chi_i(g_i))$$

vanishes on the union and is a linear combination of at most 2^n distinct characters. For a nonzero combination $f = \sum_{\chi \in T} a_\chi \chi$, orthogonality and Cauchy–Schwarz give

$$|A| \sum |a_\chi|^2 = \|f\|_2^2 \leq |\text{supp } f| \|f\|_\infty^2 \leq |\text{supp } f| |T| \sum |a_\chi|^2.$$

Thus the complement has at least $|A|/2^n$ elements.

The same deduction applies to a class-two finite group whose derived subgroup has odd order. On its underlying set define $x + y = xy[x, y]^{-1/2}$, taking the unique half only in the central odd-order derived subgroup. Class-two commutator identities show this is abelian: both triple sums equal $xyz[x, y]^{-1/2}[x, z]^{-1/2}[y, z]^{-1/2}$. Every original subgroup is additive, since the half of its commutator is a power of that commutator. For fixed g , $T_g(x) = [g, x]$ is an additive central-valued map with $T_g^2 = 0$; hence $L_g = 1 + \frac{1}{2}T_g$ is an additive automorphism, and $gx = g + L_g(x)$. Every original coset is therefore an affine coset in the same abelian group. Applying the Fourier argument proves the bound. This avoids any assumption that the whole group has odd order, but does not cover every class-two 2-group. The general theorem is credited to the earlier preprint, not inferred from these restricted cases. See `research/21.115-partial-proof.md`.

E.4 16.60: a tensor proof of an existing character bound

The full affirmative answer was stated in the May 2026 preprint of Yerrapati–Dixit–Shukla [97, Theorem 3.1]. For a finite group G and any automorphism α , the number of elements inverted by α is at most $T(G) = \sum_{\chi \in \text{Irr}(G)} \chi(1)$. The following proof supplies the needed estimate without treating a character value on a product as a product of character values.

For a unitary irreducible representation ρ on V , put

$$E = \frac{1}{|G|} \sum_{g \in G} \rho(g) \otimes \rho(\alpha(g)), \quad F(v \otimes w) = w \otimes v.$$

The operator E is the orthogonal projection onto invariants of $V \otimes (V \circ \alpha)$. Its rank is $\dim \text{Hom}_G(V^*, V \circ \alpha)$, at most one by Schur’s lemma. The trace identity $\text{tr}((A \otimes B)F) = \text{tr}(AB)$ gives

$$\iota_\alpha(\chi) = \frac{1}{|G|} \sum_g \chi(g\alpha(g)) = \text{tr}(EF), \quad |\iota_\alpha(\chi)| \leq \text{rank } E \leq 1.$$

The inequality uses an orthonormal basis of $\text{im } E$ and unitarity of F ; E need not commute with F . The regular-character identity now yields

$$|\{g : \alpha(g) = g^{-1}\}| = \sum_\chi \chi(1) \iota_\alpha(\chi) \leq T(G).$$

No involutivity of α is needed. Alternatively, every inverted element lies in $\text{Fix}(\alpha^2)$, where the restriction of α is involutive. The classical twisted indicator bound there and $T(H) \leq T(G)$ for $H \leq G$ imply the same result; the latter inequality follows by restricting the direct sum of all irreducible G -modules and using Frobenius reciprocity. The source audit identifies an unexplained scalar Cauchy–Schwarz passage in its Lemma 2.19; this tensor argument establishes the needed statement independently of that passage, without assigning priority to the answer. See `research/16.60-known-consequence.md`.

E.5 21.132: a graded radical quotient with finite centre

The experiment ultimately classified this as a deduction from prior radical theorems, rather than a new solution. Start with a two-generated, positively graded, infinite-dimensional nil $\mathbb{F}_p$ -algebra A , with finite-dimensional graded pieces, from the graded Golod construction [34]. Let $L(A)$ be its Levitzki radical, the largest locally nilpotent ideal. The centreless quotient $B = A/L(A)$ is the construction used by Sereda–Sozutov [82]. The radical is proper, since otherwise finite generation would make A nilpotent. It is homogeneous by the classical graded-radical theorem explicitly recorded in [44]. Hence B remains positively graded with finite-dimensional pieces. It is infinite-dimensional: otherwise positive grading makes it nilpotent, contradicting $L(B) = 0$.

If z commutes with all of B , then z is nilpotent and the ideal B^1z , using the unitization, is nilpotent. It must be zero, so $z = 0$. Let $H = \langle 1 + b_1, 1 + b_2 \rangle \leq 1 + B$. It is a p -group, since each of its elements is one plus a nilpotent element. Its centre is trivial: a central $1 + b$ commutes with the algebra generators, so b is central in B . It is infinite, since a finite H would have finite-dimensional linear span containing the algebra generators and hence all of B . The truncations $B/B_{\geq n}$ give finite p -group quotients separating every element, by finite homogeneous support. Thus H is residually finite. Adjoin a one-dimensional zero-multiplication algebra $\mathbb{F}_p z$. The subgroup generated by one plus the three algebra generators is $H \times C_p$, with centre C_p . It meets the Notebook’s algebraic definition of a Golod group, for every prime p , and is at most three-generated. The grading is what proves residual finiteness; it is not inferred from an arbitrary quotient preserving that property. The exact deduction under the question number was not located in a prior paper, but its inputs and classification are explicit. See `research/21.132-prior-consequence.md`.

E.6 21.42: positive gradings and self-similar actions

Every three-generated torsion-free nilpotent group of class three is self-similar, giving a negative answer to the existence question. Dekimpe–Igodt–Pouseele proved the positive-grading assertion for the corresponding Lie algebras [29]; the exact positive form is restated in [28]. Mathieu’s criterion transfers it to the original lattice, not merely an unspecified finite-index subgroup [62, Proposition 6]. The rational Lie algebra of the Malcev completion is generated by the logarithms of the three group generators, so these results apply.

The mechanism is instructive. Positive weights give dilations by q^j in weight j . For a suitable integer $q > 1$ the dilation δ preserves the given lattice G , with finite-index image. Its inverse is a virtual endomorphism $\delta(G) \rightarrow G$. If a subgroup $K \subseteq \delta(G)$ is invariant under this inverse, every $\delta^{-n}(\log k)$ remains in the discrete lattice and tends to zero. Thus $k = 1$. The virtual endomorphism has trivial normal core and gives a faithful recurrent action on the coset tree. There is no prescribed degree or finite-state requirement in the printed question. The archive also retains an independent classification argument giving positive weights for the three-generator class-three Lie algebras, with weight choices $(1, 1, 2)$, $(2, 2, 1)$, and $(3, 4, 2)$ in the nonhomogeneous cases. That proof is a rediscovery of the grading assertion; the lattice step remains an imported theorem. See `research/21.42-known-consequence.md`.

E.7 20.33: an effective universal group, with oracle bookkeeping

For any oracle X , there is a three-generated X -computably-enumerably presented group G_X containing every finitely generated group with an X -c.e. presentation. This is a standard embedding consequence; Mikaelian's explicit universal words improve three generators to two [64].

Dovetail all oracle programs enumerating relators on each finite alphabet, ignoring ill-formed outputs. Take the free product U_X of all their presented groups. It has an X -c.e. presentation on a countable computable alphabet $g_1, g_2, \dots$, and each factor embeds. For any such countably presented U , in $U * F(a, b)$ the elements $q_i = a^i b a^{-i}$ freely generate a subgroup, as part of a Schreier basis. The elements $g_i q_i$ also freely generate: the retraction killing U detects any freely reduced relation among them. The HNN extension

$$\langle U, a, b, t \mid t^{-1} q_i t = g_i q_i \ (i \geq 1) \rangle$$

therefore embeds U , and its relations express $g_i = t^{-1} q_i t q_i^{-1}$. Eliminating the g_i gives a three-generator presentation by effective substitution in the old relators. No word-problem decision is required, and the same oracle suffices.

Conversely, a finitely generated subgroup of an X -c.e. presented group has such a presentation: enumerate products of conjugates of enumerated relators and compare their freely reduced words with substitutions in the subgroup generators. Dovetailing enumerates exactly the subgroup kernel. Finite free products and adding finitely many relators preserve X -c.e. presentability as well. Thus both directions of the Notebook formulation follow; in the forward direction take one copy of G_X and the empty extra relator set. Decidable word problem and finite presentation are not additional requirements. See `research/20.33-known-consequence.md`.

E.8 20.8 and 20.11: free-group deductions with the exact quantifiers

For Problem 20.8, the Lei–Zhang construction supplies a counterexample [59]. In $F(a, b)$ put

$$u = (a^{-1}b)^2, \quad v = (a^{-2}b^2)^2, \quad H = \langle a, u, v \rangle, \quad K = \langle u, aua^{-1}, v, a^2va^{-2} \rangle.$$

The ranks are 3, 4, respectively. The homomorphism on the free basis of H sending a to b and fixing u, v differs from inclusion but fixes K : commutativity of $c_i = a^{-i}b^i$ with c_i^2 gives $a^i c_i^2 a^{-i} = b^i c_i^2 b^{-i}$. To check the additional intermediate-subgroup condition, identify H with $F(t, x, y)$. The core graph of $\langle x, txt^{-1}, y, t^2yt^{-2} \rangle$ has vertices 0, 1, 2, a t -path between them, x -loops at 0, 1, and y -loops at 0, 2. Its rank is four. Of its five vertex partitions, no identification and $0 = 2$ give rank-four folded images; the other three give the full three-loop rose. The image in the core graph of any intermediate subgroup is one of these. A rank-three image forces that subgroup to be all of H . Thus every proper intermediate subgroup has rank at least four, including infinitely generated ones. The ambient rank is two, which the question permits. Full folded graphs are retained in `research/20.8-known-consequence.md`.

For Problem 20.11, Jaikin-Zapirain's prior rigidity theorem [46] implies all four assertions, even when the initial subgroup F and ambient free group H are not finitely generated. Let $\mathcal{C}$ be the nonempty family of overgroups of F in H of least finite rank r . For $G_1, G_2 \in \mathcal{C}$, G_1 is compressed in $\langle G_1, G_2 \rangle$, hence inert by Corollary 1.2. Their intersection has rank at most r , and thus exactly r by minimality. Apply Corollary 1.5 to this now finitely generated intersection to show that the join also has rank r . A chain union has rank at most r : otherwise $r + 1$ members of a free basis generate a free factor lying in one chain member, forcing its rank too large. Zorn's lemma and pairwise joins give a greatest member. For a least member, work inside one countable G_0 . A countable subfamily has the same intersection

as all of $\mathcal{C}$, by choosing a subgroup excluding each element of G_0 outside that intersection. Finite intersections form a descending sequence of rank r . Any finite neighborhood in the intersection's Schreier graph eventually appears unchanged in those graphs. A finite subgraph of Betti number greater than r is therefore impossible. The intersection has rank r and is the least member. These limit steps are needed to preserve the printed quantifiers. See `research/20.11-known-consequence.md`.

E.9 19.9, 16.14, and 19.48: further prior answers

For Problem 19.9, both answers were already explicit in the pre-experiment GroupTheory50 preprint [38]. The archive's alternative deduction uses $G = \mathrm{Sp}(10, \mathbb{Z})$. Its nonabelian tensor square is its universal central extension, identified with the pullback of the real topological universal cover in this stable rank. Deligne's theorem makes that extension non-residually finite [31, 12]. It is finitely generated: lifts of generators generate any perfect central extension of a finitely generated group. Residual finiteness of finitely generated linear groups then excludes linearity over every field.

For braid groups B_n , $n \geq 4$, their abelianization $\mathbb{Z}$ splits the tensor square as $\mathbb{Z} \times (B_n \wedge B_n)$. The exterior square has kernel $H_2(B_n) = C_2$ over B'_n [10, 2]. In the rational Clifford algebra with $e_i^2 = 1/2$ and $e_i e_j = -e_j e_i$, put $u_i = e_i - e_{i+1}$. Then $u_i^2 = 1$, adjacent generators satisfy the braid relation, and distant generators anticommute. The induced map from the free presentation sends all relators to central signs and detects the nontrivial Hopf-kernel element $[x_1, x_3]$ as -1 . Combining it with the commutator image embeds the exterior square in $B'_n \times A_n^\times$. Left multiplication on the 2^n -dimensional Clifford algebra and the faithful Lawrence–Krammer representation [55] give a faithful tensor-square representation of dimension $1 + \binom{n}{2} + 2^n$. This detects the central kernel explicitly; it does not assume that every finite central extension of a linear group is linear. See `research/19.9-proof.md`.

For Problem 16.14, centrality of all involutions makes $E = \Omega_1(Z(G)) = \Omega_1(G)$ a maximal elementary abelian normal subgroup, of rank $r = d(Z(G))$. Sambale's earlier centralizer bound gives $|C_G(E) : \Phi(C_G(E))| \leq 2^{2r}$ [79]. Since $C_G(E) = G$ and $\Phi(G) = G^2$ for a finite 2-group, the requested inequality is $d(G/G^2) \leq 2d(Z(G))$. Products of quaternion groups attain equality. This is a specialization of the prior bound, not a new theorem; see `research/16.14-prior-result.md`.

For Problem 19.48, another pre-experiment GroupTheory50 preprint explicitly gives a nonclosed irreducible net [37]. With $R = \mathbb{Z}$, let α be a root of $X^3 - X - 1$ and $T = \mathbb{Z}[\alpha]$. Its exceptional levels are

$$A = \mathbb{Z} + \mathbb{Z}\alpha + 2\mathbb{Z}\alpha^2, \quad B = \mathbb{Z} + 2\mathbb{Z}\alpha + \mathbb{Z}(\alpha^2 - 1),$$

with other levels $2T$. The identity $\alpha^{-1} = \alpha^2 - 1$ allows root transvections to produce the diagonal $\mathrm{diag}(\alpha, \alpha^{-1})$, whose conjugation yields the forbidden upper parameter $\alpha^2 \notin A$. The common ideal $2T$ ensures the net conditions and nonzero levels. These are R -modules, as the question requires, rather than subspaces over the fraction field. The independently found characteristic-three example remains in `research/19.48-proof.md`. The earlier posted proofs are preprints; no outside human acceptance is inferred from their availability.

E.10 12.69: a literal counterexample retained as a formulation issue

The printed condition permits a nonzero element of $\mathbb{Z}G$ sending each field element into a subfield. For $F = \mathbb{Q}(i)$, $S = \mathbb{Q}$, and $G = \{1, \sigma\}$ with σ complex conjugation, the fixed element $1 + \sigma$ acts multiplicatively as the norm $x\sigma(x) \in \mathbb{Q}$. Its nonnegative exponents also give zero at $x = 0$. Additive interpretation gives the trace and the same conclusion. The extension is proper and separable. The example $F = \mathbb{C}$, $S = \mathbb{R}$ likewise contradicts the printed

claim that the uncountable analogue is affirmative. This points to a missing qualification or other formulation problem; changing from norm to trace does not repair it. The experiment retained the literal observation, proposed no invented correction, and counted no substantive resolution of the intended question. See `research/12.69-formulation-audit.md`.

F Nineteen representative carpet derivations

These tables complete the displayed finite proof of (5). A coordinate pair (m, n) denotes $ma + nb$, for simple roots a, b , with a short in B_2, G_2 . Each initial variable lies in the specified entry of A . Every subsequent line lies in B at its root, hence also in A . An application column records a Chevalley rule on earlier lines. Two applications on the same line have the stated coefficient gcd; Bezout's identity gives the element in the second column. The displayed final element has coefficient dividing the requested coefficient. Tables use one destination root of each length. Weyl transport, checked on all absolute rules, covers the other roots. The ancillary files also check all 104 targets without this reduction.

A_2 , **request 3**: $p = (1, 1)$. Requested coefficient 1. $X_1 \in A_{(1,1)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(0,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 1)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	$X_1 X_2$	$(0, 1)$	$z_2 z_1$
z_4	X_3	$(0, -1)$	initial
z_5	$X_1 X_3$	$(1, 0)$	$z_4 z_1$
z_6	$X_1^2 X_2 X_3$	$(1, 1)$	$z_5 z_3$

B_2 , **request 6**: $p = (1, 1)$. Requested coefficient 1. $X_1 \in A_{(1,0)}$, $X_2 \in A_{(1,1)}$, $X_3 \in A_{(-2,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 0)$	initial
z_2	X_2	$(1, 1)$	initial
z_3	X_3	$(-2, -1)$	initial
z_4	$X_2^2 X_3$	$(0, 1)$	$z_3 z_2^2$
z_5	$X_1 X_2^2 X_3$	$(1, 1)$	$z_4 z_1$

B_2 , **request 7**: $p = (1, 1)$. Requested coefficient 1. $X_1 \in A_{(1,1)}$, $X_2 \in A_{(0,-1)}$, $X_3 \in A_{(-1,0)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 1)$	initial
z_2	X_2	$(0, -1)$	initial
z_3	$X_1^2 X_2$	$(2, 1)$	$z_2 z_1^2$
z_4	X_3	$(-1, 0)$	initial
z_5	$X_1^2 X_2 X_3$	$(1, 1)$	$z_4 z_3$

B_2 , **request 8**: $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(0,1)}$, $X_2 \in A_{(2,1)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(0, 1)$	initial
z_2	X_2	$(2, 1)$	initial
z_3	X_3	$(-1, -1)$	initial
z_4	$X_2 X_3$	$(1, 0)$	$z_3 z_2$
z_5	$X_1 X_2^2 X_3^2$	$(2, 1)$	$z_1 z_4^2$

B_2 , **request 9**: $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(2,1)}$, $X_2 \in A_{(0,-1)}$, $X_3 \in A_{(-1,0)}$.

Line	Element	Root	Applications
z_1	X_1	$(2, 1)$	initial
z_2	X_2	$(0, -1)$	initial
z_3	X_3	$(-1, 0)$	initial
z_4	$X_1 X_3$	$(1, 1)$	$z_3 z_1$
z_5	$X_1^2 X_2 X_3^2$	$(2, 1)$	$z_2 z_4^2$

B_2 , **request 10**: $p = (2, 1)$. Requested coefficient 2. $X_1 \in A_{(2,1)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(2, 1)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	$X_1 X_2$	$(1, 1)$	$z_2 z_1$
z_4	X_3	$(-1, -1)$	initial
z_5	$X_1 X_3$	$(1, 0)$	$z_4 z_1$
z_6	$2X_1^2 X_2 X_3$	$(2, 1)$	$2z_5 z_3$

G_2 , **request 19**: $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(1,0)}$, $X_2 \in A_{(2,1)}$, $X_3 \in A_{(-3,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 0)$	initial
z_2	X_2	$(2, 1)$	initial
z_3	$3X_1 X_2$	$(3, 1)$	$3z_2 z_1$
z_4	X_3	$(-3, -1)$	initial
z_5	$X_2 X_3$	$(-1, 0)$	$z_4 z_2$
z_6	$X_2^2 X_3$	$(1, 1)$	$z_4 z_2^2$
z_7	$X_1 X_2^2 X_3$	$(2, 1)$	$z_5 z_3$; $2z_6 z_1$; $\gcd(3, 2) = 1$

G_2 , **request 20**: $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(0,1)}$, $X_2 \in A_{(2,1)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(0, 1)$	initial
z_2	X_2	$(2, 1)$	initial
z_3	X_3	$(-1, -1)$	initial
z_4	$X_1 X_3$	$(-1, 0)$	$z_3 z_1$
z_5	$2X_2 X_3$	$(1, 0)$	$2z_3 z_2$
z_6	$2X_1 X_2 X_3$	$(1, 1)$	$2z_2 z_4$
z_7	$3X_2^2 X_3$	$(3, 1)$	$3z_3 z_2^2$
z_8	$3X_1 X_2^2 X_3$	$(3, 2)$	$z_7 z_1$
z_9	$X_1 X_2^2 X_3^2$	$(2, 1)$	$z_3 z_8$; $2z_5 z_6$; $\gcd(3, 8) = 1$

G_2 , **request 21**: $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(1,1)}$, $X_2 \in A_{(2,1)}$, $X_3 \in A_{(-3,-2)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 1)$	initial
z_2	X_2	$(2, 1)$	initial
z_3	$3X_1 X_2$	$(3, 2)$	$3z_2 z_1$
z_4	X_3	$(-3, -2)$	initial
z_5	$X_2 X_3$	$(-1, -1)$	$z_4 z_2$
z_6	$X_2^2 X_3$	$(1, 0)$	$z_4 z_2^2$
z_7	$X_1 X_2^2 X_3$	$(2, 1)$	$z_5 z_3$; $2z_6 z_1$; $\gcd(3, 2) = 1$

G_2 , **request 22:** $p = (2, 1)$. Requested coefficient 2. $X_1 \in A_{(2,1)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(2, 1)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	$2X_1X_2$	$(1, 1)$	$2z_2z_1$
z_4	$3X_1^2X_2$	$(3, 2)$	$3z_2z_1^2$
z_5	X_3	$(-1, -1)$	initial
z_6	$2X_1X_3$	$(1, 0)$	$2z_5z_1$
z_7	$X_1^2X_2X_3$	$(2, 1)$	z_5z_4 ; $2z_6z_3$; $\gcd(3, 8) = 1$

G_2 , **request 23:** $p = (2, 1)$. Requested coefficient 1. $X_1 \in A_{(2,1)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(0,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(2, 1)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	$2X_1X_2$	$(1, 1)$	$2z_2z_1$
z_4	$3X_1^2X_2$	$(3, 2)$	$3z_2z_1^2$
z_5	X_3	$(0, -1)$	initial
z_6	$2X_1X_2X_3$	$(1, 0)$	z_5z_3
z_7	$3X_1^2X_2X_3$	$(3, 1)$	z_5z_4
z_8	$X_1^2X_2^2X_3$	$(2, 1)$	z_2z_7 ; $2z_3z_6$; $\gcd(3, 8) = 1$

G_2 , **request 32:** $p = (3, 2)$. Requested coefficient 3. $X_1 \in A_{(1,0)}$, $X_2 \in A_{(3,2)}$, $X_3 \in A_{(-2,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 0)$	initial
z_2	X_2	$(3, 2)$	initial
z_3	X_3	$(-2, -1)$	initial
z_4	X_2X_3	$(1, 1)$	z_3z_2
z_5	$3X_1X_2^2X_3^2$	$(3, 2)$	$3z_1z_4^2$

G_2 , **request 33:** $p = (3, 2)$. Requested coefficient 1. $X_1 \in A_{(1,0)}$, $X_2 \in A_{(3,2)}$, $X_3 \in A_{(-3,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(1, 0)$	initial
z_2	X_2	$(3, 2)$	initial
z_3	X_3	$(-3, -1)$	initial
z_4	$X_1^3X_3$	$(0, -1)$	$z_3z_1^3$
z_5	X_2X_3	$(0, 1)$	z_3z_2
z_6	$X_1^3X_2X_3$	$(3, 1)$	z_2z_4
z_7	$X_1^3X_2^2X_3^2$	$(3, 2)$	z_5z_6

G_2 , **request 34:** $p = (3, 2)$. Requested coefficient 1. $X_1 \in A_{(0,1)}$, $X_2 \in A_{(3,2)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(0, 1)$	initial
z_2	X_2	$(3, 2)$	initial
z_3	X_3	$(-1, -1)$	initial
z_4	$X_2X_3^3$	$(0, -1)$	$z_2z_3^3$
z_5	$X_2^2X_3^3$	$(3, 1)$	z_2z_4
z_6	$X_1X_2^2X_3^3$	$(3, 2)$	z_5z_1

G_2 , **request 35:** $p = (3, 2)$. Requested coefficient 1. $X_1 \in A_{(3,1)}$, $X_2 \in A_{(3,2)}$, $X_3 \in A_{(-2,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(3, 1)$	initial
z_2	X_2	$(3, 2)$	initial
z_3	X_3	$(-2, -1)$	initial
z_4	$X_2 X_3^3$	$(-3, -1)$	$z_2 z_3^3$
z_5	$X_2^2 X_3^3$	$(0, 1)$	$z_2 z_4$
z_6	$X_1 X_2^2 X_3^3$	$(3, 2)$	$z_5 z_1$

G_2 , **request 36:** $p = (3, 2)$. Requested coefficient 3. $X_1 \in A_{(3,2)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(-1,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(3, 2)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	X_3	$(-1, -1)$	initial
z_4	$X_1 X_3$	$(2, 1)$	$z_3 z_1$
z_5	$3X_1^2 X_2 X_3^2$	$(3, 2)$	$3z_2 z_4^2$

G_2 , **request 37:** $p = (3, 2)$. Requested coefficient 1. $X_1 \in A_{(3,2)}$, $X_2 \in A_{(-1,0)}$, $X_3 \in A_{(0,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(3, 2)$	initial
z_2	X_2	$(-1, 0)$	initial
z_3	X_3	$(0, -1)$	initial
z_4	$X_1 X_3$	$(3, 1)$	$z_3 z_1$
z_5	$X_2^3 X_3$	$(-3, -1)$	$z_3 z_2^3$
z_6	$X_1 X_2^3 X_3$	$(0, 1)$	$z_5 z_1$
z_7	$X_1^2 X_2^3 X_3^2$	$(3, 2)$	$z_4 z_6$

G_2 , **request 38:** $p = (3, 2)$. Requested coefficient 1. $X_1 \in A_{(3,2)}$, $X_2 \in A_{(0,-1)}$, $X_3 \in A_{(-3,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(3, 2)$	initial
z_2	X_2	$(0, -1)$	initial
z_3	$X_1 X_2$	$(3, 1)$	$z_2 z_1$
z_4	X_3	$(-3, -1)$	initial
z_5	$X_1 X_3$	$(0, 1)$	$z_4 z_1$
z_6	$X_1^2 X_2 X_3$	$(3, 2)$	$z_5 z_3$

G_2 , **request 39:** $p = (3, 2)$. Requested coefficient 3. $X_1 \in A_{(3,2)}$, $X_2 \in A_{(-1,-1)}$, $X_3 \in A_{(-2,-1)}$.

Line	Element	Root	Applications
z_1	X_1	$(3, 2)$	initial
z_2	X_2	$(-1, -1)$	initial
z_3	$X_1 X_2$	$(2, 1)$	$z_2 z_1$
z_4	X_3	$(-2, -1)$	initial
z_5	$X_1 X_3$	$(1, 1)$	$z_4 z_1$
z_6	$3X_1^2 X_2 X_3$	$(3, 2)$	$3z_5 z_3$

G Artifacts and reproduction

The configured repository address is <https://github.com/JUrban/kour1>. Anonymous access could not be established during manuscript preparation: unauthenticated API requests returned HTTP 404. A local Git query obtained a remote head different from the frozen experimental revision. These observations do not establish that either revision is anonymously available or that the two histories are equivalent. Readers need the frozen archive identified in Appendix A to reproduce repository-based measurements. The mathematical exposition in the paper does not require executing that archive.

The main entry points in the archive are `reports/FINAL_REPORT.md`, `reports/REVIEW_GUIDE.md`, and `research/complete-candidate-ledger.json`. The last file records the transcribed question, covered parts, proof paths, review paths, and evidence entry points for every candidate. The original PDF remains authoritative if an extraction differs from its typography.

The later `paper/` directory contains manuscript source, generated measurements, and a review ledger. From the repository root, the measurement command is

```
python3 paper/scripts/analyze_experiment.py --check
```

It uses Python’s standard library and Git, reads the supplied HTML and text exports, and recomputes the statistics from the frozen history. It checks agreement with committed outputs rather than modifying them. Without `--check`, it regenerates the five measurement files.

File in <code>paper/data/</code>	Meaning
<code>experiment-metrics.json</code>	Input hashes, event counts, cumulative counters, Git totals, and measurement limitations
<code>session-tokens.csv</code>	All retained token observations, including repeated and unitemized records
<code>session-visible-messages.jsonl</code>	Visible assistant, final, and human messages with event identifiers; excludes directive bodies
<code>research-commits.csv</code>	Commit identifiers, UTC timestamps, subjects, and inclusion in the research interval
<code>proof-artifact-timeline.csv</code>	First committed principal proof artifact for each problem family on the final candidate list

The full JSONL is analyzed separately:

```
python3 paper/scripts/analyze_rollout.py --check
```

This standard-library parser reads the single supplied `session/rollout-*.jsonl`, whose size is 540,278,152 bytes and whose SHA-256 digest is

```
f316fb63957f3ec50895088537786a4946a15601e54da5dac86ec0e4fcdd21d8.
```

It checks unique response and completed-item identifiers, timestamp order, usage identities, the exact compaction reconciliation, every HTML token observation, and agreement with `ccusage-supplied.json`. Its six outputs are `rollout-metrics.json`, `rollout-usage.csv`, `rollout-observations.csv`, `rollout-tasks.csv`, `rollout-hourly.csv`, and `rollout-visible-messages.jsonl`. These retain aggregate metrics, request usage, completed-operation metadata, task events, hourly activity and timestamped visible messages, respectively. Without `--check`, the parser regenerates them.

The phase boundary uses the logged request timestamp. For example, a 90,535-token request is recorded at 12 September 20:56:42.957 UTC, before the deadline, but its cumulative notification appears at 20:56:49.929 UTC after a tool completes. Using only the last

pre-deadline notification would omit that request from research usage. The separately identified compaction requests likewise cannot be reconstructed by summing the legacy “Last” display fields.

The raw session inputs remain separate from both source bundles. Their hashes and sizes identify the evidence for an authorized holder; the included projections are not substitutes for the full trace. No internal reasoning, compaction-summary or system/developer instruction bodies are included in the derived data. Mathematical scripts have different purposes: some generate examples, some check certificates, and some summarize existing results. Reproducing a summary is not rerunning a proof check. Each result’s archive entry identifies its commands and the limits of the computation.

Large computations need substantial time and storage. In particular, the $n = 7$ certificate for Problem 20.100 is distinct from smaller diagnostic and intermediate files. The retained report and manifest specify the complete input and all completed checker shards. The manuscript’s review does not silently replace that historical verification with a fresh abbreviated run.

The portable source includes the complete integer derivations for Problems 19.61–19.62, their target-coverage checkers, and the integral G_2 matrix-identity check. In the arXiv source archive these are under `anc/`; in the full manuscript project they are under `ancillary/`. Their README gives three standard-Python commands, which require neither GAP nor network access. A manifest binds the 23 original files to their byte-identical research versions. The 19 short representative derivations are also printed above.

The main file is `main.tex`. The source archives include `main.bbl` and the compiled chronology figure, so ordinary XeLaTeX/BibTeX or Tectonic can typeset the paper without regenerating the data or running mathematical computations. The full-source archive additionally contains the aggregate measurements, the frozen candidate ledger, figure source, build scripts, and manuscript review records. Recomputing the original session/Git measurements still requires the separately retained experimental archive and session inputs.

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